

Geared Rotor Dynamic Methodologies for Advancing Prognostic Modeling Capabilities
in Rotary-Wing Transmission Systems

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the faculty of the School of Engineering and Applied Science
University of Virginia

In Partial Fulfillment
of the requirements for the Degree
Doctor of Philosophy in Mechanical and Aerospace Engineering

by

David Blake Stringer

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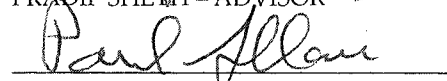
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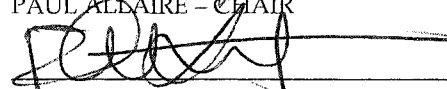


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ABSTRACT

The evolution of health management technologies for predicting component life remaining in complex mechanical systems requires modeling algorithms based upon first principles. These physics-based models provide an alternative source for comparing and interpreting electronic signals from components of interest, without necessarily resorting to expensive and time consuming experimental techniques. Physics-based models can simulate multiple operating conditions to include fault-based scenarios. The results are particularly useful in generating informational databases in support of predictive algorithms.

In rotorcraft, one such mechanical system is the drive train. Helicopters rely on the integrity of their drive trains for their airworthiness. Drive trains rely on gear technology for their integrity and function. Gears change the orientation of shaft output, transfer power between shafts, and reduce the rotational speeds between the power plant and rotor systems. Dynamics associated with geared motion are complex. Gears alter the vibration characteristics of a mechanical system and significantly contribute to noise, component fatigue, and personal discomfort prevalent in rotorcraft.

This research presents methodologies for generating a rotor dynamic model of a rotary-wing transmission based on first principles, through the following steps: development of a three-dimensional gear-mesh stiffness model for helical and spur gears, linear and nonlinear analyses of a geared system for comparison and validation of the gear-mesh model, development of a modal synthesis technique for potentially

providing model reduction and faster analysis capabilities for geared systems, and extension of the gear-mesh model to bevel and epicyclic configurations.

The overarching objective of this research is the development of a robust, rotor dynamic model of a helicopter drive train using finite elements (Chapter 9). In addition to model construction and validation, faults indigenous to geared systems are presented. Two such faults are simulated. Diagnostic vibration parameters are presented and used as damage indicators in the analysis. The fault models produce results consistent with damage experienced during experimental testing.

The results demonstrate the robustness of the physics-based approach in simulating multiple normal and abnormal conditions. The advantages of this physics-based approach, when combined with conventional probabilistic and time-series techniques, provide a useful method for improving health monitoring technologies in mechanical systems.

DEDICATION

To Mom, I wish you were still with us. Thank you for your love and support and helping me to believe. I dedicate this to you. Rest in peace.

"For the Lord gives wisdom,
from His mouth come knowledge
and understanding;"

Proverbs 2:6

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ACRONYMS

ARDS	Analysis of Rotor Dynamic Systems
BPFI	ball pass frequency, inner race
BPFO	ball pass frequency, outer race
BSF	ball spin frequency
CBM	condition-based maintenance
CF	crest factor
DOF	degrees of freedom
EHD	elastohydrodynamic
FEA	finite element analysis
FFT	fast Fourier transform
FTF	fundamental train frequency
GUI	graphical user interface
HUMS	health usage and monitoring system
LOA	line-of-action
NASA	National Aeronautics and Space Administration
NOTAR	no tail rotor
ODE	ordinary differential equation
PHMS	prognostic and health management system
RMS	root mean square

SYMBOLS

A	cross-sectional area
A_o	outer cone distance
$[\mathbf{A}_i]$	original state-space matrix
a_n, b_n	Fourier coefficients
B_d	ball or roller diameter
$[\mathbf{B}_i]$	transformed state-space matrix
$\begin{bmatrix} & \end{bmatrix}_b$	bearing matrix
C_g	gear-mesh damping coefficient
$[\mathbf{C}]$	general damping matrix
c_{xx}, c_{yy}	bearing damping values in x and y -directions, respectively
D	generalized force
D_o	distance from apex to midpoint width of gear
$[\mathbf{D}]$	sum of damping and gyroscopic matrix
d	difference signal
$\bar{d}$	mean difference
d_i	diagonal damping matrix component; diameter of gear (i)
d_g	gear diameter
d_{si}	shaft inner diameter
d_{so}	shaft outer diameter
$\begin{bmatrix} & \end{bmatrix}_d$	disk (gear) matrix
E	elastic (Young's) modulus
$\begin{bmatrix} & \end{bmatrix}^{(e)}$	element matrix
e_u	eccentricity of unbalance

eq	equivalent shaft
F	forcing amplitude
$\{\mathbf{F}\}$	force/moment vector
$\bar{\mathbf{F}}_n$	normal force
$\{\mathbf{F}_0\}$	excitation force amplitude vector
$\{\mathbf{F}(t)\}$	excitation force vector
f	gear facewidth
G	shear modulus
$[]^{(G)}$	global matrix
$G(\omega_f)$	frequency response function
$[\mathbf{G}]$	gyroscopic matrix
h	gear height
I	area moment of inertia
I_r	inertial ratio
I_t	transverse mass moment of inertia
$[\mathbf{I}]$	identity matrix
j	imaginary component, $\sqrt{-1}$
K_g	gear-mesh stiffness coefficient
K_{go}	average gear-mesh stiffness coefficient
ΔK_g	periodic gear-mesh stiffness coefficient
$[\mathbf{K}]$	stiffness matrix
$[\mathbf{K}_{ii}]$	gear-mesh stiffness matrix sub-quadrant
$[\mathbf{K}]_{mesh}$	gear-mesh stiffness matrix
$[\mathbf{K}_\Omega]$	stiffness hardening matrix due to high-speed carrier rotation
k	stiffness

k_h	epicyclic bearing direct stiffness, $h = c, r, s$
k_{hu}	epicyclic bearing rotational stiffness, $h = c, r, s$
k_i	diagonal stiffness matrix component
k_{pn}	planet (n) bearing direct stiffness
k_q, K_q	translational and rotational coordinate elastic stiffnesses, respectively
k_{rn}	ring-planet mesh stiffness
k_{sn}	sun-planet mesh stiffness
k_{xx}, k_{yy}	bearing stiffness values in x and y -directions, respectively
L	Lagrangian function; length
$[L]$	transformation sub-matrix designation
l	element length
M	moment amplitude
$[M]$	mass matrix
m	mass
m_i	diagonal mass matrix component
N	number of degrees of freedom; number of data points
N_b	number of balls or rollers
N_{vi}	number of teeth of equivalent spur gear (i)
P	pitch point; number of reduced degrees of freedom
P_d	bearing pitch diameter
P_i	diametral pitch of gear (i)
$\{Q_o\}$	harmonic response amplitude vector
$\{q\}$	generalized coordinate vector
R_{mi}	mean radius of gear (i) at the pitch point
R_{vi}	equivalent spur gear radius of gear (i)
r_g	radius of gyration

r_h	radius of carrier, ring, or sun gears,, $h = c, r, s$
r_i	radius of gear (i)
S	signal data-point value
$\bar{S}$	mean signal value
s	displacement along line-of-action
$[\]_s$	shaft matrix
T	kinetic energy
$[T]$	transformation matrix
Δt	numerical integration time step
$\{u\}$	displacement vector; first derivative of principal coordinate vector
V	potential energy
$\{v\}$	principal coordinate vector
w	periodic mesh-stiffness frequency scaling factor
x, y, z	translational degrees of freedom
x_h, y_h, u_h	epicyclic lateral and rotational degrees of freedom, $h = c, r, s$
Z	orbit
α	real eigenvalue component
α_n	spur or helical gear pressure angle
α_r, α_s	epicyclic pressure angles
β	imaginary eigenvalue component; helical angle
Γ_i	pitch angle of gear (i)
$[\Gamma]$	diagonalizing matrix
ζ	damping ratio
η_n, ζ_n, u_n	tangential, radial, and rotational degrees of freedom of planet (n)
θ	contact angle

$\theta_x, \theta_y, \theta_z$	rotational degrees of freedom
κ	shape factor
$[\Lambda]$	diagonal eigenvalue matrix
λ	eigenvalue
ν	Poisson's ratio
$[\Xi]$	elimination matrix
ρ	density
Φ	transverse shear parameter
$[\Phi]$	reduced system modal matrix
ϕ	phase angle
ϕ_x, ϕ_y, ϕ_z	direction cosines
φ	orientation angle
$\{\Phi\}$	eigenvector
$[\Psi]$	eigenvector modal matrix
ψ	spiral angle
ψ_n	circumferential angle of planet (n)
Ω	shaft rotational speed
ω	system frequency
ω_d	damped natural frequency
ω_f	forcing frequency
ω_{mesh}	gear-mesh frequency
ω_n	natural frequency

Subscripts

c	carrier
D	diagonal
Dm	transformed mesh
l	left
m	number of substructures of global system
r	ring; right
s	sun
sys	global system to include coupling effects

Superscripts

T	transpose
$'$	first-order vector, scaled or normalized
$*$	global system excluding coupling effects

Chapter 1

Introduction, Scope, and Contributions

This chapter provides the motivational framework and technological relevance of this research effort. A brief introduction to rotorcraft technology and its resulting maintenance considerations are presented. The importance of mathematical models to predictive capabilities for emerging maintenance processes is discussed. Focusing on the transmission as a unique and critical helicopter subsystem, the current state-of-the-art in drive train technology is presented. Using this framework, this chapter ultimately defines the scope of work, research objectives, and contributions to the state-of-the-art.

1.1 Rotorcraft Dynamic Considerations

Unlike their graceful fixed-wing counterparts, rotorcraft, some of which are depicted in Figure 1.1, are complex dynamic systems unlike any other. A conventional airplane separates the lift and thrust-producing mechanisms of the aircraft. Wings provide lift. A propeller or turbine engine provides thrust. The empennage (tail) provides stability. Control surfaces, such as elevators, ailerons, and rudder, provide controllability.

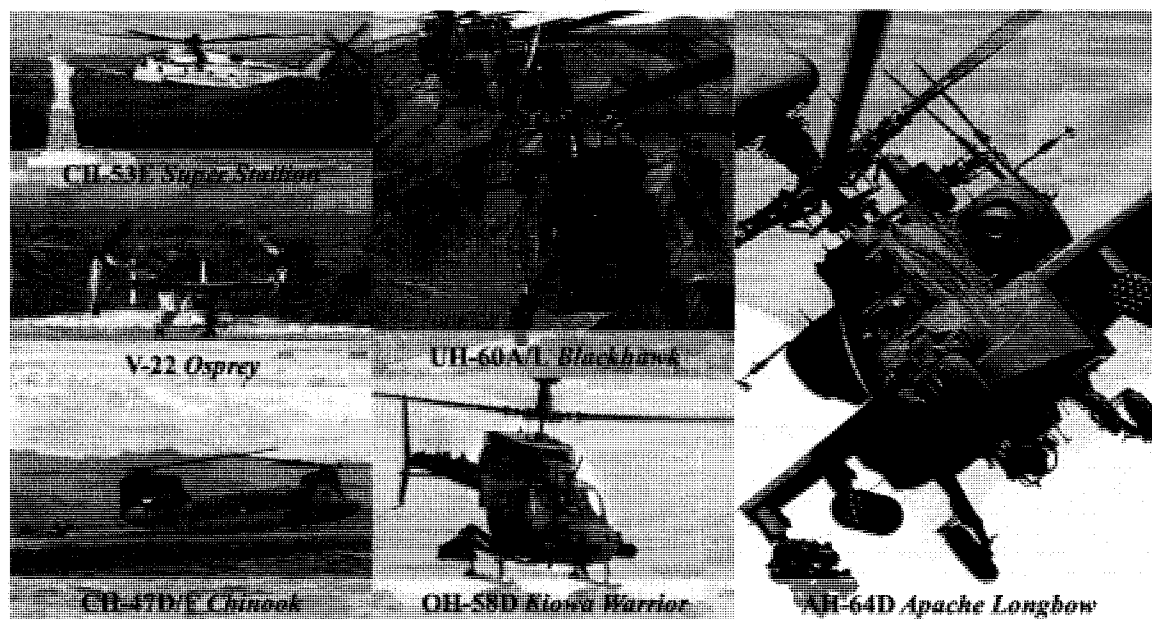


Figure 1.1: Contemporary Variants of Military Rotorcraft

In contrast, a conventional rotorcraft uses a spinning main rotor and tail rotor to provide all these functions simultaneously. A turboshaft engine(s) powers the rotor system, producing torques of large magnitude as a by-product. Conventional tail rotors, the newer NOTAR (NO Tail Rotor) systems, or counter-rotating main rotors provide the necessary anti-torque mechanisms. The rotor also introduces gyroscopic motion into the system, making stability and control considerations significantly more complex than in fixed-wing aircraft.

These are just some of the factors which create a myriad of structural and dynamic effects and lead to excessive noise, vibration, personal discomfort, and component fatigue. Consequently, rotorcraft maintenance and direct operating costs are significantly higher than fixed-wing aircraft. As a result, rotorcraft require a thirtyfold increase in safety to equal that of conventional fixed-wing aircraft (Zakrajsek, 1993). This has resulted in efforts to incorporate new practices aimed at reducing life-cycle

costs, and increasing rotorcraft affordability and overall aircraft safety. Since 1998, one technological effort has taken aim at lowering maintenance costs, specifically, through the incorporation of a new maintenance practice, the condition-based maintenance initiative.

1.1.1 Condition-Based Maintenance Initiative

The condition-based maintenance initiative (CBM) is a “maintenance action based on actual condition ... obtained from in-situ, non-invasive tests, operating and condition measurements,” to optimize lifetime ownership of a system (Mitchell, 1998). CBM practices lead to increased safety through lower failure rates, higher availability, greater reliability, and decreased maintenance costs.

CBM represents the most advanced of a three-tiered maintenance posture, as depicted in Figure 1.2 (Bengtsson, 2004). The first tier is the familiar corrective action technique. Maintenance occurs when a system breaks down. The second tier is pre-determined maintenance, also known as schedule-based maintenance, and is in popular use today on automobiles and aircraft. Maintenance occurs on a fixed schedule. The third tier is the CBM initiative. Each tier represents the evolution of the maintenance process, characterized by decreasing failure rates with each progressive shift in maintenance philosophy. This evolution is captured in Figure 1.2, with associated advantages, (+), and disadvantages, (-), of each (Bengtsson, 2004).

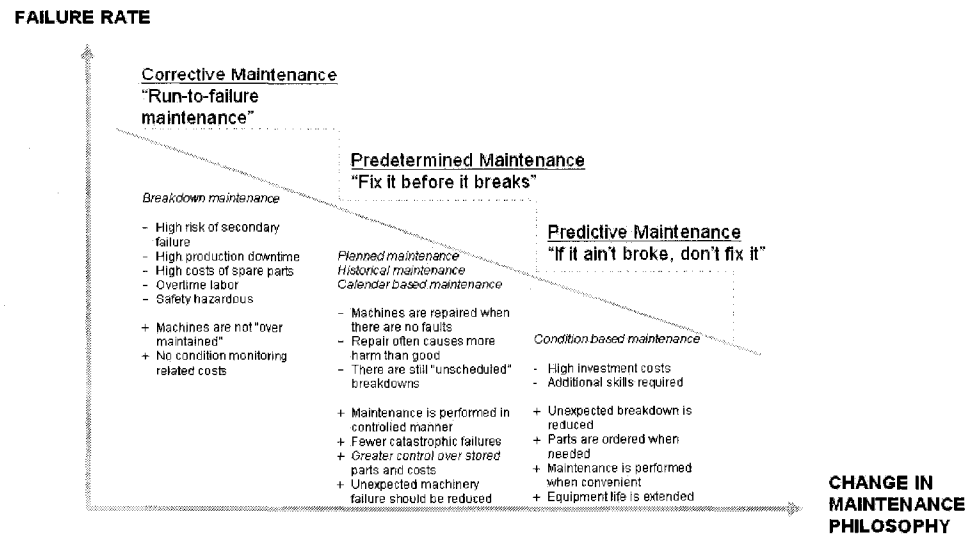


Figure 1.2: Maintenance Postures with Pros (+) and Cons (-) of Each (Bengtsson, 2004)

Rotorcraft maintenance has traditionally followed conservative, schedule-based maintenance techniques to ensure safety and reliability. For example, a main rotor blade may have a schedule of 500 flight hours. At the end of 500 hours, the blade is replaced, regardless of condition. All major components such as rotor blades, engines, transmissions, tail rotor gearboxes, etc., follow this conservative schedule.

Over the past twenty years, maintenance and diagnostic procedures and the quality of robustly engineered systems have continued to improve. These effects, combined with the constant need for cost reductions have created a paradigm shift toward CBM maintenance practices. A CBM maintenance criterion might require rotor blade replacement when the wear on the blade's leading edge exceeds a certain tolerance. This tolerance may not be exceeded until 1,000 flight hours. Conversely, it may be exceeded at 275 hours, although the latter is highly unlikely. The CBM initiative enables the blades to be replaced on condition, preventing unnecessary and costly maintenance. Thus, CBM practices provide the promise of reduced maintenance downtime and increased aircraft availability, which significantly reduce operating costs. Furthermore,

by constantly monitoring component condition, CBM technology greatly increases overall safety and reliability of the aircraft.

1.1.2 Health Monitoring Systems

A critical milestone to CBM implementation is the maturation of health management technologies, specifically in Health Usage and Monitoring Systems (HUMS) and Prognostic and Health Management Systems (PHMS). One goal of CBM technology is to monitor system operation and provide the operator with an accurate assessment of the system's current status. This capability is performed by HUMS systems. More importantly, the technology should provide an accurate prediction of the remaining useful life of the system. Such is the primary function of the PHMS (Samuel, 2005, Garga, 1997). The literature provides several methods for the diagnostic evaluation of mechanical systems. However, the accuracy of the diagnostics and what they report are still in question. While much research is being devoted to this problem, health monitoring systems with reliable prognostic capabilities are still an emerging technology and have yet to be fully realized.

1.1.3 Modeling System Health

Many damage detection methods have been proposed using different techniques to include statistical metrics, neural network applications, time-frequency analysis, wavelet analysis, adaptive signal processing, and mathematical modeling. However, until recently, mathematical modeling has been relatively overlooked (Samuel, 2005). To date, the primary techniques used in these mathematical models involve either time-series analysis, such as the autoregressive or autoregressive moving average modeling methods (Box, 1976), or dynamic data systems analysis (Wu, 1977).

However, the physics-based approach, or first principles analysis, is an important alternative for investigation, because time-series analysis is limited in the information it can provide. Chapter 2 compares the first principles and time-series approaches as analysis techniques and discusses the advantages and disadvantages of each.

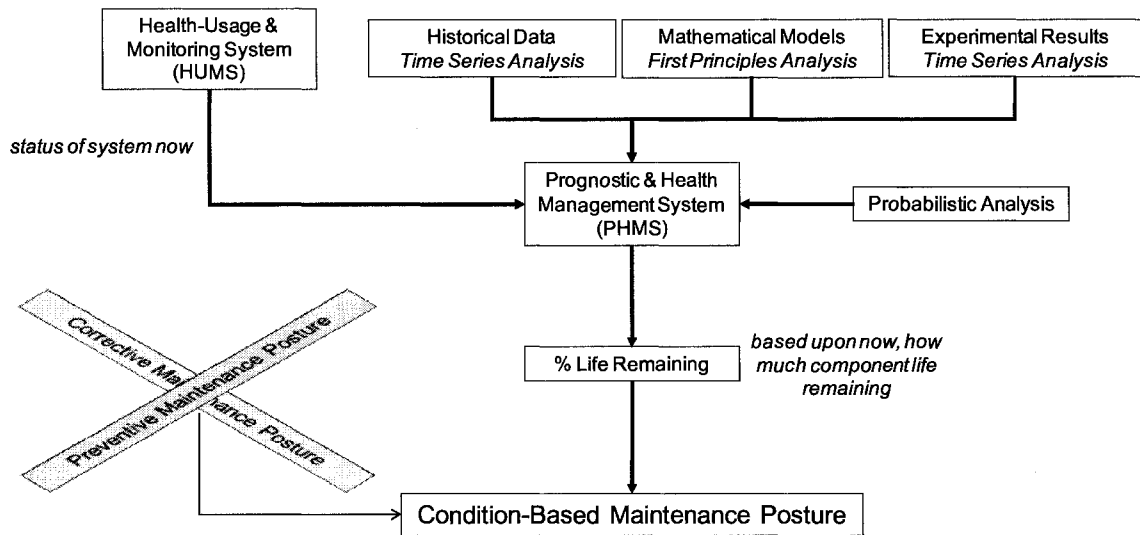


Figure 1.3: Prognostic and Health Management Information Sources (Stringer, 2008)

The CBM initiative requires an advanced digital architecture, similar to the CBM prognostic system presented by Lebold (2003). Mathematical models play an important role in the digital structure, providing critical information to the prognostic algorithm, as illustrated in Figure 1.3 (Stringer, 2008). For health monitoring capability, algorithms require a comprehensive database of system information to analyze and compare with actual signals gathered from system components. The sources of this database information include historical data from aircraft components, experimental data from test facilities, and numerical data generated from mathematical models. Mathematical models are particularly useful in allowing the user to simulate several different operating conditions and populate a database for a variety of conditions and faults.

After model validation, it is also possible to simulate conditions for which historical data or experimental results may be sparse or non-existent (Stringer, 2008). Finally, mathematical modeling also provides a means of conducting other types of system analysis, such as system identification for design or optimization studies.

1.2 Rotorcraft Transmission and Gearing Technology

Current aerodynamic and structural limitations in rotorcraft require a drive system for reducing the high-speed output of the engine to the lower speeds of the rotor system. Such a transmission system is illustrated in Figure 1.4 (Lewicki, 1987). The power train performs many functions. It transfers and reduces the output of the propulsion system to the main rotor and tail rotor components. It physically attaches the rotor system to the fuselage. Consequently, the transmission is subjected to, among others, dynamics from the power plant and the unsteady aerodynamic forces acting on the main rotor and tail rotor systems.

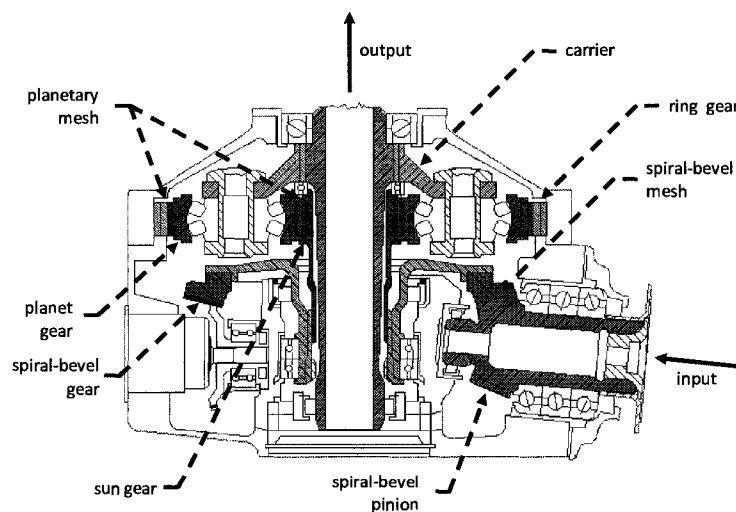


Figure 1.4: OH-58A Helicopter Transmission (Lewicki, 1987)

The transmission system is therefore perhaps the most critical component to safe rotorcraft operation. Rotorcraft rely on the integrity of their drive trains for their airworthiness (Valco, 1992). Unlike a fixed-wing aircraft which can glide in the event of an engine failure, rotary-wing aircraft depend upon their autorotational capabilities for power-off landings. Maintaining adequate rotor speed for safe autorotation requires the drive train clutch to disengage from the engine shaft in the event of power failure. The most threatening in-flight emergencies, other than fire, in the CH-47D *Chinook* helicopter operator manual deal with the transmission system and include the step, "Land as soon as possible" (US Army, 2003). Similarly worded procedures exist for other rotary-wing airframes as well (US Army, 1988, 2002).

The overarching mechanism in the transmission for power transfer and speed reduction is gear technology. In addition to external dynamic influences, the transmission is exposed to dynamic influences from the internal forces caused by gear-mesh interactions. The transmission is a primary source of the noise, vibration, and component fatigue produced in rotorcraft. These effects are further discussed in the next section in terms of their contribution to rotary-wing accidents and safety.

1.2.1 Rotorcraft Accident Data – Transmission Related Accidents

The transmission has been the cause of several accidents in both military and civil rotorcraft over the past 40 years. For example, the United States Navy reports that a majority of the Navy's Class A accidents (complete destruction of aircraft and/or loss of personnel) have been due to failures in the power plant or drive train (Hess, 2002).

In 2000, a similar study was conducted for civil rotorcraft accidents reported between 1963 and 1997 (Harris, 2000). The largest number of accidents fell into the

category of human error, approximately 59% of the total. The second highest category was “loss of engine power” at 29%. The remaining 12% were due to mechanical malfunctions.

Focusing on the 12% of accidents due to mechanical malfunction, deficiencies in the drive train accounted for 40% of all mechanical problems, the most of any other mechanical system on board to include certain power plant systems. Between 1964 and 1997, this equates to 13 accidents per year. From these statistics, the transmission’s significance is readily apparent.

1.2.2 Drive Train Technology – Post Year 2000 State-of-the-Art Research

The focus of rotary-wing drive train technological research in this decade has been oriented toward the development of smaller and lighter rotorcraft transmissions with low noise and vibrations, and increased reliability and affordability. Technology enablers include increased component loading, fewer gear reduction stages, high temperature operation, analytical model development, and diagnostic/prognostic tool implementation. As such, the current state-of-the-art in rotorcraft drive research falls into one of the following three areas (Handschuh, 2006):

1. Analysis code and algorithm development.
2. Advanced gear materials, lubrication, and design.
3. Drive system component fault detection and prognostics.

1.3 Research Objectives and Scope

The objective of this research is to present the results of a physics-based model of a helicopter transmission as a basis for fault detection and prediction in support of CBM initiatives. Modeling methodologies for different gearing types are discussed in detail.

The model is a finite element, rotor dynamic model incorporating gyroscopic, damping, and gear-interaction effects. Modal analysis and substructure synthesis techniques are presented and discussed. Fault conditions, fault-seeding methodologies, and health monitoring capabilities of the model are demonstrated and discussed.

The following details subsequent chapter topics in terms of the scope of work:

1. Chapter 2 compares and contrasts the physics-based and time-series analysis approaches as individual techniques of dynamic analysis.
2. Chapter 3 presents a comprehensive literature review on the evolution of gear research and rotorcraft technology over the last 150 years. The review illustrates the helicopter's role in furthering gear and transmission research and the linkage between the two.
3. Chapter 4 establishes the basic rotor dynamic modeling and finite element principles used in the analysis. It provides the analytical foundation for the developments in this research.
4. Chapter 5 provides a detailed discussion of the development of both 12 and 6 degree-of-freedom (DOF) gear-mesh stiffness models for a spur and helical gear pair. The discussion includes application of the 12-DOF model to a geared shaft system for validation with published data in the literature.
5. Chapter 6 proposes a preliminary methodology for determining the instantaneous gear-mesh stiffness, an inherently nonlinear problem. The nonlinear equations of motion are solved using numerical integration. The frequency content is determined using a Fast Fourier Transform (FFT) of the time data. The results of the linear and nonlinear analyses are compared and discussed.

6. Chapter 7 presents a method of performing modal synthesis on a geared rotor dynamic system that includes non-proportional damping and gyroscopic effects, utilizing the state-space formulation of the equations of motion.

7. Chapter 8 extends the gear-mesh methodology of Chapter 5 to bevel and spiral-bevel gears. An epicyclic gear train model is introduced.

8. Chapter 9 combines the previous chapters through the construction of the rotorcraft transmission model. A discussion of transmission faults is presented. The model as a health indicator is demonstrated through two simulations of system damage.

1.4 Contributions

This effort provides the following contributions in two of the three focus areas of the drive system state-of-the-art: analysis code and algorithm development, and drive system component fault detection and prognostics.

1. Gear-Mesh Model

- a. Spatial decomposition of the forces and moments acting on a spur or helical gear set, resolved into a general 12-DOF, lumped parameter mesh stiffness matrix.
- b. Incorporation of linear and nonlinear gear-mesh stiffness effects.
- c. Extension of approach to bevel gear configurations. Incorporation of epicyclic gear modeling techniques.
- d. Extension to a general 6-DOF mesh stiffness matrix in the context of a distributed parameter, structural dynamic analysis.

2. Modal Synthesis

- a. An accurate technique for modal reduction of a geared system model, accounting for both general damping and gyroscopic effects.

- b. Identification of the limits to available DOF reduction imposed by gearing and gyroscopic effects.

- c. Method of modal control for use in design, optimization, and other system identification schemes.

- d. Possible alternative for the quick computation of Campbell Diagrams.

3. System Transmission Model.

- a. Integration of the methodologies described above into a physics-based, finite element model of a rotorcraft power train system. This provides a complex system model of multiple shafts connected by multiple gear pairs for incorporating the methodologies presented, and determining its effectiveness for health monitoring and prognostic applications

- b. A platform for developing and characterizing the nature of faults/failures within the drive train.

Chapter 2

System Dynamic Analysis Techniques

Due to cost and time constraints, modeling has become a preferred alternative to experimentation and classical analytical techniques. Classic analytical modeling techniques have only been successful at the most basic levels, restricting their use to the classroom environment. Consequently, complex problems have traditionally been solved through empirical means, at the sacrifice of increased cost and time.

The evolution of computers at all levels has far exceeded the pace of any other development in the last 20 years. The high-end computing systems of today are over 100,000 times faster than those used in 1990 (Information Technology Advisory Committee, 2005). The result has been an increase in modeling capability, establishing computational techniques as an equal discipline of science and engineering, complimenting theoretical and experimental techniques.

Over the years, different modeling techniques have developed. This discussion focuses on the physics-based and time-series approaches, comparing and contrasting the two, and discussing the importance of each to prognostic technology.

The primary difference between the two approaches is the type of model, deterministic or stochastic. A model developed using a set of laws or relationships such that the same input always yields the same output is a deterministic model. A common example is a missile launched with known velocity and trajectory. The position, velocity, and acceleration of the missile at any given time can be calculated exactly.

A model where inputs do not necessarily result in the same output is stochastic or probabilistic in nature. These models include a dependence on probability distributions associated with certain parameters. A common example is a prediction model for the number of insurance claims filed for the upcoming fiscal year. All prognostic models are stochastic.

2.1 Physics-Based Models / First Principles Analysis

A physics-based, first principles analysis (FPA) model provides a framework based upon physical laws. These models describe the physical behavior of the system using the first principles of mass, momentum, energy, and Newton's laws of motion. FPA models usually take the form of a differential equation, such as in Equation 2.1:

$$m\ddot{x} + kx = Fe^{j\omega t} \quad (2.1)$$

Equation 2.1 is recognizable as the equation of motion for an undamped, single DOF system. The mass, (m) , and stiffness, (k) , variables are the state variables for a given system. The right-hand-side of the equation is a forcing function, (F) , providing input to the system. The displacement variable, (x) , represents the output of the model analyzed over time. The FPA approach is deterministic. The only way to obtain different results is to change the input or state variables.

2.2 Time-Series Analysis

Analysis of time-series data was first presented in 1927 (Tsay, 2000). The ultimate goal of this approach is forecasting the future based upon previous data, a stochastic process. A time-series is a set of continuous or discrete observations, ordered in time.

Although statistical analysis of time-series has existed since the early 20th Century, it was not until the 1970's that Box and Jenkins (1976) developed and published a systematic approach for applying time-series analysis to forecasting. The fundamental premise of their methodology asserted that the data presented in time-series are dependent on past occurrences and that the dependence itself is significant.

There are three steps in time-series modeling: system identification, parameter estimation, and diagnostic checking. System identification determines the type of model that best represents the time-series data. Parameter estimation consists of obtaining system parameters and performing goodness-of-fit tests. Many techniques are available to include likelihood estimation, least squares estimation, nonlinear estimation, Bayesian estimation, etc. Lastly, diagnostic checking validates the model as an accurate representation of the data.

2.3 Comparison of Approaches

Both approaches are necessary for prognostic health monitoring capabilities. Predictive models are stochastic. However, achieving accuracy in predictions requires a FPA approach to provide the necessary physical information on which to base predictive and decision-making algorithms.

2.3.1 Physics-Based/First Principles Approach

The physics-based, FPA approach is based upon natural laws and provides a quantitative description of the physical phenomena in a system. As such, it has several intrinsic advantages over time-series models. First, FPA are applicable to any problem, since the laws of physics theoretically always apply. These models are well-suited to

general problems and provide very precise descriptions of nature (Olson, 2006, Blankenship, 1998).

Furthermore, the FPA approach does not rely on comparisons to experimental data or a table of interpolated values to produce results. As a result, these models produce better trends outside the database of available experimental results and reduce the error associated with extrapolation (Olson, 2006).

FPA modeling possesses another advantage with new technologies. Physics-based models can directly simulate the insertion of new technology and produce results for analysis. In non-deterministic models, new technology is applied through a correction factor that attempts to predict how the new technology is going to affect the system. An example is the effect of new materials on the structural dynamics of a given system.

FPA also provides the capability for modeling a host of different conditions by rapidly changing the input or state parameters. The FPA model can simulate realistic conditions, ideal conditions, as well as damage or fault conditions. Finally, the First Principles Analysis generates detailed and comprehensive data, which can be used to help explain differences between expectations and results (Blankenship, 1998).

The FPA modeling approach does have its disadvantages, many of which revolve around complexity, cost, and time. A first principles model can become very complex, very quickly. The engineer must also develop the computational structure required to model the system. The complexity of the system dictates the time involved in developing the model.

FPA models find their most significant disadvantage in the area of prediction technologies. FPA analysis has not been successful in demonstrating this capability for a number of reasons. One reason stems from the inability of the FPA model to account for

all physical processes or the outcomes of a physical process. For example, in a battle damage scenario, a vehicle may be hit by a small-arms missile. The FPA model can predict the amount of penetration and energy transfer. However, it cannot predict those processes resulting from the initial damage such as fire or the ability of the vehicle to function after the strike (Blankenship, 1998).

The processing speed of a first principles model is slower than that of a stochastic model and is also a function of complexity. However, the computing cost and time factors are becoming less of an issue with the continuing leaps in the evolution of computer processing power. The processing speed of a first principles model may not be conducive to decision-making processes, where the time between initiation of a problem and component failure may be shorter than the ability of FPA to diagnose and alert the operator to the problem (Blankenship, 1998). This can be mitigated by assembling model results in a database for predictive algorithms to scan, compare, and interpret information.

Despite these disadvantages, the physics-based modeling approach has evolved into an integral part of the engineering process, increasing the engineer's understanding of the design, operation, and physical processes of complex systems (Moin, 2006).

2.3.2 Time-Series Analysis Approach

The disadvantages of the FPA model become advantages in a time-series environment. The model is much less complex, reducing the processing time and cost. The development of time-series models does not require the time or complexity of the FPA model. Model fitting and parameter estimation techniques are available using many commercial statistical software packages.

For a specific problem or set of conditions, time-series analysis can provide a good approximation of the system, depending upon the level of precision and accuracy desired. Finally, the stochastic nature of the time-series model makes it a convenient and necessary method for use in prognostic and health management technologies (Blankenship, 1998, Samuel, 2005).

However, the use of time-series also implies several limitations (Tsay, 2000, Yaffee, 2000). The first depends upon the amount and type of data. An insufficient or excessive amount of data can greatly reduce the forecasting capability of the model. Additionally, time-series analysis does not facilitate the mixture of discrete and continuous variables. Missing data from a time-series complicates the model, as well as unequally-spaced data series over time.

Another limitation of time-series analysis is temporal focus. If the user does not have the proper temporal focus, the underlying trends needed for the model could be missed (Yaffee, 2000). For example, if a data series contains information on a monthly basis, but the trends of the series occur on a yearly basis, the model may not produce the desired output.

Finally, the most significant disadvantage with a time-series model is the lack of robustness, the inability to apply the model to any desired condition (Olson, 2006). A time-series model may be very accurate and precise when modeling the operating conditions for which it was intended. When departing from that condition, the stochastic model becomes much less accurate.

2.4 Summary Comparison

Figure 2.1 provides a summary comparison of the FPA and time-series modeling approaches. Both approaches are important to engineering analysis and play a key role in the development of system diagnostics and prognostics.

First Principles Analysis

- Deterministic analysis
- Model derived from principles of physics; applies to all physical processes
- Model generates data
- Increased generality: applicable to many conditions: ideal, realistic, and unnatural or undesirable conditions
- Ability to simulate new technologies within the model
- Slower processing speeds and increased computing costs*
- Reduced capability of predicting performance (prognostics)
- Generates complete information
- Increased complexity; model takes time to set up and develop.

Time Series Analysis

- Stochastic Analysis
- Parameter estimation derived from model fitting and system identification
- Existing data generates the model
- Decreased generality: applicable to specific problems or configurations
- Infusion of new technologies requires correction factors to measure predicted deviation of model
- Faster processing speeds and lower computing costs*
- Increased capability of predicting performance (prognostics)
- Generates limited information
- Reduced complexity; easier to set up

**Evolution of computers and falling computing costs are quickly negating this as a factor.*

Figure 2.1: Summary Comparison of Modeling Approaches

The usefulness of time-series model is its potential to forecast future states of the system. Its ability to do that successfully depends upon the system data available, which is where the synergistic relationship between first principles and time-series modeling becomes important. In the schematic presented in Figure 1.4, the FPA model is a source of data encompassing many different operating conditions. It provides a basis for a comprehensive database of information, based upon several scenarios,

simulating the entire range of the system's operation. The decision-making algorithms, in turn, compare and verify real-time data with the database to determine component life remaining. Through the time-series approach, the PHMS performs its prognostic algorithms to estimate and predict the remaining useful life. The time-series component provides predictive capability. The physics-based, first principles model ensures the accuracy of the prediction.

Chapter 3

A History of Rotorcraft and Gearing Theory

Rotorcraft and gear technology have been linked for over seven decades. Advances in one have often led to developments in the other. The purpose of this chapter is to provide both a historical perspective of rotorcraft development as well as an in-depth look into the evolution of gearing analysis methods over the past 150 years.

3.1 A Brief History of Rotorcraft

The first roots of vertical lift technology date from antiquity, beginning with Chinese tops (*circa.* 400 B.C.) and Archimedes' water screw (*circa.* 225 B.C.). Leonardo da Vinci sketched his aerial screw around 1483 (Leishman, 2000). Some have even suggested that da Vinci was perhaps the first to understand and begin man's study of and quest for vertical flight (Wood, 1956).

At the turn of the Nineteenth Century, Sir George Cayley, the "father" of modern aerodynamics, experimented with model helicopters and a rotating whirl stand to study and determine the aerodynamic forces acting on a lifting wing (Anderson, 2005). Although ultimately regarded as the inventor of the modern airplane, Cayley's 1843 aerial carriage detailed a vertical flight configuration, consisting of two pairs of tandem rotors. Cayley's contributions also marked a turning point in the quest for flight, from an inventive pursuit to a scientific one (Leishman, 2000).

The Frenchman, Ponton d'Amécourt, built many helicopter models in the 1860s, and coined the term *helicopter*, derived from the Greek *elikoeioas* for "spiral" or "winding," and *pteron* for "wing" or "feather" (Leishman, 2000). Others also experimented with rotorcraft models between 1880 and 1920, achieving various degrees of success, to include Thomas Edison (Wood, 1956).

Full-size prototypes were less successful, due to fundamental technical problems, stemming from a misunderstanding of rotorcraft aerodynamics, appropriate power plants, weight considerations, anti-torque, stability and control, and vibration issues (Leishman, 2000, Sikorsky, 1938). However, although these early attempts failed, some did provide beneficial results. In the early 1920s, the Argentinean, Raul Pescara, first successfully applied cyclic pitch to periodically warp the rotor blades for controllability. In the late 1920s, Austrian Raoul Hafner first incorporated a swashplate for pitch control (Leishman, 2000).

3.1.1 First Successes: de Bothezat, de La Cierva, and Sikorsky

Military interest in vertical flight technology began in 1918 at the end of World War I (Wood, 1956, Hardy, 1956). The United States Army first contracted with Russian-born Georges de Bothezat to produce its first helicopter. On December 18, 1922, his quad-rotor helicopter made its first flight demonstration and hovered a few inches off the ground for just under two minutes (Wood, 1956). However, the complexity of Bothezat's system and the Army's desire for a single-rotor configuration resulted in the cancellation of the project. Interest in the further development of the helicopter waned as the Army turned its attention to the Spaniard, Juan de la Cierva, and his Autogiro (Wood, 1956, Leishman, 2000).

The autogiro, while not a vertical flight machine, used a propeller and main rotor combination. The propeller provided thrust to move the aircraft forward, directing air through the rotor blades, causing them to autorotate and produce lift (de La Cierva, 1931). Although the aircraft could not hover, it required a very short space to takeoff. The autogiro very nearly matched the performance expected from a helicopter. It was safe, stall-proof, with the ability to land in confined areas. It would take another pioneer, Igor Sikorsky, to prove that vertical flight was actually possible.

The Russian-born Sikorsky's interest in aviation grew from the pioneering efforts of von Zeppelin and the Wright Brothers. Influenced by da Vinci and Jules Verne, he envisioned a vertical takeoff and landing machine powered by a lifting propeller. His first attempts at a helicopter were in 1909, but ultimately the machine could not produce enough lift. Sikorsky then turned to fixed-wing aviation. In 1913, he developed and became the first person to pilot a four-engine aircraft, the Russian S-21 *Grand*, which was also used as a bomber during World War I (Sikorsky Historical Archives).

With the Communist takeover of Russia, Sikorsky fled and arrived in the United States in 1919. By 1923, he had established the Sikorsky Aero Engineering Corporation. The company successfully produced passenger aircraft for the evolving airline business to include the 1931 S-40 *American Clipper* amphibious airplane (Sikorsky Historical Archives).

In 1931, Sikorsky returned to helicopters and patented what became the conventional helicopter, a single main rotor and tail rotor configuration (Sikorsky Historical Archives). He received permission to develop the helicopter in 1937, and on September 14, 1939, the VS-300 left the ground for the first time. The first free flight occurred on May 13, 1940 (Sikorsky Aircraft, 1956).

Sikorsky received contracts to develop the helicopter for military use. On May 13, 1942, the XR-4 helicopter completed the first helicopter cross-country flight, delivering the first helicopter to the United States Army. Four hundred Sikorsky R-series helicopters would serve in World War II (Sikorsky Aircraft, 1956).

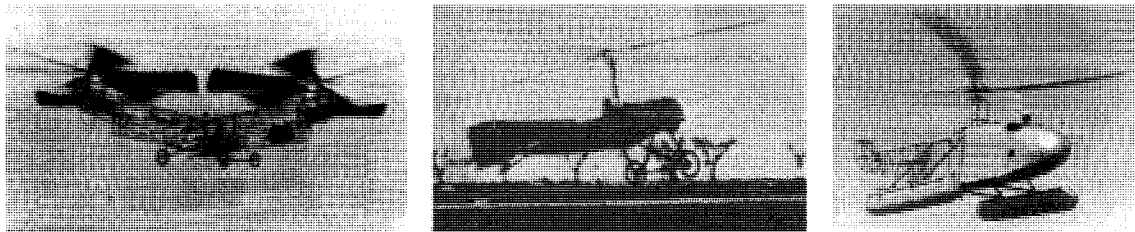


Figure 3.1: Rotorcraft Pioneers de Bothezat (left), de La Cierva (center), and Sikorsky (right)

3.1.2 Later Developments

The Navy was one of the first organizations to dissent from Sikorsky's conventional helicopter as the only "practical" configuration (Knapp, 1956). Understanding the need for higher payload capacities, the Navy researched other configurations. The Piasecki Helicopter Corporation, which had produced the United States' second commercial helicopter, contracted with the Navy to build the first large transport helicopter. In 1945, the XHRP-1 was the first successful American tandem helicopter, a configuration using two main rotor systems. The Piasecki Corporation (now Boeing Helicopters) would become very successful through their development and upgrades of the CH-46 *Sea Knight* and CH-47 *Chinook* series helicopters (Leishman, 2000).

In 1946, the commercial helicopter transportation market achieved a significant milestone. The Bell Helicopter Corporation received the world's first commercial helicopter certificate (Butterbaugh, 1956). Bell would later become extremely successful with the UH-1 and OH-58 series helicopters and civilian counterparts.

The Kaman Helicopter Corporation pioneered the use of gas turbine engines as its power plants. In 1951, Kaman produced the world's first turbine-engine helicopter by adding a turbine to the successful K-225. Later, Kaman developed the first twin turbine-engine helicopter, the HTK-1 (Leishman, 2000, Kirchner, 1956).

In later years, rotorcraft technology continued to mature with notable aircraft such as the Sikorsky H-60 series of helicopters (*Blackhawk*, *Seahawk*, etc.), Boeing's AH-64 *Apache* attack helicopter, and Bell/Boeing's most recent V-22 *Osprey* tilt-rotor aircraft. The past seventy years of rotorcraft development have witnessed their development from "noisy, unstable, rickety, vibrating contraptions" to complex, sophisticated, and versatile aircraft (Leishman, 2000). Over the last twenty years, the primary thrust of effort has been at reducing the undesirable by-products of rotary-wing use to include aerodynamic limitations, vibration, and noise.

3.2 The Evolution of Gear Analysis

The emergence of the turbine engine as the primary rotorcraft power plant in the 1950s required a re-design of helicopter drive trains. Operating speeds of turbine engines were significantly higher than their piston engine counterparts, while the rotating outputs of the rotor system remained relatively unchanged. This required transmissions to provide much higher gear reduction ratios, resulting in increased operating speeds, noise, fatigue, and vibration (Valco, 1992). The introduction of multi-engine technology complicated the gearbox design even further. Since gear technology is critical to drive train integrity, it is important to look at the evolution and types of gear analysis and modeling developed over the last 150 years.

The first question is how to categorize the development of geared research. There are two ways. The first is an approach used in an extensive review of gear dynamics models published by Özgüven and Houser (1988). They loosely grouped mathematical models into five general classifications: (1) simple dynamic factor models, (2) tooth compliance models, (3) gear dynamics models, (4) geared rotor dynamic models, and (5) torsional vibration models. These classifications do not apply for every model, and a particular model may have more than one classification.

The second is an easier method which discusses the evolution of geared analysis through periods in the development timeline. The different periods can be characterized by a primary focus or theme of scientific effort and more easily show the development of analysis techniques through the decades.

The use of gears as a means of energy and power transfer also dates back to antiquity. Writings attributed to Aristotle from the Fourth Century, B.C. refer to gears or circles in friction, transferring energy (Winter, 2007). The Antikythera Mechanism, the earliest known surviving historical geared machine, dates back to Ancient Greece (Price, 1974).

Gearing methods by skilled artisans continued through the middle ages, most notably in astrolabes for astronomy or for the intricate clock towers dotting the European landscape. It was not until the Industrial Revolution of the 19th Century that the demand for precision gear performance emerged, requiring gear technology became to become solidly founded on scientific principles.

The earliest models dealt with loading and stress calculations in gear teeth. In 1868, Walker first suggested a "speed factor," which would later become the dynamic factor, a penalty factor on gears based upon their rotational speeds (Özgüven, 1988). In 1892,

Wilfred Lewis first calculated the flexural strength of gear teeth with reasonable accuracy. He approximated a gear tooth as a cantilever beam with uniform load acting over the entire face width of the tooth. His Lewis Equation is still used in contemporary gear manuals (Coy, 1985, Dudley, 1984).

In the early 1900s, studies revealed shortcomings in the Lewis Equation, namely that it was too conservative (Valco, 1992). Timoshenko and Baud (1926) published the first article referencing the calculation of gear tooth deformation. Modifying Lewis' work, they modeled the gear tooth as a tapered cantilever beam. They also discussed the implications of manufacturing inconsistencies and vibration on gear loading (Ingram, 2006).

Baud and Peterson (1929) investigated the load sharing of multiple teeth in contact, observing that when two or more pairs of gear teeth were in contact, the load was not equally distributed between tooth pairs. Noting that the deflection of gear teeth was, for all purposes, linear with respect to loading conditions, they introduced an "elasticity factor," along the line-of-action. This represented, in essence, the first use of a spring constant to model the gear-mesh itself (Valco, 1992).

In 1931, as a research committee head in the American Society of Mechanical Engineers, Earle Buckingham published the first authoritative work on dynamic tooth loading. One of the results of this analysis was the Buckingham Equation for calculating dynamic tooth load, assuming a single pair of gear teeth in contact (Coy, 1985, Dudley, 1984). Buckingham would continue to improve upon his equation in his successive works (1949) and extend it to accommodate helical gears, bevel gears, and other types of configurations as well (Valco, 1992).

Similar types of studies continued through the 1930s and into the 1940s. The theme of these early decades could be characterized as studies oriented toward stress, load factor, and tooth deformation calculations (Özgüven, 1988, Valco, 1992).

In the 1950s, vibration models first appeared in gear analysis with Tuplin's (1950, 1953, 1958) introduction of the first spring-mass model of a gear pair. Tuplin's model assumed constant gear-mesh stiffness and studied the effects of pitch error, the positional error of a tooth in mesh (Tuplin, 1950). Tuplin modeled pitch error by inserting a wedge moving vertically at a constant speed, as illustrated in Figure 3.2. Through this simple model, Tuplin was able to approximate dynamic loads for various errors. However, his model was limited to speeds well below resonance (Özgüven, 1988).

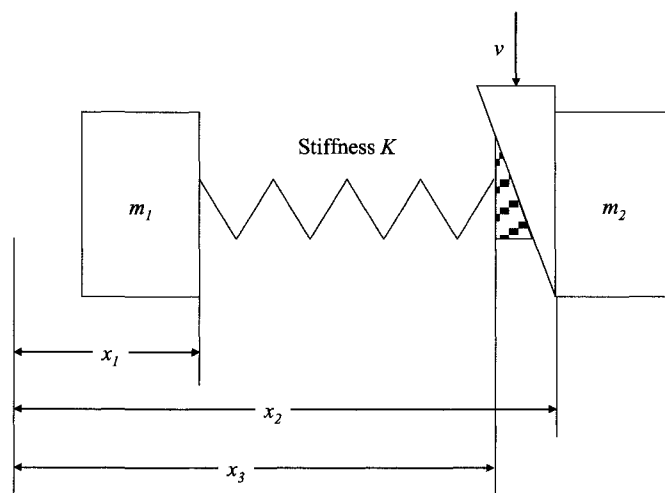


Figure 3.2: Tuplin's Gear Tooth Model (Tuplin, 1950).

The 1950s saw greater interest develop in the periodic nature of the gear-mesh stiffness. The first in-depth study into periodic fluctuation of gear-mesh stiffness is credited to the German engineer, Strauch (Özgüven, 1988). Strauch (1953) studied the changes in mesh stiffness due to the transition between the number of tooth pairs in

contact. Nakada and Utagawa (1956) introduced another periodic waveform. Their model approximated the mesh stiffness as a time-varying rectangular waveform using a vibrating model of mating gear teeth (Özgüven, 1988).

Harris (1958) attributed the gear dynamics of spur gears to internal vibration stemming from variations due to manufacturing errors, periodic stiffness variations, and nonlinearities in the tooth stiffness. Harris' work was the first to detail the importance of transmission error and instabilities arising from excitations of the gear-mesh (Özgüven, 1988).

A more important observation from Harris' study dealt with the periodic variation of the gear-mesh stiffness. In his analysis, Harris used a spatially varying mesh stiffness. With a periodic stiffness, the natural frequencies from the gear-mesh were not constant. Harris was able to show that the gears tended to vibrate at a natural frequency corresponding to the "mean stiffness" of the teeth (Harris, 1958). This became, in essence, the basis for modeling the dynamics of the gear-mesh using a time-averaged stiffness.

In the early 1960s, Johnson (1962) became one of the first to use mesh stiffness in modeling geared shaft systems. Until then, most spring-mass models still dealt with determining stresses, fatigue, and the dynamics associated with the gears themselves. Johnson's work produced a system representation. His model was the first in a new classification of gear models, those for geared rotor dynamics (Özgüven, 1988). Similar gear-shaft studies such as Seirig (1966) would continue to develop into the 1960s and beyond.

In the 1970s, Mitchell and Mellen (1975) provided experimental evidence of the cross-coupling between geared rotor shafts as a result of the gear-mesh interface. Their

conclusions called for the development of coupled, lateral-torsional modeling techniques for geared systems in order to correctly design, analyze, and interpret vibratory characteristics. Additionally, Kiyono (1978a, 1978b), conducted some early theoretical and experimental vibration analysis investigations into helical gears. His research showed the torsional vibration component to be dominant in helical gear dynamics.

Iida conducted a series of investigations (1980, 1984, 1985, 1986) using a 3-DOF model to calculate the first three modes of vibration and the response due to imbalance on a rigid shaft supported by hydrodynamic bearings. Eventually, he would apply this to a three-shaft geared system coupled by two gear meshes.

Transfer matrix method techniques were used extensively in the geared system studies of Daws (1979), David (1984, 1986, 1987), and Blanding (1985). They studied both linear and nonlinear dynamic coupling effects in geared rotor systems for noise reduction studies. Using this method, they illustrated the usefulness of transfer matrices in analyzing parametric systems with large coefficient variations.

Rotor dynamic modeling has typically employed either the transfer matrix method or the finite element method (Younan, 2004). The transfer matrix method was computationally efficient compared to the finite element method, but was less accurate (Zu, 2002). Moreover, in the 1980s, advances in computing power gave rise to the increasing popularity of finite element analysis.

Early finite element analysis methods coupled the lateral and torsional dynamics of geared rotor systems. Neriya (1984, 1985a, 1985b), applied finite element analysis to Iida's model (1980) using a two-mass, two-spring, and two-damper gear model with constant mesh stiffness. Kahraman (1990a) used a finite element model to model the

lateral-torsional vibrations of gear shafts and the lateral vibration of rotor bearings. The study preferred linear systems for large system models, because they provided accurate results at a fraction of the computing time. However, Kahraman also conducted studies in nonlinear gear dynamic analysis (Kahraman, 1990b, 1991a).

Singh (1990a) presented a steady-state analysis investigating the nonlinear behavior of a spur gear pair. Singh generated a finite element model to investigate the nonlinear frequency responses of the gear pair with backlash, and the response of the rotor-bearing system. The investigation focused on the effects of lightly and heavily loaded gear systems.

One of the first attempts to develop a system model of a rotorcraft transmission system began in the late 1980s. Choy (1988) presented a dynamic analysis of multi-mesh gearboxes. The analysis consisted of comparing experimental data with two analytical computer codes developed under NASA and Army co-sponsorship. He also published a method for predicting the overall system dynamics of a gearbox (Choy, 1989). Choy's model provided the first three modes of vibration and included effects such as geometric complexity, support flexibility, mass imbalance, and shaft bow.

Later, Choy (1990, 1991) further improved the previous model and presented an analytic procedure to simulate vibrations in a geared transmission system. This technique built upon his previous efforts through a combined approach of modal synthesis and finite element analysis. Modal characteristics were determined using the transfer matrix method. The gearbox structure was modeled using finite elements. A nonlinear gear-mesh coupled the gear stages together. The bearing supports coupled the gearbox structure to the gear system (Choy, 1993).

Rao (1995) provided another finite element formulation for determining the coupled vibration characteristics of a geared rotor system. His work produced a sensitivity analysis for determining the effects of pressure angle, gear-mesh stiffness, and type of bearing on the system natural frequencies and mode shapes.

Özgüven and Houser published their extensive review of the various types of gear dynamics modeling techniques in 1988. Blankenship and Singh (1992b) also published a comparative study of selected gear-mesh models. These and other techniques presented prior to 1995 considered the force generated in the gear-mesh to be a scalar quantity rather than a vector. The primary sources of noise and vibration were attributed to kinematic transmission error and variation in mesh stiffness, while discounting coupled motions caused by gear manufacturing errors, etc. (Blankenship, 1992a). Many geared modeling analyses up to this time, dealt almost exclusively with spur gears, with only few exceptions, such as Kiyono (1978a, 1978b).

In the early and mid-1990s, Blankenship (1992a,b, 1995a,b) presented a 6-DOF gear-mesh model employing both helical and spur gears. His model analyzed a multidimensional vector of forces and moments transmitted through the gear mesh. This was one of the first instances of a comprehensive gear-mesh model appearing in the literature, although Iida and Neriya had called for such methods previously (Iida, 1985, Neriya, 1984).

Blankenship assumed rigid gear bodies, except for the elastic compliance of the meshing gear teeth, and neglected any tooth separation. Additionally, all energy dissipative effects were assumed to occur only in the gear-mesh. Sliding friction in the helical gears was neglected, ensuring that all resultant forces acted normal to the gear teeth. Blankenship completed a rigorous and complex development of the system

equations. The model was then compared to selected linear time-varying models from the literature. A linear time-invariant simplification was also presented.

The results of Blankenship's approach (1995a, 1995b) concluded that the forces and moments generated and transmitted through the gear-mesh interface required additional DOFs in complex geared models. Acknowledged drawbacks of this approach were the inability to reduce the model in general and the complexity required for analytical prediction. The number of parameters required for model accuracy also added to the complexity.

Luo (1996) extended Choy's transmission modeling approach. Choy's work dealt with a spur gear train, which did not include axial vibration coupling effects as in a helical gear system. Luo proposed a more general approach coupling the lateral-torsional-axial vibration of the system. In this approach, each shaft was modeled as an individual substructure. The global system matrix was then developed using the modal synthesis technique. The lateral-torsional-axial coupling was accounted for in the form of an increment of the stiffness matrix. Non-zero elements of the incremental stiffness matrix were used to calculate the global characteristics of the system.

Choi (1999) modeled dynamic gear loading in response to unexpected vibration characteristics occurring on a geared turbo set, specifically severe vibrations occurring in the axial direction of a helical geared set. In this approach, Choi modeled the gear pair as a set of rigid disks with linearly distributed springs over the contact line. Through transmission error analysis, he was able to determine the source of the instability and re-design the turbo set to eliminate the high vibration.

One of the more recent modeling approaches was presented by Wang (2002). This approach modeled a geared system using the Delft Finite Element Theory (Werff, 1977).

In this formulation, the definition of an element consists of both the nodal coordinates and deformation coordinates. The gear element was shown to account for time-varying mesh stiffness, transmission errors, and lateral-torsional vibration coupling.

Finally, Ebrahimi (2006) presented an approach for a contact analysis of meshing gear teeth, by considering the elastic effects between the gear teeth and body. Applicable to both spur and helical gear teeth, both tooth and body were assumed to be rigid with an elastic element between them. Through a rigid gear tooth contact analysis, the resulting gear forces and moments are determined. The contact analysis was then extended to the elastic elements between the gear body and gear teeth.

3.3 Gear-Mesh Stiffness Approximations

One of the primary differences in the gearing analyses discussed in this chapter has been the approach to modeling the gear-mesh stiffness. These methodologies for gear-mesh approximations have been described as both a major source of error in dynamic models (David, 1986), and a valuable source of simplification without undesirable consequences (Kiyono, 1978). The mesh stiffness itself is a combination of (1) localized contact stiffness influenced by the elastohydrodynamic (EHD) behavior, (2) the elastic beam deflections of the tooth itself, and (3) the stiffness of the gear disk or the ring (Stringer, 2007).

The mesh interaction between two gears has traditionally been modeled using a spring-dashpot combination as pictured below in Figure 3.3.

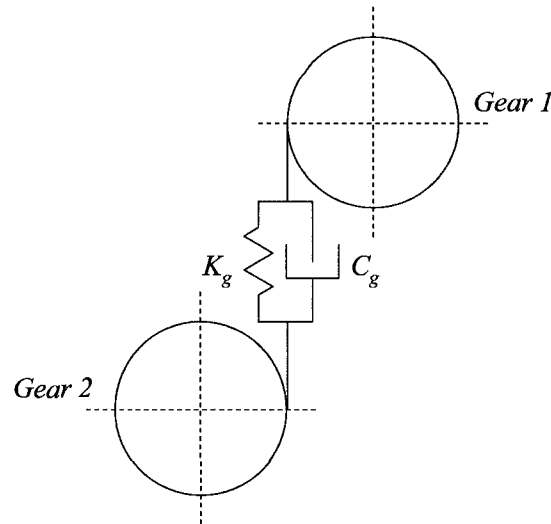


Figure 3.3: Typical Spur or Helical Gear-Mesh Model

where (K_g) and (C_g) are the mesh stiffness and damping coefficients, respectively.

Some techniques also account for the friction consistent with the rolling motion of gear teeth within the mesh through a friction coefficient, such as in Choy (1993) or Ebrahimi (2006).

The stiffness magnitude is not constant through the gear-mesh cycle, due to the number of teeth in contact at any given moment. A full contact analysis of the mesh cycle is complex and cannot be effectively performed in the context of a contemporary dynamic analysis, because of the complexity of the instantaneous meshing parameters (Blankenship, 1995a). As such, closed form solutions of the mesh cycle do not exist.

The dynamics of the gear tooth through a meshing sequence fall into three specific periods. The first corresponds to initial tooth pair engagement while the preceding tooth pair is still engaged. The second period occurs when the single tooth pair itself is engaged. The final period begins when the succeeding tooth pair engages and ends with the initial tooth pair's disengagement. Consequently, with a cyclical variation of

the number of teeth in contact, the mesh stiffness fluctuates periodically. An example of this is presented by Lin (1987) in Figure 3.4 below.

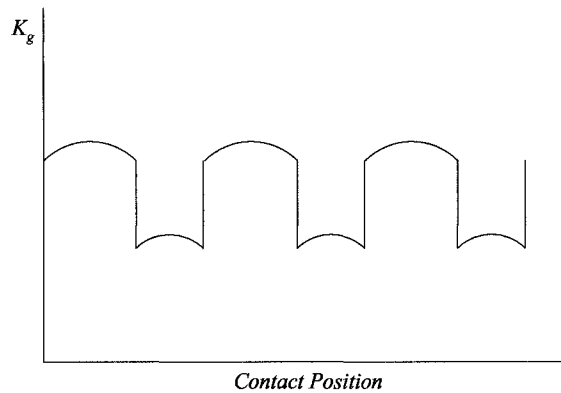


Figure 3.4: Sample Gear-Mesh Stiffness Variation Pattern (Lin, 1987).

The literature generally presents three mesh stiffness techniques: spatially-varying, time-varying, or linear time invariant approaches. An example of a spatially-varying expression was given in Umezawa (1986) for the mesh stiffness of narrow facewidth helical gears. A time-varying stiffness formulation of a spur gear-mesh was presented by Kahraman (1991b). In both instances, the periodic coefficients were based upon a curve-fit of experimentally observed data. The time-varying and spatially-varying formulations are related and are further discussed in Chapter 6.

The linear time-invariant mesh stiffness has been the most common modeling approach. It is represented as the time-averaged mesh stiffness, as first demonstrated and discussed by Harris (1956). Lim (1991) presented a 4-DOF linear time invariant stiffness matrix for a spur gear, mounted on a flexible shaft-bearing rotor. Blankenship and Singh (1995a, 1995b) proposed a linear time invariant simplification to their complex mesh development. The use of the time invariant approximation invokes additional considerations, which will be presented in Chapter 6.

The magnitude of the average gear-mesh stiffness can be either an assumed value or the results of a curve-fit approximation. The literature contains several variations of the magnitude, but the order of the magnitude is relatively the same. Previous assumed magnitudes in the literature have been $1.0\text{E}8 \text{ N/m}$ (Kahraman, 1990a, Lim, 1991), $2.0\text{E}8 \text{ N/m}$ (Neriya, 1985b), and $3.0\text{E}8 \text{ N/m}$ (Blankenship, 1995b), and some higher (Kahraman, 1991b). A common “rule of thumb” introduced by Houser and Singh (2006) presents the average mesh stiffness as $\sim 2\text{E}6 \text{ lb/in}$ per inch (*in*) of facewidth or $\sim 1.38\text{E}7 \text{ N/m}$ per millimeter (*mm*) of facewidth. These rule-of-thumb values match well with those presented from the literature.

Experimental results show average mesh stiffness approximation to be valid, especially for highly loaded gear systems (Harris, 1958). This is illustrated in Figure 3.5 (Stringer, 2008) by observing the difference in the same periodic stiffness profile at different operating speeds. The time averaged stiffness assumption becomes more and more sufficient as the operating speed increases and the gears become more highly loaded.

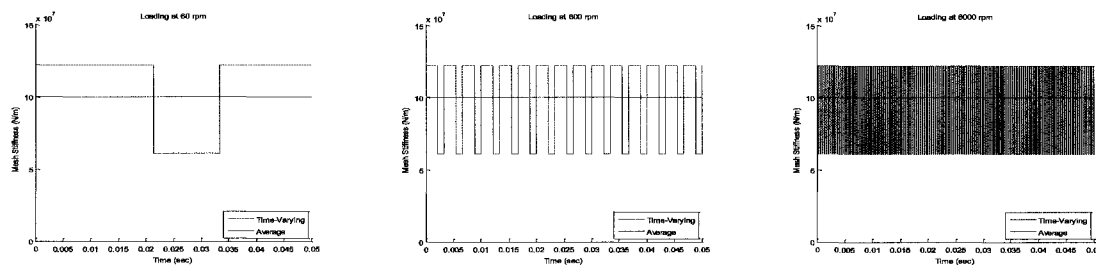


Figure 3.5: Mesh Stiffness Profiles at 60, 600, and 6,000 *rpm*, respectively (Stringer, 2008).

3.4 Summary

This chapter has provided an in-depth look into the evolution of the helicopter, modern gear theory and its application to rotor dynamics, and the approaches to

modeling the gear-mesh stiffness. In concluding, the following is a brief highlight of the key events and contributions to gearing science and rotorcraft technology.

Antiquity provides the first evidence of gears and rotary-wing technology with the writings of Aristotle, the Antikythera Mechanism of Ancient Greece, and the invention of Chinese tops in the First Millennium, B.C. Leonardo da Vinci produces sketches of the aerial screw in the Fifteenth Century. In the late Eighteenth Century, Sir George Cayley performs aerodynamic experiments using a rotating wing whirl stand. His aerial carriage uses rotors for providing lift.

During the Industrial Revolution, the theory of gearing becomes more organized and scientific as the need for efficient gearing systems emerges. The dynamic factor is attributed to Walker in 1868. The Lewis Equation is developed around 1898 for determining tooth stress. The Buckingham Equation is introduced in 1931. The first use of an equivalent spring for modeling the gear-mesh occurs in the 1920s by Baud and Peterson. These first and subsequent studies through the 1940s are primarily focused on tooth stress and dynamic factor analyses.

With the Wright Brothers first successful flight in 1903, interest in rotary-wing and fixed-wing development explodes. The military in particular invests in rotary-wing research, beginning in 1918. Igor Sikorsky and others build helicopter prototypes in the early 1900s with little success. Rotary-wing technology continues to improve with de la Cierva's autogiros in the 1920s and de Bothezat's first helicopter in 1922. Sikorsky returns to rotary-wing development and produces the first successful helicopter in 1939. The new aircraft first see action in World War II, where the United States employs over 400 helicopters. Helicopter development continues, and in 1954, Kaman produces the world's first turbine engine helicopter. With the advent of the turbine engine comes the

need for strong, reliable, and lightweight transmission systems, and therefore, better engineered gears.

In the 1950s, the first vibration gear models appear with Tuplin's work, modeling the mesh stiffness as a linear spring. Strauch introduces the fluctuating mesh stiffness in 1953. In 1958, Harris introduces the time-averaged mesh stiffness. The essence of 1950s gear research is characterized by the emergence of spring-mass stiffness models and periodic mesh stiffness studies.

The 1960s see an increase in geared system studies to produce more comprehensive models. Additionally, this decade sees the emergence of geared rotor dynamic models, with the works of Johnson and Seirig. In 1975, Mitchell and Mellen demonstrate the lateral-torsional vibration coupling in geared systems, and investigative studies of geared rotor systems continue.

Finite element analysis and transfer matrix method techniques are used extensively in the 1980s. With the growth of computing power, the finite element method becomes dominant, as used by the geared studies of Neriya, Choy, and Kahraman. Since 1990, the past two decades have seen an increase in rotor dynamic studies, incorporating more sophisticated mesh modeling techniques.

In studying gear-mesh approximations, the dynamics of the gear-mesh are complex and nonlinear. All nonlinear mesh models are approximations based upon experimental data. Closed form analytical solutions do not exist. The most common mesh stiffness approximation is the linear time-invariant approach, although it does require additional considerations. Rules of thumb exist for estimating the magnitude of the average mesh stiffness for a given gear pair. Investigative studies confirm its validity, especially for highly loaded gear systems.

Chapter 4

Basic Rotor Dynamic Analysis

This chapter provides background information into some of the basic techniques of rotor dynamic analysis techniques. These systems can be simple, such as the single-rotor Jeffcott Rotor depicted in Figure 4.1, or complex, such as the multi-rotor industrial turbine of Figure 4.2.

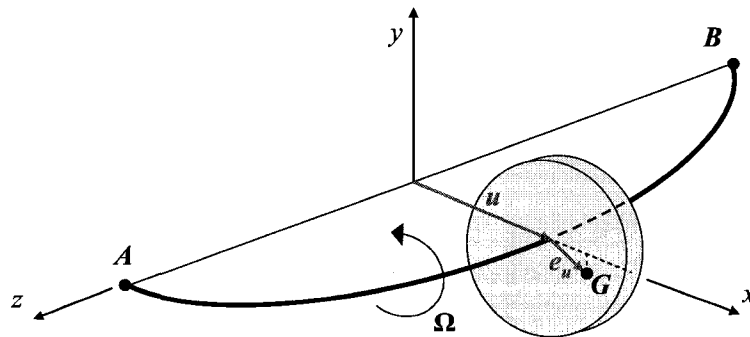


Figure 4.1: Jeffcott Rotor (Allaire, 2007)



Figure 4.2: Industrial Compressor Rotor (Adams, 2001)

4.1 Historical Perspective

Rotating machines produce vibrations. If the system's rotational speed operates at or near one of its natural frequencies, excessive vibration and often catastrophic results

can occur. Figure 4.3 shows such a failure of two steam-turbine generator sets in the 1970s (Adams, 2001). Consequently, careful analysis must go into the design of such systems to ensure safe operation.

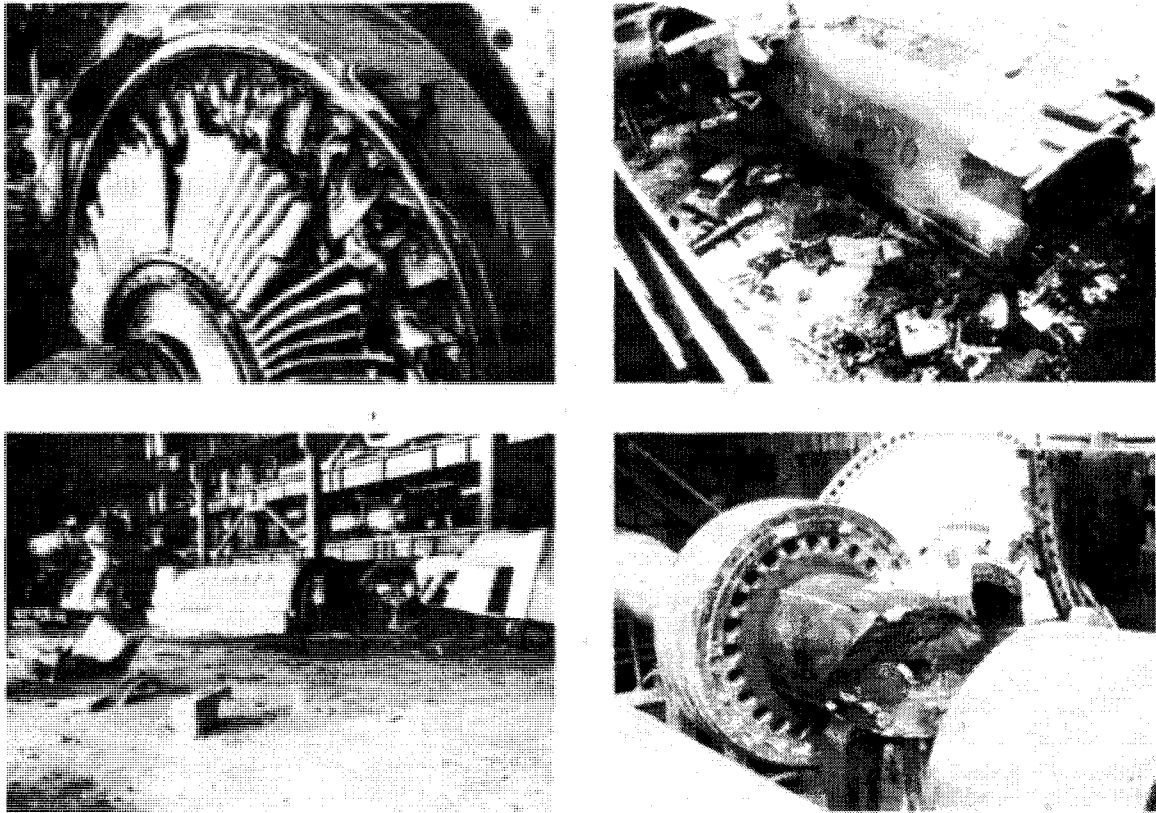


Figure 4.3: Catastrophic Failure of Steam-Turbine Generator Sets (Adams, 2001)

Rotor dynamic theory dates back to the Industrial Revolution as inventors and industrialists designed rotating machines for various applications, using long, slender shafts operating at high speeds. Longer shafts resulted in lower natural frequencies, generally meaning that a shaft passed through at least one natural frequency in order to reach its nominal operating speed. This development required a theory and method of analysis for determining whether systems could operate above their first, second, or

even third natural frequencies. Initial investigations were conducted by Foppl and Jeffcott, and the result was the Jeffcott Rotor (1919).

The Jeffcott Rotor was a circular disk at the center of a thin, massless shaft attached to two rigid bearings, as illustrated in Figure 4.1. The Jeffcott Rotor was a milestone in rotor dynamic analysis, illustrating that operation above the first critical speed was possible (Adams, 2001). The familiar harmonic frequency response is presented below in Figure 4.4.

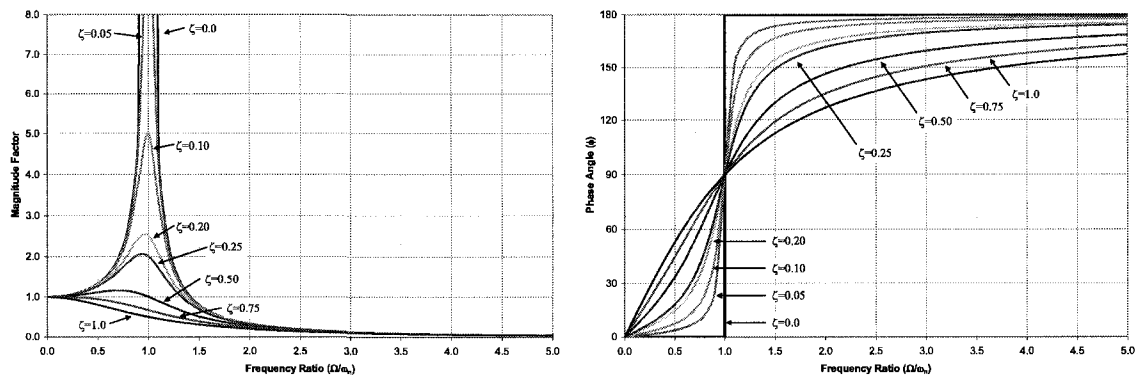


Figure 4.4: Jeffcott Rotor Response – Magnitude (left) and Phase Angle (right)

One of the main drawbacks of the Jeffcott Rotor was its simplicity. It provided a good starting point for analysis, but for a true representation of the dynamics in rotor-bearing systems, more comprehensive models became necessary.

4.2 First Principles in Rotor Dynamics

4.2.1 Equations of Motion

The equations of motion for a rotor dynamic system are easily available from any text (Childs, 1993). Since this approach promotes the use of first principles, the

Lagrangian development is presented for completeness. The Lagrangian formulation is given by Equation 4.1 (José, 1998).

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) - \frac{\partial L}{\partial q^i} &= D_i \\ L &= T - V \\ \{\mathbf{q}\} &= \{x, y, z, \theta_x, \theta_y, \theta_z\}^T \end{aligned} \tag{4.1}$$

where

- D = generalized forces
- L = Lagrangian function
- q = generalized coordinates
- T = kinetic energy
- V = potential energy

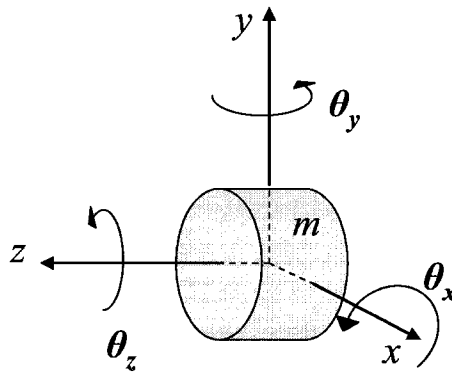


Figure 4.5: Simple Rotating Cylindrical Mass System

A singular cylindrical mass contains 6 DOFs, as illustrated in Figure 4.5. Many models use only 4 DOFs, neglecting axial, (z), and torsional, (θ_z), motion. Other models simply focus on torsional motion only. The justification for 6 DOFs is twofold. Incorporating all DOFs provides the most general model, applicable to any system.

Secondly, rotating shafts may not share the same orientation. Some shafts may be vertical, with others horizontal or at a slight angle. Consequently, motion in the global system may be a combination of lateral motion of one shaft, combined with axial motion of another.

The expressions for the total kinetic energy, (T) , and potential energy, (V) , are given by Equation 4.3 and 4.4:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) - \frac{\partial L}{\partial q^i} = D_i \quad (4.2)$$

$$L = T - V$$

$$T = \frac{1}{2} \left[m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) + I_t(\dot{\theta}_x^2 + \dot{\theta}_y^2) + I_p\dot{\theta}_z^2 + I_p(\Omega^2 - 2\Omega\dot{\theta}_y\theta_x) \right] \quad (4.3)$$

$$V = \frac{1}{2} \left[k_x x^2 + k_y y^2 + k_z z^2 + K_{\theta_x} \theta_x^2 + K_{\theta_y} \theta_y^2 + K_{\theta_z} \theta_z^2 \right] \quad (4.4)$$

where

I_T = transverse mass moment of inertia

I_p = polar mass moment of inertia

m = mass of cylindrical shape

K_q = rotational elastic stiffness about x , y , and z axes

k_q = translational elastic stiffness along x , y , and z axes

Ω = cylindrical mass rotating speed

This results in the following equations of motion for the 6-DOF system.

$$\begin{aligned}
m\ddot{x} + k_x x &= F_x \\
m\ddot{y} + k_y y &= F_y \\
m\ddot{z} + k_z z &= F_z \\
I_t \ddot{\theta}_x + \Omega I_p \dot{\theta}_y + K_{\theta x} \theta_x &= M_x \\
I_t \ddot{\theta}_y - \Omega I_p \dot{\theta}_x + K_{\theta y} \theta_y &= M_y \\
I_p \ddot{\theta}_z + K_{\theta z} \theta_z &= M_z
\end{aligned} \tag{4.5}$$

or in matrix form:

$$[\mathbf{M}]\{\ddot{\mathbf{q}}\} + [\mathbf{G}]\{\dot{\mathbf{q}}\} + [\mathbf{K}]\{\mathbf{q}\} = \{\mathbf{F}\}$$

$$[\mathbf{M}] = \begin{bmatrix} m & 0 & 0 & 0 & 0 & 0 \\ 0 & m & 0 & 0 & 0 & 0 \\ 0 & 0 & m & 0 & 0 & 0 \\ 0 & 0 & 0 & I_t & 0 & 0 \\ 0 & 0 & 0 & 0 & I_t & 0 \\ 0 & 0 & 0 & 0 & 0 & I_p \end{bmatrix}, [\mathbf{K}] = \begin{bmatrix} k_x & 0 & 0 & 0 & 0 & 0 \\ 0 & k_y & 0 & 0 & 0 & 0 \\ 0 & 0 & k_z & 0 & 0 & 0 \\ 0 & 0 & 0 & K_{\theta x} & 0 & 0 \\ 0 & 0 & 0 & 0 & K_{\theta y} & 0 \\ 0 & 0 & 0 & 0 & 0 & K_{\theta z} \end{bmatrix}$$

$$[\mathbf{G}] = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \Omega I_p & 0 \\ 0 & 0 & 0 & -\Omega I_p & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{4.6}$$

where

$[\mathbf{M}]$ = mass matrix

$[\mathbf{K}]$ = stiffness matrix

$[\mathbf{G}]$ = gyroscopic matrix

$\{\mathbf{F}\}$ = force/moment vector

The coupling of the (θ_x) and (θ_y) directions represent the effect of gyroscopic moments on the system. Gyroscopic effects occur when rotating masses tilt relative to the axis of rotation (Allaire, 2007). Although these terms look similar to damping effects, gyroscopic moments are conservative and involve no energy loss. The magnitudes of these effects are proportional to the rotational speed of the mass.

4.2.2 Eigenvalue Analysis

From the equations of motion, a simple analysis can be performed on the 6-DOF rotor in Figure 4.5 (Younan, 2004). The equations of motion in matrix form are given in Equation 4.7.

$$\begin{bmatrix} m & 0 & 0 & 0 & 0 & 0 \\ 0 & m & 0 & 0 & 0 & 0 \\ 0 & 0 & m & 0 & 0 & 0 \\ 0 & 0 & 0 & I_t & 0 & 0 \\ 0 & 0 & 0 & 0 & I_t & 0 \\ 0 & 0 & 0 & 0 & 0 & I_p \end{bmatrix} \begin{Bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \\ \ddot{\theta}_x \\ \ddot{\theta}_y \\ \ddot{\theta}_z \end{Bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \Omega I_p & 0 \\ 0 & 0 & 0 & -\Omega I_p & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{\theta}_x \\ \dot{\theta}_y \\ \dot{\theta}_z \end{Bmatrix} + \begin{bmatrix} k_x & 0 & 0 & 0 & 0 & 0 \\ 0 & k_y & 0 & 0 & 0 & 0 \\ 0 & 0 & k_z & 0 & 0 & 0 \\ 0 & 0 & 0 & K_{\theta_x} & 0 & 0 \\ 0 & 0 & 0 & 0 & K_{\theta_y} & 0 \\ 0 & 0 & 0 & 0 & 0 & K_{\theta_z} \end{bmatrix} \begin{Bmatrix} x \\ y \\ z \\ \theta_x \\ \theta_y \\ \theta_z \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix} \quad (4.7)$$

Manipulating such that the mass matrix, $[\mathbf{M}]$, becomes the identity matrix yields

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \\ \ddot{\theta}_x \\ \ddot{\theta}_y \\ \ddot{\theta}_z \end{Bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \Omega I_r & 0 \\ 0 & 0 & 0 & -\Omega I_r & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{\theta}_x \\ \dot{\theta}_y \\ \dot{\theta}_z \end{Bmatrix} + \begin{bmatrix} \omega_x^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \omega_y^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \omega_z^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \omega_{\theta_x}^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \omega_{\theta_y}^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \omega_{\theta_z}^2 \end{bmatrix} \begin{Bmatrix} x \\ y \\ z \\ \theta_x \\ \theta_y \\ \theta_z \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix} \quad (4.8)$$

where

$$I_r = I_p / I_t$$

$$\omega_x^2 = k_x / m \quad \omega_y^2 = k_y / m \quad \omega_z^2 = k_z / m \quad \omega_{\theta x}^2 = K_{\theta x} / m$$

$$\omega_{\theta y}^2 = K_{\theta y} / m \quad \omega_{\theta z}^2 = K_{\theta z} / m$$

Assuming a harmonic solution to the equations of motion,

$$\{\mathbf{q}\} = \{\mathbf{Q}_o\} e^{(j\omega t)} = \{X_o \quad Y_o \quad Z_o \quad \Theta_{ox} \quad \Theta_{oy} \quad \Theta_{oz}\}^T e^{(j\omega t)} \quad (4.9)$$

and substituting into Equation 4.8 yields the following:

$$\begin{bmatrix} \omega_x^2 - \omega^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \omega_y^2 - \omega^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \omega_z^2 - \omega^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \omega_{\theta x}^2 - \omega^2 & j\Omega I_r \omega & 0 \\ 0 & 0 & 0 & -j\Omega I_r \omega & \omega_{\theta y}^2 - \omega^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \omega_{\theta z}^2 - \omega^2 \end{bmatrix} \begin{Bmatrix} X_o \\ Y_o \\ Z_o \\ \Theta_{ox} \\ \Theta_{oy} \\ \Theta_{oz} \end{Bmatrix} = \{\mathbf{0}\} \quad (4.10)$$

When set equal to zero, the determinant of the matrix in Equation 4.10 represents the characteristic equation of the rotor system. This is the traditional eigenvalue problem.

$$\begin{aligned} & (\omega^2 + \omega - \omega_x^2)(\omega^2 + \omega - \omega_y^2)(\omega^2 + \omega - \omega_z^2)(\omega^2 + \omega - \omega_{\theta z}^2) \\ & \left(-\omega^4 - 2\omega^3 + (\omega_{\theta x}^2 + \omega_{\theta y}^2 + \Omega^2 I_r^2 - 1)\omega^2 \right. \\ & \quad \left. + (\omega_{\theta x}^2 + \omega_{\theta y}^2)\omega - \omega_{\theta x}^2 \omega_{\theta y}^2 \right) = 0 \end{aligned} \quad (4.11)$$

Solving for (ω) yields twelve natural frequencies, which are the square roots of the system eigenvalues.

$$\begin{aligned}
\omega_{1,2} &= \pm \omega_x \\
\omega_{3,4} &= \pm \omega_y \\
\omega_{5,6} &= \pm \omega_z \\
\omega_{7,8} &= \pm \omega_{\theta z} \\
\omega_{9,10} &= \pm \frac{1}{2} \sqrt{2(\Omega I_r)^2 + 2(\omega_{\theta x}^2 + \omega_{\theta y}^2) + 2 \sqrt{\left(\omega_{\theta x}^2 + \omega_{\theta y}^2 + 2\omega_{\theta x}\omega_{\theta y} + (\Omega I_r)^2\right) \left(\omega_{\theta x}^2 + \omega_{\theta y}^2 - 2\omega_{\theta x}\omega_{\theta y} + (\Omega I_r)^2\right)}} \\
\omega_{11,12} &= \pm \frac{1}{2} \sqrt{2(\Omega I_r)^2 + 2(\omega_{\theta x}^2 + \omega_{\theta y}^2) - 2 \sqrt{\left(\omega_{\theta x}^2 + \omega_{\theta y}^2 + 2\omega_{\theta x}\omega_{\theta y} + (\Omega I_r)^2\right) \left(\omega_{\theta x}^2 + \omega_{\theta y}^2 - 2\omega_{\theta x}\omega_{\theta y} + (\Omega I_r)^2\right)}}
\end{aligned} \tag{4.12}$$

Using the simplifications

$$S_1 = \frac{(\omega_{\theta x}^2 + \omega_{\theta y}^2)}{2}, S_2 = \frac{(\omega_{\theta x}^2 - \omega_{\theta y}^2)}{2} \tag{4.13}$$

Equation 4.12 then results in a simpler expression:

$$\begin{aligned}
\omega_{1,2} &= \pm \omega_x \\
\omega_{3,4} &= \pm \omega_y \\
\omega_{5,6} &= \pm \omega_z \\
\omega_{7,8} &= \pm \omega_{\theta z} \\
\omega_{9,10} &= \pm \sqrt{\frac{(\Omega I_r)^2}{2} + S_1 + \sqrt{\left(\frac{\Omega I_r}{\sqrt{2}}\right)^4 + S_1(\Omega I_r)^2 + S_2^2}} \\
\omega_{11,12} &= \pm \sqrt{\frac{(\Omega I_r)^2}{2} + S_1 - \sqrt{\left(\frac{\Omega I_r}{\sqrt{2}}\right)^4 + S_1(\Omega I_r)^2 + S_2^2}}
\end{aligned} \tag{4.14}$$

The last four two expressions in Equation 4.14, $(\omega_{9,10})$ and $(\omega_{11,12})$, correspond to the rotational modes, (θ_x, θ_y) . These frequencies are a function of the gyroscopics and are

speed dependent. If the rotational speed reduces to zero, the coupling disappears and the frequencies become

$$\begin{aligned}\omega_{9,10} &= \pm \omega_{\theta x} \\ \omega_{11,12} &= \pm \omega_{\theta y}\end{aligned}\tag{4.15}$$

The mode shapes can be determined by substituting each natural frequency value into Equation 4.10 and determining the corresponding eigenvector. For the simple model, the eigenvectors are

$$\{\Phi\} = \begin{Bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}_{1,2}, \begin{Bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}_{3,4}, \begin{Bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{Bmatrix}_{5,6}, \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{Bmatrix}_{7,8}, \begin{Bmatrix} 0 \\ 0 \\ 0 \\ \frac{-j\Omega I_r \omega}{\omega_{\theta x}^2 - \omega^2} \\ 1 \\ 0 \end{Bmatrix}_{9,10}, \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ \frac{j\Omega I_r \omega}{\omega_{\theta y}^2 - \omega^2} \\ 0 \end{Bmatrix}_{11,12}\tag{4.16}$$

The rotational eigenvectors themselves are not planar, as evidenced by the presence of complex numbers in the vectors. One should expect this due to the gyroscopic nature of rotating systems that rotation about one lateral axis causes movement about the other. Additionally, the existing natural frequencies are both positive and negative with the same eigenvectors for either the positive or negative natural frequency. In the physical system, the frequencies are positive. Therefore, it is entirely proper to deal with the positive frequencies and ignore those that are negative.

4.2.3 Campbell Diagram

The Campbell Diagram is a useful tool for visualizing the sensitivity of a system to rotational speed and gyroscopic effects. The Campbell Diagram shows the relationship between frequency and shaft rotational speed. It provides a visual reference for the gyroscopic-frequency relationship of the system.

An example Campbell Diagram follows in Figure 4.6 for the single mass problem. Only the positive frequencies have been plotted. Note the two frequencies that are sensitive to the rotational speed and their behavior. One increases as the operating speed increases, while the other decreases as a result of the increase in operating speed. These are defined as the forward and backward whirl frequencies respectively. The system parameters used to construct Figure 4.6 are

$$\begin{aligned}
 m &= 1 \text{ kg}, I_t = 0.0008 \text{ kg-m}^4, I_p = 0.0004 \text{ kg-m}^4, \\
 k_x &= 12000, k_y = 12000, k_z = 500 \text{ N/m}, \\
 K_{\theta_x} &= 25000, K_{\theta_y} = 25000, K_{\theta_z} = 10000 \text{ N-m/rad}
 \end{aligned}$$

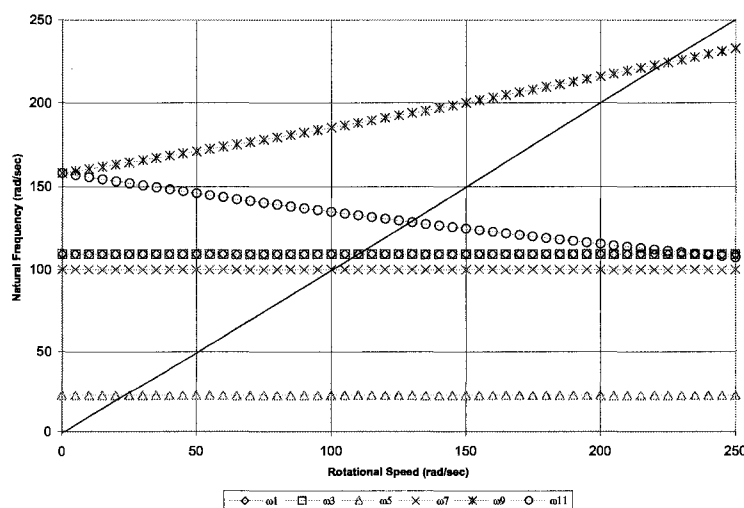


Figure 4.6: Campbell Diagram from Simple System Setup

In forward whirl, the orbit of vibration around the centerline of the shaft moves in the same direction as the rotation of the shaft. Backward whirl implies that the shaft vibration opposes the direction of shaft rotation. From the Campbell Diagram in Figure 4.6, the gyroscopic effects will “stiffen” the rotor in forward whirl, while “softening” the rotor in backward whirl (Younan, 2004).

4.2.4 Frequency Response Analysis

The final technique discussed involves determining the frequency response of the system to a harmonic excitation, $\{\mathbf{F}_o\}$, at some forcing frequency, (ω_f) .

$$\{\mathbf{F}_o\} e^{j(\omega_f t)} = \begin{Bmatrix} F_{ox} & F_{oy} & F_{oz} & M_{ox} & M_{oy} & M_{oz} \end{Bmatrix}^T e^{j(\omega_f t)} \quad (4.17)$$

Substituting into Equation 4.8 and solving for $\{\mathbf{Q}_o\}$ yields

$$\{\mathbf{Q}_o\} = G(\omega_f) \{\mathbf{F}_o\} \quad (4.18)$$

such that $G(\omega_f)$ is the frequency response function of the system due to the excitation force, $\{\mathbf{F}_o\}$. In this case, the frequency response is

$$G(\omega_f) = \begin{bmatrix} \frac{1}{\omega_x^2 - \omega_f^2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{\omega_y^2 - \omega_f^2} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{\omega_z^2 - \omega_f^2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\omega_{\theta y}^2 - \omega_f^2}{D} & \frac{j\Omega I_r \omega_f}{D} & 0 \\ 0 & 0 & 0 & \frac{-j\Omega I_r \omega_f}{D} & \frac{\omega_{\theta x}^2 - \omega_f^2}{D} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{\omega_{\theta z}^2 - \omega_f^2} \end{bmatrix} \quad (4.19)$$

where

$$D = \left[(\omega_{\theta x}^2 - \omega_f^2)(\omega_{\theta y}^2 - \omega_f^2) + (j\Omega I_r \omega_f)^2 \right] \quad (4.20)$$

Given the forcing function, $\{\mathbf{F}_o\}$, the response, $\{\mathbf{Q}_o\}$, can be determined from Equation 4.19 for any range of frequencies desired. Plotting these results creates a chart similar to Figure 4.4.

4.3 Finite Element Rotor Dynamic Analysis

4.3.1 Formulation and Assumptions

The finite element formulation for a multi-DOF system is well documented and can be found in any undergraduate text (Allaire, 1985). In general, the finite element method consists of formulating the mass, damping, and stiffness matrices for each element. These are then used to construct the global matrices in the following manner:

$$[\mathbf{M}]^{(G)} = \begin{bmatrix} m_{1,1}^{(1)} & & & & & \\ & \mathbf{M}^{(1)} & & & & \\ & & m_{n,n}^{(1)} + m_{1,1}^{(2)} & & & \\ & & & \mathbf{M}^{(2)} & & \\ & & & & m_{n,n}^{(2)} + m_{1,1}^{(3)} & \\ & & & & & \ddots \\ & 0 & & & & m_{n,n}^{(N-1)} + m_{1,1}^{(N)} \\ & & & & & & \mathbf{M}^{(N)} \\ & & & & & & & m_{n,n}^{(N)} \end{bmatrix} \quad (4.21)$$

Each element, (e) , consists of two nodes, (i) and (j) , each with 6 DOFs. The displacement vector of each element contains 12 DOFs.

$$\{\mathbf{q}\}^{(e)} = \{x_i, y_i, z_i, \theta_{xi}, \theta_{yi}, \theta_{zi}, x_j, y_j, z_j, \theta_{xj}, \theta_{yj}, \theta_{zj}\}^T \quad (4.22)$$

In the finite element formulation, the following assumptions are made.

1. The rotors (gears) are thin, rigid disks. As such, they are modeled as point masses at the node corresponding to their location on the shaft.
2. The rotating shafts are modeled as frictionless, Timoshenko beams.
3. The consistent mass matrix is used in modeling the mass of the rotating shaft, rather than a lumped mass matrix.
4. Damping occurs only through the bearings or the gear-mesh.

4.3.2 State-Space Methodology

For a rotor dynamic system undergoing general damping and gyroscopic effects, the general equation of motion is

$$\begin{aligned} [\mathbf{M}]\{\ddot{\mathbf{q}}\} + [\mathbf{D}]\{\dot{\mathbf{q}}\} + [\mathbf{K}]\{\mathbf{q}\} &= \{\mathbf{F}\} \\ [\mathbf{D}] &= [\mathbf{C}] + \Omega[\mathbf{G}] \end{aligned} \quad (4.23)$$

The presence of damping, $[\mathbf{C}]$, or gyroscopics, $[\mathbf{G}]$, presents a problem for traditional eigenvalue analysis, due to the presence of a third matrix, $[\mathbf{D}]$. In many structural dynamic applications, proportional damping is assumed, and the familiar method of eigenvalue analysis still applies. In rotor dynamics, the damping is rarely proportional, and the effect of gyroscopics can become significant, especially at higher speeds. A general eigenvalue problem can be formulated by converting the problem into a first-order state-space equation.

$$-\begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K} \end{bmatrix} \begin{Bmatrix} \dot{\mathbf{q}} \\ \mathbf{q} \end{Bmatrix} + \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{M} & \mathbf{D} \end{bmatrix} \begin{Bmatrix} \ddot{\mathbf{q}} \\ \dot{\mathbf{q}} \end{Bmatrix} = \begin{Bmatrix} \mathbf{0} \\ \mathbf{0} \end{Bmatrix} \quad (4.24)$$

or

$$-[\mathbf{A}]\{\mathbf{q}'\} + [\mathbf{B}]\{\dot{\mathbf{q}}'\} = \{\mathbf{0}\} \quad (4.25)$$

where

$$[\mathbf{A}] = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K} \end{bmatrix}, [\mathbf{B}] = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{M} & \mathbf{D} \end{bmatrix}, \{\mathbf{q}'\} = \begin{Bmatrix} \dot{\mathbf{q}} \\ \mathbf{q} \end{Bmatrix}, \{\dot{\mathbf{q}}'\} = \begin{Bmatrix} \ddot{\mathbf{q}} \\ \dot{\mathbf{q}} \end{Bmatrix} \quad (4.26)$$

The matrices, $[\mathbf{A}]$ and $[\mathbf{B}]$, are defined as such to follow the convention used in MATLAB® notation. Assuming the displacement and velocity vectors in Equation 4.26 to be of the form:

$$\begin{aligned} \{\mathbf{q}'\} &= \{\mathbf{Q}_o\} e^{j\omega t} \\ \{\dot{\mathbf{q}}'\} &= j\omega \{\mathbf{Q}_o\} e^{j\omega t} \end{aligned} \quad (4.27)$$

and letting $\lambda = j\omega$, the generalized eigenvalue problem for the state-space formulation becomes

$$[\mathbf{A}]\{\Phi_r\} = \lambda[\mathbf{B}]\{\Phi_r\} \quad (4.28)$$

Using this formulation, (λ) is a complex eigenvalue, and $\{\Phi_r\}$ is the corresponding eigenvector. Eigenvalues occur in complex conjugate pairs. The imaginary component of each eigenvalue represents a frequency of the system. If damping is present, the imaginary component represents the damped natural frequency, (ω_d) . The real component of the eigenvalue will be negative. If the system is undamped, then the eigenvalue's imaginary component represents the natural frequency of the system (ω_n) , and the real component of the eigenvalue will equal zero.

The generalized eigenvalue problem of Equation 4.28 can be solved using traditional methods. The upper half of the resulting eigenvectors represents velocities, and the lower half represents displacements. The primary disadvantages of this formulation are that visibility of forward and backward whirl modes is lost using the state-space (Kessler, 2002), and the size of the model is doubled, $(2N)$.

The global matrices are constructed in the format of Equation 4.21. The matrices break down further into system components as follows:

$$\begin{aligned} \mathbf{M} &= \mathbf{M}_s + \mathbf{M}_d \\ \mathbf{C} &= \mathbf{C}_s + \mathbf{C}_b + \mathbf{C}_{mesh} \\ \mathbf{K} &= \mathbf{K}_s + \mathbf{K}_b + \mathbf{K}_{mesh} \\ \mathbf{G} &= \mathbf{G}_s + \mathbf{G}_d \end{aligned} \quad (4.29)$$

with the following subscript designations. The subscript, (s) , designates the matrices corresponding to the rotor shaft; the subscript, (d) , designates the disk or gear. The subscript, (b) , designates a bearing matrix; and finally, the subscript, $(mesh)$, identifies the gear-mesh matrix. Appendix A lists all matrix formulations except for the mesh matrix. Its derivation is the topic of Chapter 5.

4.3.3 Frequency Response

The frequency response of the state-space equations of motion to an exciting force can be derived. Applying an excitation force to the right hand side of Equation 4.25:

$$-[\mathbf{A}]\{\mathbf{q}'\} + [\mathbf{B}]\{\dot{\mathbf{q}}'\} = \{\mathbf{F}_o\} e^{j\omega_f t} \quad (4.30)$$

and assuming a harmonic response to the forcing frequency, (ω_f) .

$$\begin{aligned} \{\mathbf{q}'\} &= \{\mathbf{Q}_o\} e^{j\omega_f t} \\ \{\dot{\mathbf{q}}'\} &= j\omega_f \{\mathbf{Q}_o\} e^{j\omega_f t} \end{aligned} \quad (4.31)$$

Substituting into Equation 4.30 and simplifying yields

$$\begin{aligned} -[\mathbf{A}]\{\mathbf{Q}_o\} + j\omega_f [\mathbf{B}]\{\mathbf{Q}_o\} &= \{\mathbf{F}_o\} \\ (j\omega_f [\mathbf{B}] - [\mathbf{A}])\{\mathbf{Q}_o\} &= \{\mathbf{F}_o\} \end{aligned} \quad (4.32)$$

Solving for the response, $\{\mathbf{Q}_o\}$, results in

$$\{\mathbf{Q}_o\} = (j\omega_f [\mathbf{B}] - [\mathbf{A}])^{-1} \{\mathbf{F}_o\} \quad (4.33)$$

The frequency response function, $G(\omega_f)$, is

$$G(\omega_f) = (j\omega_f [\mathbf{B}] - [\mathbf{A}])^{-1} \quad (4.34)$$

Another useful tool is the speed-sweep response as presented in Lin (1999). This is particularly useful for systems operating over a range of rotational speeds. The speed-sweep technique substitutes the rotational speed, (Ω) , for the forcing frequency, (ω_f) . The matrix, $[\mathbf{B}]$, contains the system gyroscopics, which are a function of speed. In calculating the speed-sweep response, the matrix, $[\mathbf{B}]$, is re-calculated for each speed step. Equation 4.34 can be re-written as a speed-sweep equation.

$$G(\Omega) = (j\Omega [\mathbf{B}(\Omega)] - [\mathbf{A}])^{-1} \quad (4.35)$$

4.3.4 Plotting Complex Mode Shapes

The mode shapes of the system consist of complex numbers. If no damping is present, the mode shapes will always be planar, although the eigenvectors may exist in complex form. If damping is present, the resulting mode shapes are not planar. For analysis purposes, plotting the mode shapes of the shaft produces an orbital mode-shape of the rotating system using the lateral (x) and (y) displacements at each node (Nelson, 1992, Lee, 1993). From the eigenvectors, the displacements at any node can be expressed as

$$\begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{Bmatrix} x_r + jx_i \\ y_r + jy_i \end{Bmatrix} \quad (4.36)$$

where (r) and (i) designate the real and imaginary components, respectively.

The complex displacements can also be expressed in terms of a magnitude and phase angle as follows:

$$\begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{Bmatrix} x_r + jx_i \\ y_r + jy_i \end{Bmatrix} = \begin{Bmatrix} Xe^{j\phi_x} \\ Ye^{j\phi_y} \end{Bmatrix} \quad (4.37)$$

where the magnitudes equal

$$\begin{Bmatrix} X \\ Y \end{Bmatrix} = \begin{Bmatrix} \sqrt{x_r^2 + x_i^2} \\ \sqrt{y_r^2 + y_i^2} \end{Bmatrix} \quad (4.38)$$

and the phase angles are

$$\begin{Bmatrix} \phi_x \\ \phi_y \end{Bmatrix} = \begin{Bmatrix} \tan^{-1}(x_i/x_r) \\ \tan^{-1}(y_i/y_r) \end{Bmatrix} \quad (4.39)$$

The orbit, (Z) , at a given node can be expressed by either Equation 4.40 or 4.41.

$$Z = X \cos(\omega t + \phi_x) + jY \cos(\omega t + \phi_y) \quad (4.40)$$

or

$$Z = X \sin(\omega t + \phi_x) + jY \sin(\omega t + \phi_y) \quad (4.41)$$

Equations 4.40 and 4.41 produce the same orbit as shown to the left chart in Figure 4.7. The data points marked by squares represent the starting location using Equation

4.40 and Equation 4.41. If the displacements in Equation 4.37 are complex conjugates, the orbit is a circle as pictured to the right in Figure 4.7.

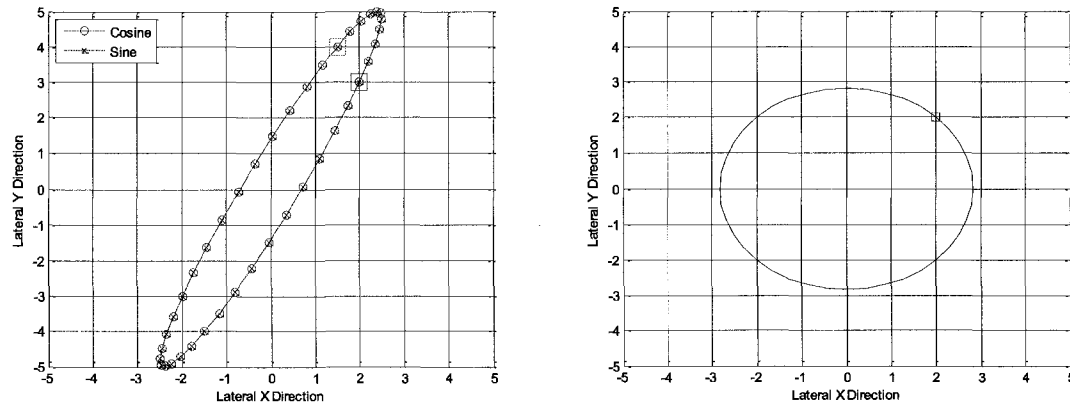


Figure 4.7: Elliptical (Left) and Circular (Right) Lateral Orbits

When plotted at each node along the shaft, the mode shape emerges as pictured in Figure 4.8. The line represents the shape of the shaft at a single time, allowing a visual of the mode shape in its orbit.

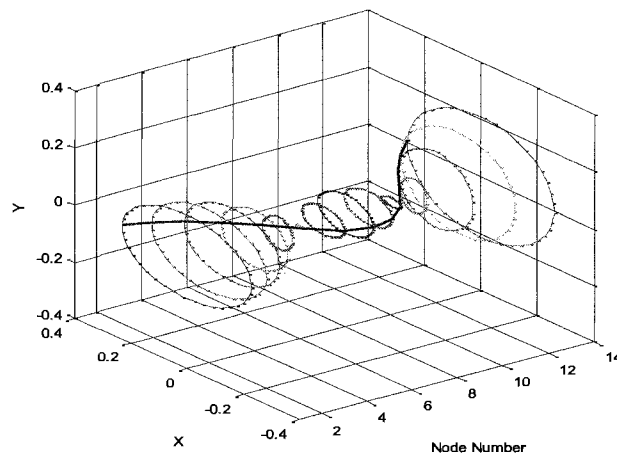


Figure 4.8: Lateral Mode Shape

4.4 Summary

In summary, Chapter 4 has presented a brief overview of some of the fundamental techniques of rotor dynamic analysis. The Lagrangian formulation of the rotor dynamic equations of motion was presented. The analysis of a simple single-mass system illustrated the principles of the eigenvalue problem. The Campbell Diagram was introduced, illustrating the effects of shaft rotational speed on the natural frequencies of the system. The frequency response function for the single-mass system undergoing harmonic excitation was also introduced.

The finite element formulation of the equations of motion was presented. The equations of motion were broken down into the system matrices and their components. The state-space methodology was introduced, reducing the equations of motion to a system of first-order differential equations, from which the general eigenvalue problem was formulated. The eigenvalues using this methodology occurred in complex conjugate pairs, representing the natural frequencies of the system. The right eigenvectors represented the mode shapes. The system frequency response was presented. Lastly, the procedure for plotting the orbit of the shaft was presented. The analysis techniques discussed in this chapter provide the framework for the subsequent analysis and results presented in the remaining chapters.

Chapter 5

Gear-Mesh Model Methodology with Application

This chapter presents a methodology for resolving the forces and moments transmitted through the gear mesh into a 12-DOF mesh stiffness matrix for a spur or helical gear pair. The methodology is generally based on a technique proposed by Luo (1996). It also incorporates some aspects presented by Choy (1993) and Blankenship (1995a). The mesh matrix is validated using a simple geared shaft system. A simplified 6-DOF mesh model is also discussed.

5.1 Background

Blankenship (1995a) and Choy's (1993) approaches provided results tailored to specific applications. Choy's work was limited to spur gear systems. Blankenship's final model assumed a certain arrangement of the gears which reduced the model. This simplification may not always be possible when dealing with a large system and can render such models unusable for general application.

Another drawback to Blankenship's approach is that the gear pressure angle was not directly addressed. Only the helical angle was considered. The pressure angle, (α_n) , is the angle between the line-of-action and the line tangent to the pitch circle. It represents the effective direction of force transmission from one gear to another. Blankenship (1995a) was able to omit it due to his choice of coordinate axes. As such, the model loses its generality.

5.2 General Mesh Forcing Function Development

Figure 5.1 below shows a finite element representation of a gear-shaft pair as illustrated by Luo (1996).

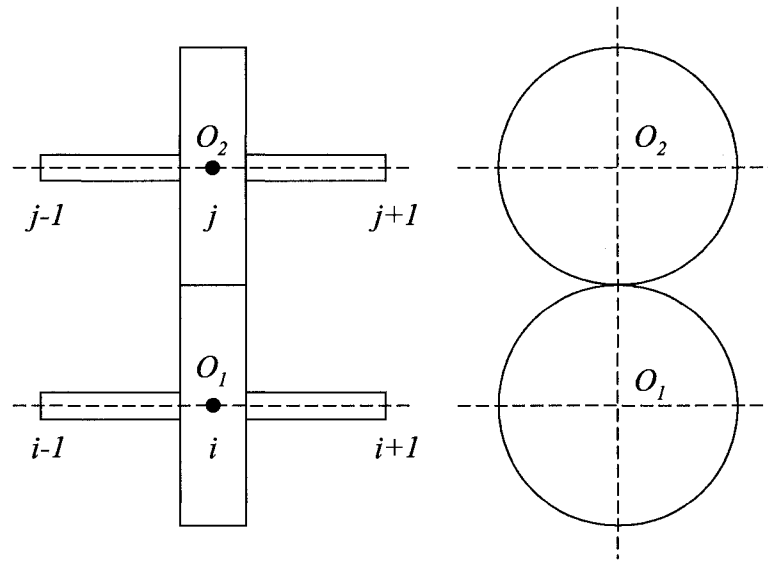


Figure 5.1: Finite Element Representation of a Gear Pair

The gears act as rigid disks on flexible beam elements. The center points, (O_1) and (O_2) , of the gear are assumed to be nodes (i) and (j) of the finite element analysis. A force and moment vector, $\{\mathbf{F}\}$, can be modeled as a function of a mesh stiffness matrix, $[\mathbf{K}]_{mesh}$, and a displacement vector, $\{\mathbf{q}\}$, acting on nodes (i) and (j) .

$$\begin{Bmatrix} \mathbf{F}_i \\ \mathbf{F}_j \end{Bmatrix} = [\mathbf{K}]_{mesh} \begin{Bmatrix} \mathbf{q}_i \\ \mathbf{q}_j \end{Bmatrix} = K_g \begin{bmatrix} [\mathbf{K}_{ii}] & [\mathbf{K}_{ji}] \\ [\mathbf{K}_{ij}] & [\mathbf{K}_{jj}] \end{bmatrix} \begin{Bmatrix} \mathbf{q}_i \\ \mathbf{q}_j \end{Bmatrix} \quad (5.1)$$

5.2.1 Geometry and Loading Condition

As depicted in Figure 5.2, the mesh initially is a spring-mass system, with the spring acting along the line-of-action (LOA) between the two gears. The z -axis lies in the axial

direction. The LOA is the path of force transmission between the two gears, and is represented by a line tangent to both gear disks.

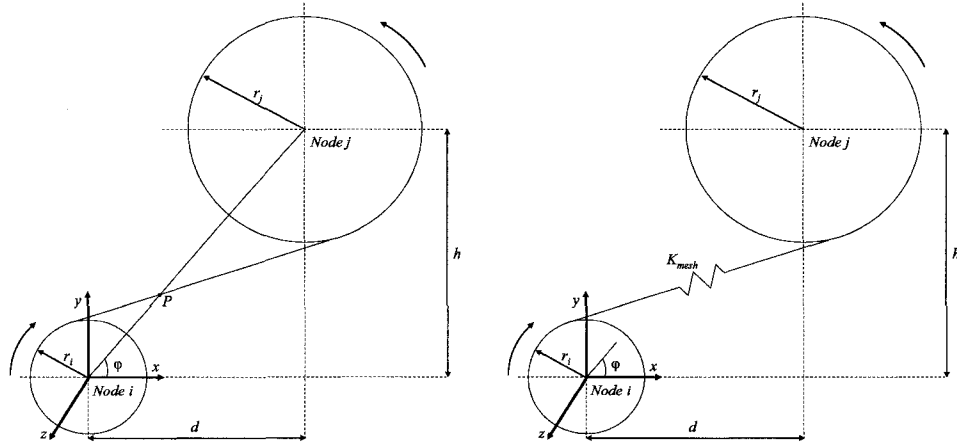


Figure 5.2: Gear-Mesh Spring-Mass Model

The intersection of the line-of-action and the line connecting the center of the two gears is defined as the pitch point, (P) , the effective point of force transmission between gears. In spur gears, this is the point of concentrated load based on a uniform load distribution of the gear tooth while in contact. In helical gears, the load distribution is not uniform, as illustrated in Figure 5.3. Since the teeth are canted at an angle, (β) , the amount of the tooth in contact varies during the mesh cycle. The beginning of the mesh cycle is characterized by areas of increasing contact along the width of the tooth until it reaches the point of maximum contact. The contact load then decreases along the width of the tooth during the last stage of the mesh cycle. For purposes of this analysis, the force transmissibility of helical gears is assumed to occur at the pitch point, (P) .

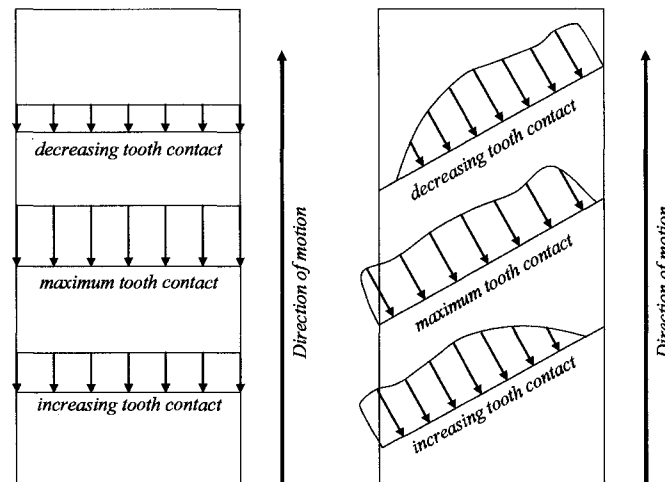


Figure 5.3: Load Distributions: Spur (left) versus Helical (right) Gears

The location of the gears relative to each other must also be taken into consideration, which is designated by a height variable, (h) , in Figure 5.2. This is required for compatibility between the gear coordinate axes with those of a global system. For example, a gear system might employ gears horizontally stationed side-by-side ($h=0$), vertically oriented above and below ($h=r_i+r_j$), or at some other generic orientation angle, (φ) , to each as depicted in Figure 5.2. The distance, (h) , is the vertical distance between the centers of the gear pair.

This approach assumes a mesh stiffness, (K_g) , which in most cases is an average mesh value derived from experimental results and depends upon the material parameters of the particular gears in the system. A nonlinear stiffness, such as time or spatially-varying, can also be used. The stiffness matrix is related to the geometry of the system as a function of three variables: the helical angle, (β) , the pressure angle, (α_n) ,

and the angle of orientation, (φ) , between the two gears. The orientation angle, (φ) , is a direct function of the height, (h) .

Many previous studies only consider spur gears, where all axial and lateral rocking and bending motions are neglected. This keeps the problem two-dimensional. The transmitted force through the gears acts in the xy -plane and is a function of the pressure angle, (α_n) , only. When dealing with helical gears, the problem becomes three-dimensional due to the added effect of force components acting out of the xy -plane, as shown in Figure 5.4.

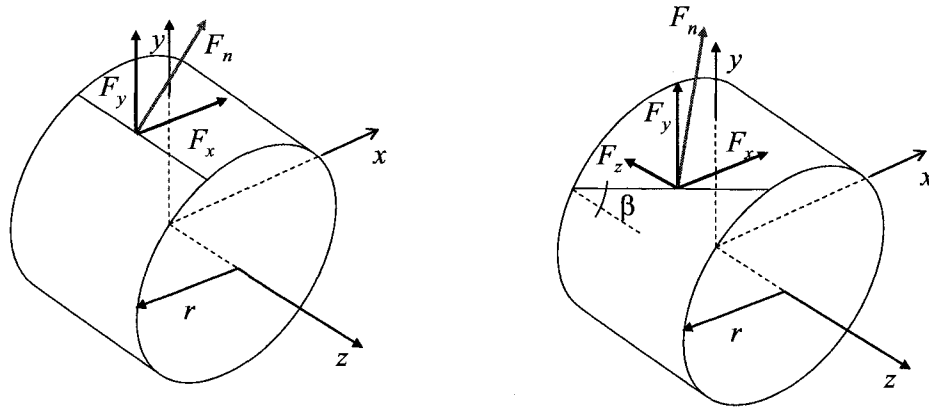


Figure 5.4: Spur (left) and Helical (right) Gear Force Contributions

The three-dimensional force components are then a function of the direction cosines, illustrated in Figure 5.5. The direction cosine angles can be derived in terms of the pressure angle, (α_n) , and helical angle, (β) .

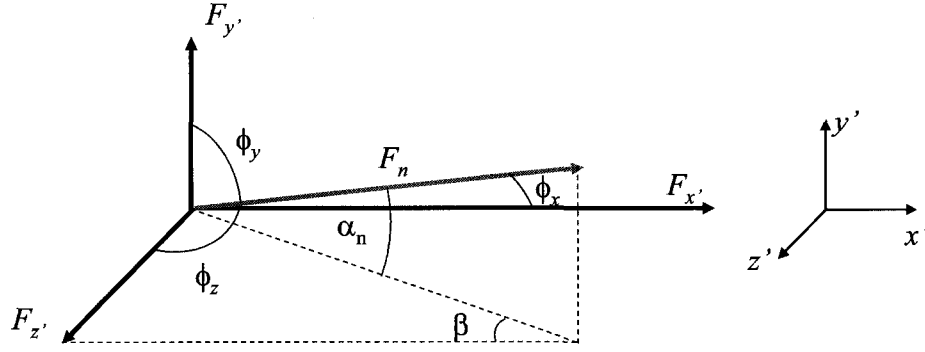


Figure 5.5: Mesh Force Components and Direction Cosines

The force elements of the x' , y' , and z' axes are determined by Equation 5.3. The significance of the prime coordinates (x' , y' , and z') will be discussed later.

$$\begin{aligned} F_{x'} &= F_n \cos \phi_x \\ F_{y'} &= F_n \cos \phi_y \\ F_{z'} &= F_n \cos \phi_z \end{aligned} \quad (5.2)$$

where the direction cosines take the form:

$$\begin{aligned} \cos \phi_x &= \cos \beta \cos \alpha_n \\ \cos \phi_y &= \sin \alpha_n \\ \cos \phi_z &= \sin \beta \sin \alpha_n \end{aligned} \quad (5.3)$$

The equation relating the force vector to displacement is

$$\{\mathbf{F}\}_{mesh} = -[\mathbf{K}]_{mesh} \{\mathbf{u}\}_{LOA} \quad (5.4)$$

The negative sign here implies that the forces and moments acting on the gear body are in the direction of positive displacement, $\{\mathbf{u}\}_{LOA}$ (Blankenship, 1995b). In other words, Equation 5.4 is a forcing function, which can be brought to the left hand side of the equations of motion as a stiffness matrix. Building the matrix requires an expression

for the displacement of the gear body as a function of the 12 DOFs. This can be accomplished by examining the contribution of each DOF to the total displacement along the line-of-action. First, the appropriate coordinate transformation between the global coordinates and the prime coordinates of Equation 5.2 must be defined.

5.2.2 Gear-Mesh Coordinate System

The prime coordinate axes in Figure 5.5, (x') , (y') , and (z') , are local to the pitch point and not necessarily parallel with that of the global coordinate system. The prime coordinates are dependent upon the orientation angle, (φ) , of the gear pair. The pressure angle and helical angle fix the force vector with respect to the gear tooth. The geometry of the gear tooth fixes the prime axes at the pitch point, such that the prime coordinate system is fixed and unchanging. The force vector must also be fixed in terms of the orientation angle to account for the height of the gears relative to one another. If the height of the gears was allowed to vary such that the orientation angle was allowed to "sweep" through a full revolution, the prime axes would appear to "rotate" about the pitch circle, as depicted in Figure 5.6. For example, at a value of $\varphi = \pi/2$ and $\varphi = 3\pi/2$, the two coordinate systems are parallel, although the directions of the axes between the two are reversed. The only axes remaining parallel at any orientation are the (z) and (z') axes. At any other configuration, the mesh axes and global axes are not parallel.

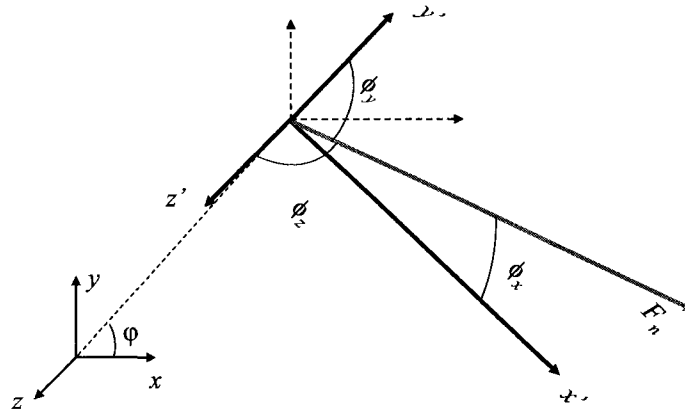


Figure 5.6: Relation of Local and Global Axes with φ

As seen in Figure 5.6, a force component in the (x') direction of the prime coordinate system has components in both the (x) and (y) directions of the global coordinate system. Similarly, the force component acting in the (y') direction has components in both the (x) and (y) directions of the global coordinate system. In the global coordinate system, the force components of gear (i) can be written as

$$\begin{aligned} F_{xi} &= F_{x'} \sin \varphi + F_{y'} \cos \varphi \\ F_{yi} &= -F_{x'} \cos \varphi + F_{y'} \sin \varphi \\ F_{zi} &= F_{z'} \end{aligned} \quad (5.5)$$

5.2.3 Displacement Method

The contribution of each DOF to the total displacement along the line-of-action, (s) , is the result of a displacement that attempts to shift the pitch point in the corresponding

direction. The effect of a translation along the x -axis is considered in Figure 5.7. A positive x -displacement provides a positive component along the line-of-action, (s_{xi}) .

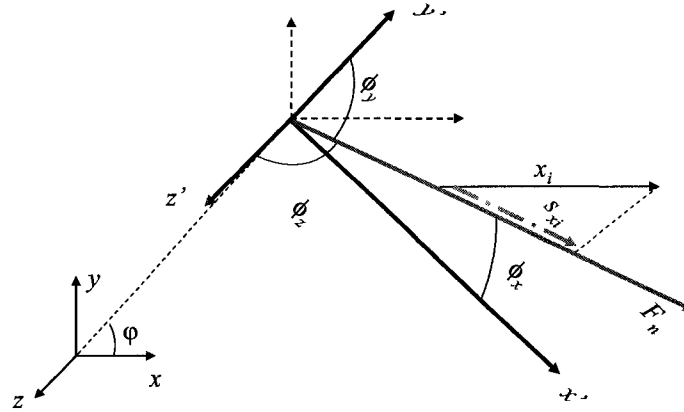


Figure 5.7: Contribution of x -displacement to Line-of-Action

The effect of translational motion along the three axes attempts to shift the pitch point in the direction of motion. Rotational motion, however, is not as apparent. A rotational motion about an axis will attempt to rotate the pitch point in a circular arc about the respective axis. The circular arc incorporates motion in two directions. Figure 5.8 depicts the effect of rotation about the y -axis, (θ_y) . A positive rotation about the y -axis yields effective displacements in the xz -plane. These can be analyzed as translational displacements in (x) and (z) .

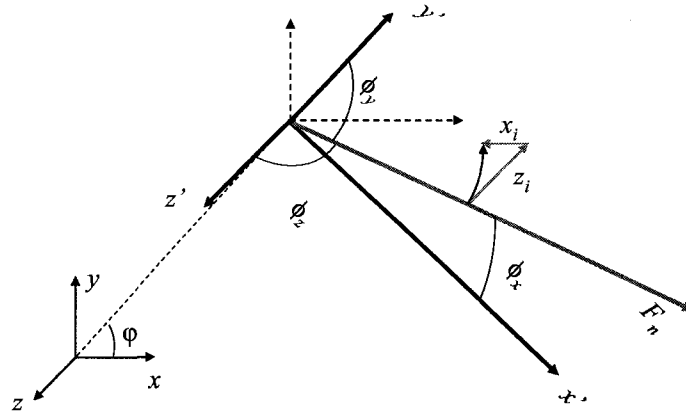


Figure 5.8: (θ_y) Contribution to Line-of-Action

This displacement method is applied to each gear, resulting in a 12-DOF displacement model. The pitch point displacements are resolved into global coordinates through two coordinate transformations. The pressure and helical angles fix the components to the prime coordinate system located on the pitch circle. The angle of orientation fixes the prime coordinate system to the global coordinate system, resulting in Equations 5.6 through 5.8.

$$\begin{aligned} s_{xi} &= x_i (\sin \varphi \cos \phi_x + \cos \varphi \cos \phi_y) \\ s_{xj} &= x_j (\sin \varphi \cos \phi_x + \cos \varphi \cos \phi_y) \end{aligned} \quad (5.6)$$

$$\begin{aligned} s_{yi} &= y_i (\sin \varphi \cos \phi_y - \cos \varphi \cos \phi_x) \\ s_{yj} &= y_j (\sin \varphi \cos \phi_y - \cos \varphi \cos \phi_x) \end{aligned} \quad (5.7)$$

$$\begin{aligned} s_{zi} &= z_i \cos \phi_z \\ s_{zj} &= z_j \cos \phi_z \end{aligned} \quad (5.8)$$

The rotational motion displacements are presented in terms of (x) , (y) , and (z) .

For rotation about (x) :

$$\begin{aligned} y_i &= r_i (\sin \varphi) (\cos \theta_{xi} - 1) \\ y_j &= r_j (\sin \varphi) (1 - \cos \theta_{xj}) \end{aligned} \quad (5.9)$$

$$\begin{aligned} z_i &= r_i \sin \varphi \sin \theta_{xi} \\ z_j &= -r_j \sin \varphi \sin \theta_{xj} \end{aligned} \quad (5.10)$$

For rotation about (y) :

$$\begin{aligned} x_i &= r_i (\cos \varphi) (\cos \theta_{yi} - 1) \\ x_j &= r_j (\cos \varphi) (1 - \cos \theta_{yj}) \end{aligned} \quad (5.11)$$

$$\begin{aligned} z_i &= -r_i \cos \varphi \sin \theta_{yi} \\ z_j &= r_j \sin \varphi \sin \theta_{yj} \end{aligned} \quad (5.12)$$

For rotation about (z) :

$$\begin{aligned} x_i &= -r_i \cos \varphi + r_i \cos(\varphi + \theta_{zi}) \\ x_j &= r_j \cos \varphi - r_j \cos(\varphi + \theta_{zj}) \end{aligned} \quad (5.13)$$

$$\begin{aligned} y_i &= -r_i \sin \varphi + r_i \sin(\varphi + \theta_{zi}) \\ y_j &= r_j \sin \varphi - r_j \sin(\varphi + \theta_{zj}) \end{aligned} \quad (5.14)$$

Equations 5.9 through 5.14 are linearized with the small angle assumption.

For rotation about (x) :

$$\begin{aligned} z_i &= r_i \theta_{xi} \sin \varphi \\ z_j &= -r_j \theta_{xj} \sin \varphi \end{aligned} \quad (5.15)$$

For rotation about (y) :

$$\begin{aligned} z_i &= -r_i \theta_{yi} \cos \varphi \\ z_j &= r_j \theta_{yj} \sin \varphi \end{aligned} \quad (5.16)$$

For rotation about (z) :

$$\begin{aligned} x_i &= -r_i \theta_{zi} \sin(\varphi) \\ x_j &= r_j \theta_{zj} \sin(\varphi) \end{aligned} \quad (5.17)$$

$$\begin{aligned} y_i &= r_i \theta_{zi} \cos(\varphi) \\ y_j &= -r_j \theta_{zj} \cos(\varphi) \end{aligned} \quad (5.18)$$

Substituting into the translational displacements of Equations 5.6 through 5.8 yields the following displacement equations.

$$\begin{aligned} s_{xi} &= x_i (\sin \varphi \cos \phi_x + \cos \varphi \cos \phi_y) \\ s_{xj} &= x_j (\sin \varphi \cos \phi_x + \cos \varphi \cos \phi_y) \end{aligned} \quad (5.19)$$

$$\begin{aligned} s_{yi} &= y_i (\sin \varphi \cos \phi_y - \cos \varphi \cos \phi_x) \\ s_{yj} &= y_j (\sin \varphi \cos \phi_y - \cos \varphi \cos \phi_x) \end{aligned} \quad (5.20)$$

$$\begin{aligned} s_{zi} &= z_i \cos \phi_z \\ s_{zj} &= z_j \cos \phi_z \end{aligned} \quad (5.21)$$

$$\begin{aligned} s_{\theta_{xi}} &= r_i \theta_{xi} \sin \varphi \cos \phi_z \\ s_{\theta_{xj}} &= -r_j \theta_{xj} \sin \varphi \cos \phi_z \end{aligned} \quad (5.22)$$

$$\begin{aligned} s_{\theta_{yi}} &= -r_i \theta_{yi} \cos \varphi \cos \phi_z \\ s_{\theta_{yj}} &= r_j \theta_{yj} \sin \varphi \cos \phi_z \end{aligned} \quad (5.23)$$

$$\begin{aligned} s_{\theta_{zi}} &= -r_i \theta_{zi} \sin(\varphi) (\sin \varphi \cos \phi_x + \cos \varphi \cos \phi_y) + r_i \theta_{zi} \cos(\varphi) (\sin \varphi \cos \phi_y - \cos \varphi \cos \phi_x) \\ s_{\theta_{zj}} &= r_j \theta_{zj} \sin(\varphi) (\sin \varphi \cos \phi_x + \cos \varphi \cos \phi_y) - r_j \theta_{zj} \cos(\varphi) (\sin \varphi \cos \phi_y - \cos \varphi \cos \phi_x) \end{aligned} \quad (5.24)$$

The displacement equation can now be assembled, which describes the relative motion between the displacements of node (*i*) and node (*j*). For the translational components, this equates to

$$u_{xyz} = (s_{xj} - s_{xi}) + (s_{yj} - s_{yi}) + (s_{zj} - s_{zi}) \quad (5.25)$$

The angular displacement terms must be treated differently due to the effect of force coupling. For the derivation of Equations 5.15 through 5.18, a positive angular displacement was assumed about each axis. In reality, however, the angular displacements are negatively coupled. A positive displacement of (θ_{xi}) results in a negative displacement in (θ_{xj}) and vice versa. Consequently, the angular displacement relative motion must take the form:

$$u_{\theta} = (-s_{\theta_{xj}} - s_{\theta_{xi}}) + (-s_{\theta_{yj}} - s_{\theta_{yi}}) + (-s_{\theta_{zj}} - s_{\theta_{zi}}) \quad (5.26)$$

The total displacement along the line-of-action can be written as

$$u_{LOA} = (s_{xj} - s_{xi}) + (s_{yj} - s_{yi}) + (s_{zj} - s_{zi}) + (-s_{\theta_{xj}} - s_{\theta_{xi}}) + (-s_{\theta_{yj}} - s_{\theta_{yi}}) + (-s_{\theta_{zj}} - s_{\theta_{zi}}) \quad (5.27)$$

The force along the line-of-action becomes

$$F_{mesh_{LOA}} = -K \left((s_{xj} - s_{xi}) + (s_{yj} - s_{yi}) + (s_{zj} - s_{zi}) + (-s_{\theta xj} - s_{\theta xi}) + (-s_{\theta yj} - s_{\theta yi}) + (-s_{\theta zj} - s_{\theta zi}) \right) \quad (5.28)$$

where (K) is a component of the mesh stiffness matrix.

When the helical angle equals zero, Equation 5.28 becomes the equation of a spur gear, similar to that presented by Choy (1988).

5.2.4 Force and Moment Nodal Equations

The components of the force acting on node (i) in the cardinal directions are given in Equation 5.5. The moment equations about node (j) are determined by multiplying the force components and moment arms about the respective axis.

$$\begin{aligned} M_{xi} &= -F_{zi} r_i \cos \varphi \\ M_{yi} &= -F_{zi} r_i \sin \varphi \\ M_{zi} &= -F_{yi} r_i \sin \varphi + F_{xi} r_i \cos \varphi \end{aligned} \quad (5.29)$$

The force components acting on node (j) are equal and opposite to those acting on node (i) , since the forces acting on the gear teeth must be in equilibrium. The moments about node (j) are not equal and opposite to the moments about node (i) , unless the gear ratio is unity. The forces and moments for node (j) can be written as

$$\begin{aligned}
F_{xj} &= -F_{xi} \\
F_{yj} &= -F_{yi} \\
F_{zj} &= -F_{zi} \\
M_{xj} &= -F_{zj}r_j \cos \varphi \\
M_{yj} &= -F_{zj}r_j \sin \varphi \\
M_{zj} &= -F_{yj}r_{ij} \sin \varphi + F_{xj}r_{ij} \cos \varphi
\end{aligned} \tag{5.30}$$

5.2.5 Influence Coefficient Method

Each element of the mesh stiffness matrix, $[\mathbf{K}]_{mesh}$, can be calculated by determining the components of a force vector, $\{\mathbf{F}_i\}$, acting under the condition that only one of the components of the displacement vector, $\{\mathbf{q}_i\}$, is a unit displacement, and all other components in $\{\mathbf{q}_i\}$ and $\{\mathbf{q}_j\}$ are zero. This is the basic principle of the influence coefficient method. The mesh stiffness matrix can be assembled by applying a unit displacement for each variable in $\{\mathbf{q}_i\}$ sequentially with zero displacement in all other variables to include the $\{\mathbf{q}_j\}$ vector. Applying to each of the six different unit displacements in $\{\mathbf{q}_i\}$ will yield a (6×12) matrix. Similarly, applying the same method to each of the variables in $\{\mathbf{q}_j\}$ yields another (6×12) matrix. When combined, these form a symmetric mesh stiffness matrix between nodes (i) and (j) .

Returning to the form of Equation 5.1, the mesh stiffness matrix can be written as

$$[\mathbf{K}]_{mesh} = K_g \begin{bmatrix} [\mathbf{K}_{ii}] & [\mathbf{K}_{ij}] \\ [\mathbf{K}_{ji}] & [\mathbf{K}_{jj}] \end{bmatrix} = K_g \begin{bmatrix} [\mathbf{K}_{ii}] & [\mathbf{K}_{ji}]^T \\ [\mathbf{K}_{ji}] & [\mathbf{K}_{jj}] \end{bmatrix} \tag{5.31}$$

The elements of the four sub-matrices in Equation 5.31 are presented in Appendix B.

5.3 Insertion into the Finite Element Model

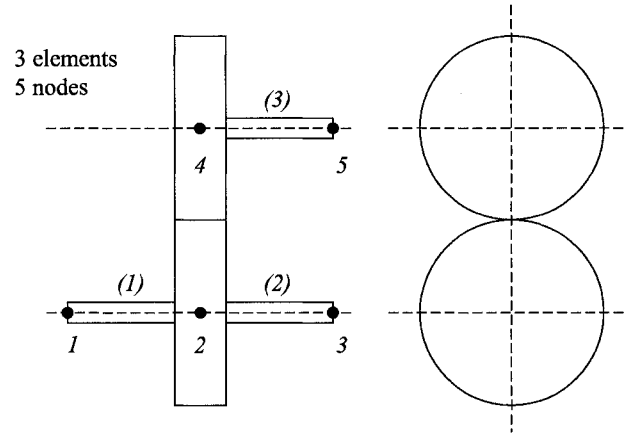


Figure 5.9: Finite Element Implementation of Mesh Stiffness

The mesh stiffness is implemented into a finite element analysis in the following way. Nodes of shaft elements are generally numbered sequentially for ease of analysis. However, the coupled nodes of gear shafts are usually non-sequential, as shown in Figure 5.9. The mesh coupling between nodes 2 and 4, results in the mesh model

$$[\mathbf{K}]_{mesh} \begin{Bmatrix} \mathbf{q}_2 \\ \mathbf{q}_4 \end{Bmatrix} = \begin{bmatrix} [\mathbf{k}_{2,2}]_{mesh} & [\mathbf{k}_{2,4}]_{mesh} \\ [\mathbf{k}_{4,2}]_{mesh} & [\mathbf{k}_{4,4}]_{mesh} \end{bmatrix} \begin{Bmatrix} \mathbf{q}_2 \\ \mathbf{q}_4 \end{Bmatrix} \quad (5.32)$$

When assembling the global matrix, the mesh matrix “breaks apart” to place the corresponding sub-matrices in the appropriate blocks of the global matrix. For Figure 5.9, the mesh stiffness matrix is substituted into the global matrix as follows.

$$[\mathbf{K}]^{(G)} = \begin{bmatrix} [\mathbf{k}_{1,1}^{(1)}] & [\mathbf{k}_{1,2}^{(1)}] & 0 & 0 & 0 \\ [\mathbf{k}_{2,1}^{(1)}] & [\mathbf{k}_{2,2}^{(1)} + \mathbf{k}_{2,2}^{(2)}] + [\mathbf{k}_{2,2}]_{mesh} & [\mathbf{k}_{2,3}^{(2)}] & [\mathbf{k}_{2,4}]_{mesh} & 0 \\ 0 & [\mathbf{k}_{3,2}^{(2)}] & [\mathbf{k}_{3,3}^{(2)} + \mathbf{k}_{3,3}^{(3)}] & [\mathbf{k}_{3,4}^{(3)}] & 0 \\ 0 & [\mathbf{k}_{4,2}]_{mesh} & [\mathbf{k}_{4,3}^{(3)}] & [\mathbf{k}_{4,4}^{(3)} + \mathbf{k}_{4,4}^{(4)}] + [\mathbf{k}_{4,4}]_{mesh} & [\mathbf{k}_{4,5}^{(4)}] \\ 0 & 0 & 0 & [\mathbf{k}_{5,4}^{(4)}] & [\mathbf{k}_{5,5}^{(4)}] \end{bmatrix} \quad (5.33)$$

5.4 Application

5.4.1 Approach and System Parameters

The application selected for testing this methodology is a basic spur gear shaft pair similar to the Spur Gear Test Rig at the NASA Glenn Research Center and shown in Figure 5.10. This application was selected because of the ease of availability of system parameters from the literature.

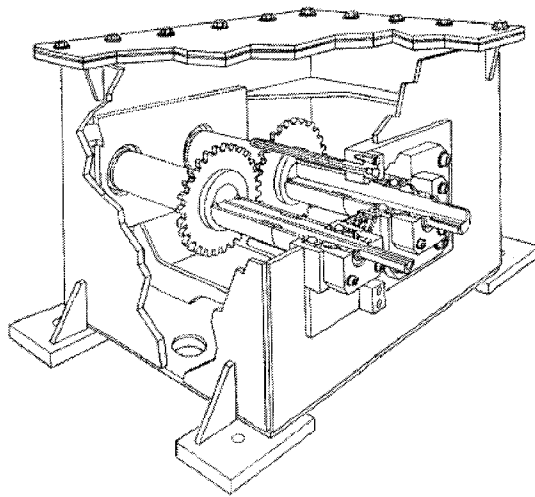


Figure 5.10: NASA Glenn Research Center Spur Gear Test Rig (Lim, 1991)

The system consists of two identical gears resting on two identical shafts, connected to the gearbox by a pair of rolling element bearings per shaft. The parameters of the shafts and the gears are provided in Table 5.1, as documented in Singh (1990b).

Table 5.1: Spur Gear Application System Parameters

Material Parameters

Young's Modulus (E) (N/m^2)	2.03E11
Shear Modulus (G) (N/m^2)	8.0E10
Poisson's Ratio (ν)	0.27
Density (ρ) (kg/m^3)	7750

Shaft Parameters

Length (L_s) (m)	0.254
Mass (M_s) (kg)	1.96
Rigidity (EI) ($N-m^2$)	18576
Outer Diameter (d_{so}) (m)	0.037
Inner Diameter (d_{si}) (m)	0.01

Gear Parameters

Mass (M_g) (kg)	1.84
Moment of Inertia (I_t) ($kg-m^2$)	1.8E-3
Polar Moment of Inertia (I_p) ($kg-m^2$)	3.6E-3
Diameter (d_g) (m)	0.089
Pressure Angle (α_n) (deg)	20
Helical Angle (β) (deg)	0.0
Orientation Angle (ϕ) (deg)	0.0
Average Mesh Stiffness (K_g) (N/m)	1.0E8
Number of Teeth	28

Bearing Parameters

k_{xx}, k_{yy} (N/m)	1.0E9
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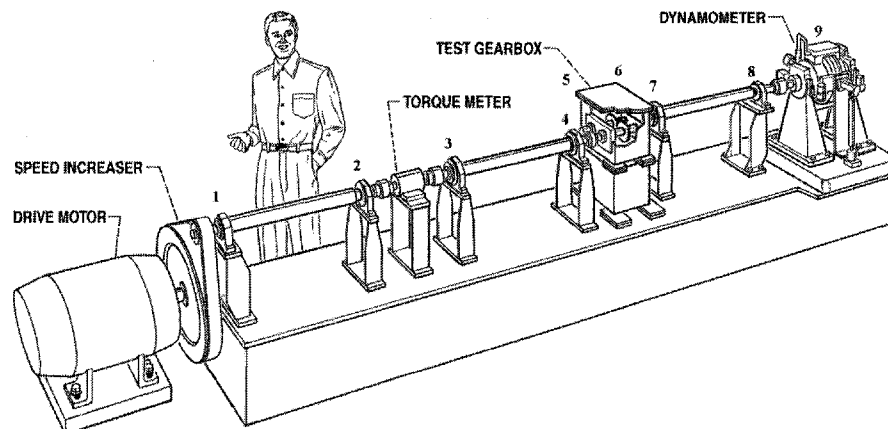


Figure 5.11: Gear Noise Test Rig - NASA Glenn Research Center (Lim, 1991)

In the analysis, two scenarios have been compared and analyzed. The first is one of the rotating gear shafts as a single system, with the gear acting as a disk only. The second examines the complete geared system.

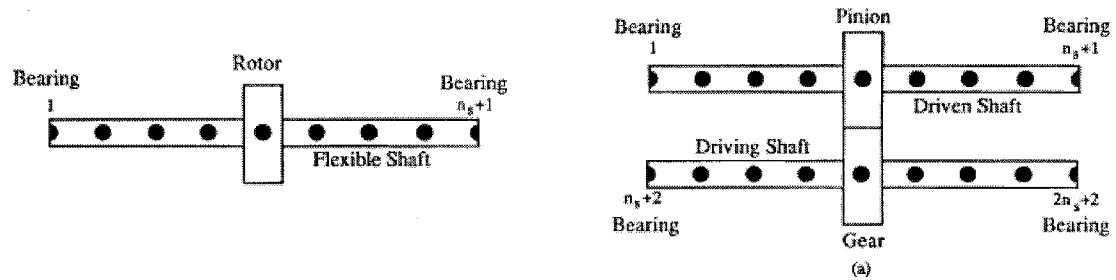


Figure 5.12: Single Shaft (Left) and Geared Shaft (Right) Scenarios (Lim, 1991)

5.4.2 Results – Single Shaft System

Table 5.2 lists the first eight natural frequencies of the single shaft system with their corresponding descriptions.

Table 5.2: Natural Frequencies (Hz) of Single Shaft

Frequency No.	Mode	Single Rotor
1	Axial, Torsional	0
2	1 st Lateral	684
3	2 nd Lateral	2838,2858
4	3 rd Lateral	6677
5	4 th Lateral	7138
6	2 nd Torsional	12669
7	3 rd Torsional	12795
8	4 th Lateral	14467,14469
9	5 th Lateral	14714
10	2 nd Axial	20180

The first 10 mode shapes are plotted in Figures 5.13 through 5.21 for two conditions, the first at 0 *rpm* and the second at 1,000 *rpm*. At zero rotational speed, the mode shapes are planar. The shaft behaves as a vibrating beam. At 1,000 *rpm*, the mode shapes are no

longer planar, and the orbits in the mode shapes become evident. The axial and torsional frequencies are not speed dependent.

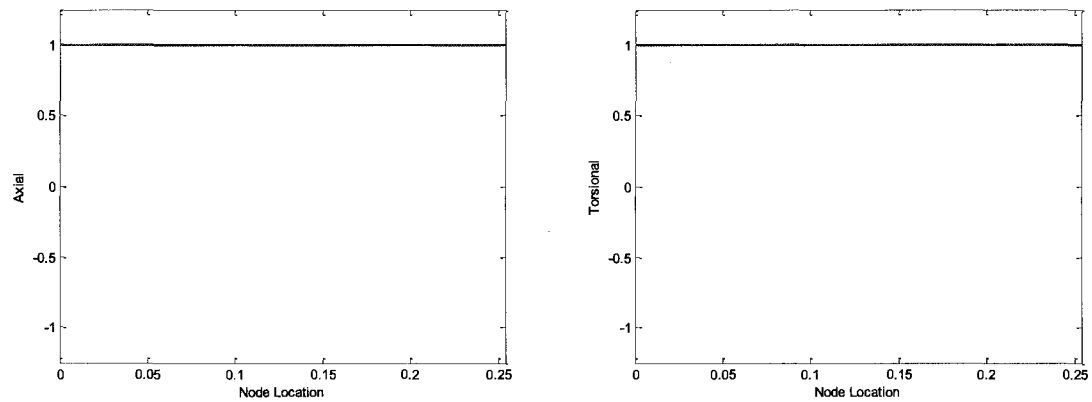


Figure 5.13: Axial (left) and Torsional (right) Rigid Body Modes - 0 Hz

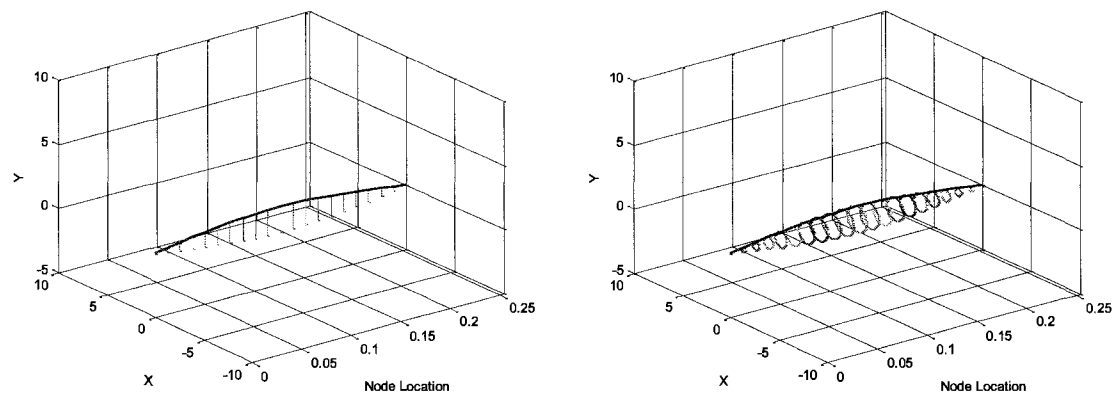


Figure 5.14: 1st Lateral Mode Shape - 684 Hz

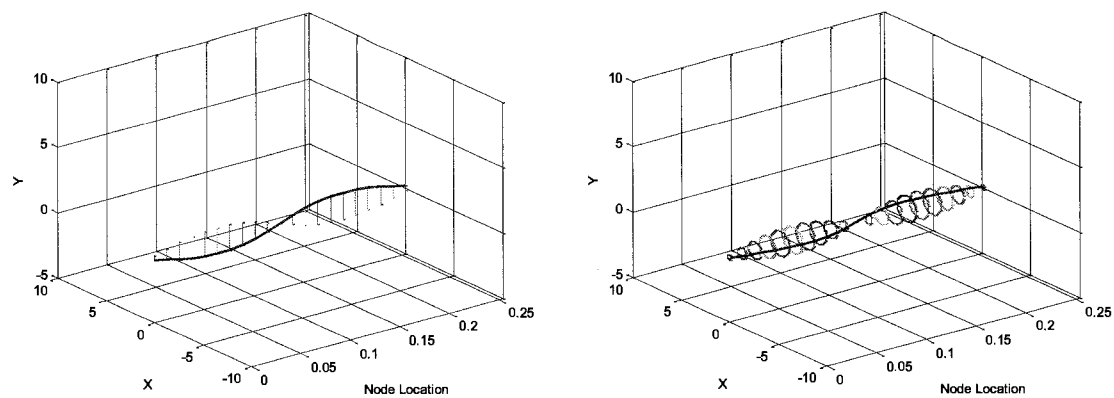


Figure 5.15: 2nd Lateral Mode Shape - 2,848 Hz

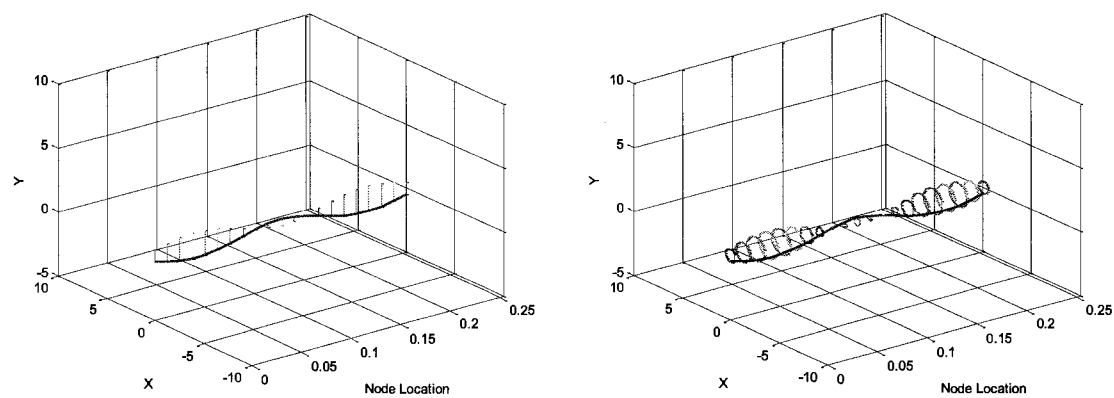


Figure 5.16: 3rd Lateral Mode Shape - 6,677 Hz

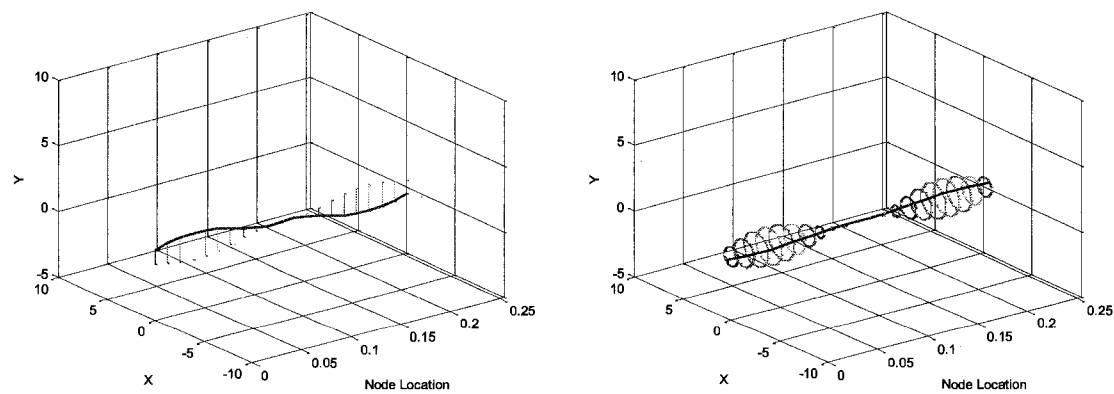


Figure 5.17: 4th Lateral Mode Shape - 7,138 Hz

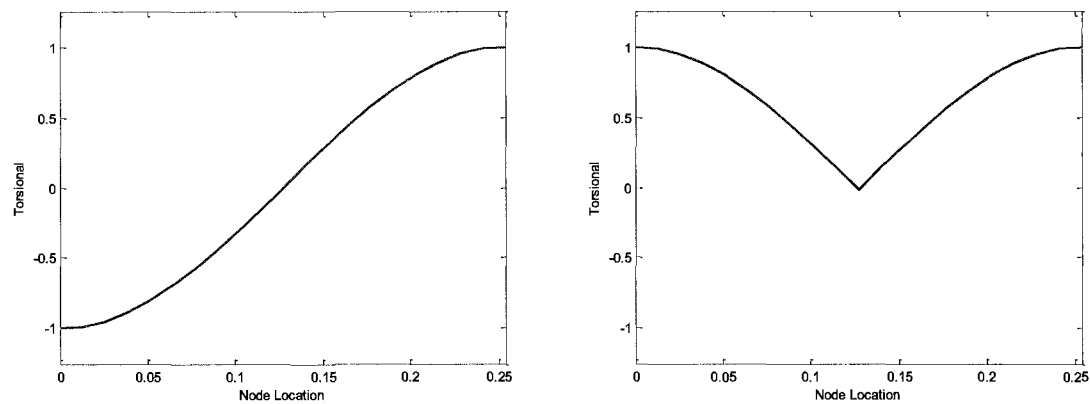


Figure 5.18: 2nd (Left) and 3rd (Right) Torsional Frequencies – 12,668 and 12,795 Hz, respectively

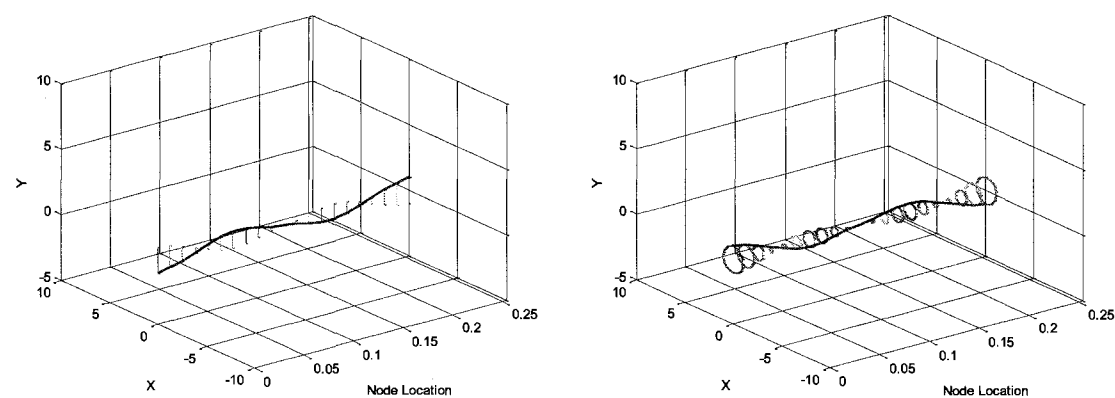


Figure 5.19: 4th Lateral Mode Shape – 14,468 Hz

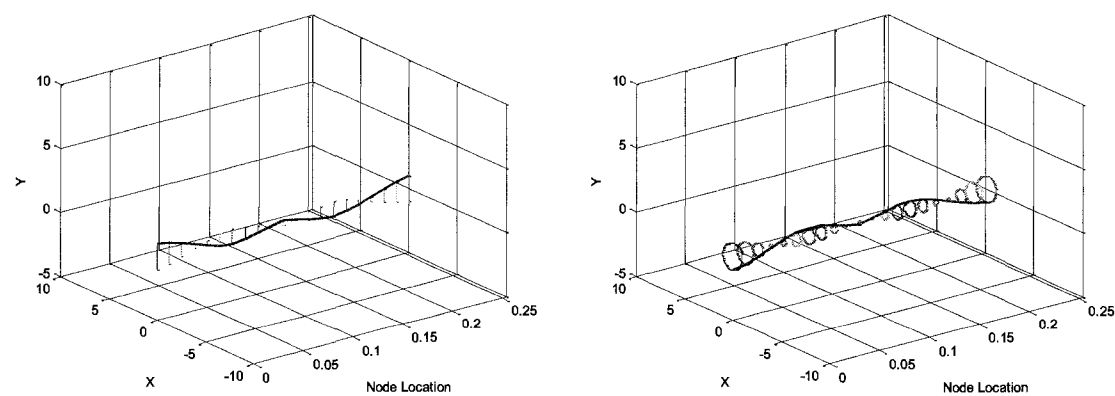


Figure 5.20: 5th Lateral Mode Shape – 14,714 Hz

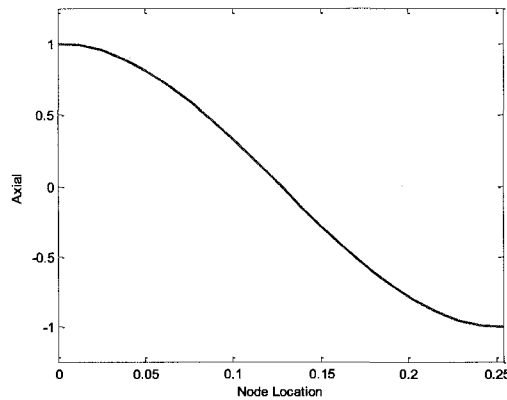


Figure 5.21: 2nd Axial Mode Shape – 20,180 Hz

The natural frequencies were determined for speeds between 0 *rpm* and 10,000 *rpm* in order to determine the effects of the gyroscopic moments on the system. The natural frequencies presented above are quite large. The shaft itself is short, with a length of only 10.0 *in*. One would expect the frequencies to be relatively insensitive to changes in rotational speed. A Campbell Diagram verifies this.

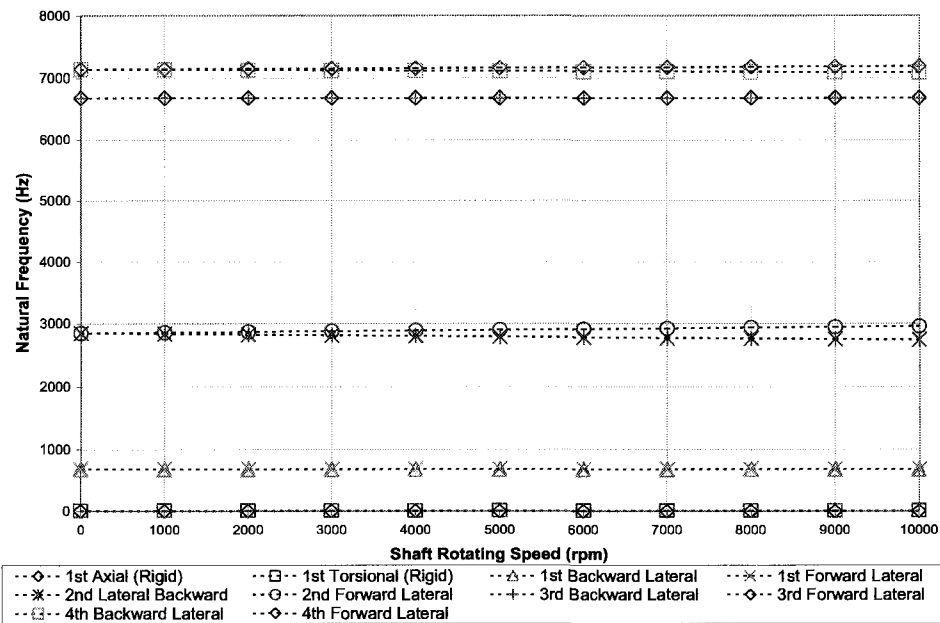


Figure 5.22: Campbell Diagram - Single Shaft Scenario

5.4.3 Results – Dual Shaft System

Table 5.2 lists the frequencies of the geared system and compares them to the frequencies of the single-shaft results.

Table 5.3: Comparison of Natural Frequencies (Hz) at 1,000 *rpm*

Frequency No.	Mode	Single Rotor	Gear Pair
1	Axial, Torsional	0	0
	Coupled, 1 st Lateral	--	505
2	1 st Lateral	684	684
	Coupled, 1 st Lateral	--	2090
3	2 nd Lateral	2838, 2858	2838, 2858
4	3 rd Lateral	6677	6677
	Coupled, 3 rd Lateral	--	6706
5	4 th Lateral	7133, 7143	7133, 7146
6	2 nd Torsional	12669	12669
7	3 rd Torsional	12795	
	Coupled, 5 th Lateral	--	12798
8	4 th Lateral	14467, 14469	14467, 14469
9	5 th Lateral	14714	14714
	Coupled, 5 th Lateral	--	14720
10	2 nd Axial	20180	20180

For comparison, the first two lateral mode shapes of the single and dual shaft systems at a rotational speed of 1,000 *rpm* are presented in Figures 5.23 and 5.24.

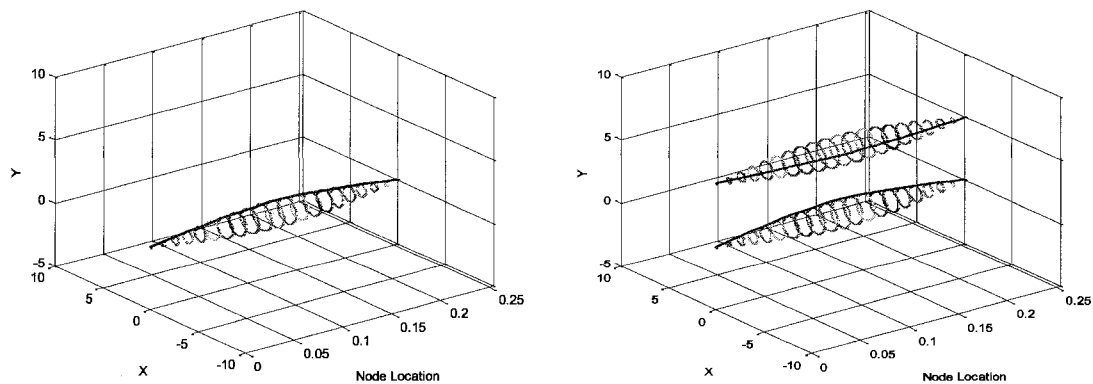


Figure 5.23: Comparison of 1st Lateral Mode Shape – 684 Hz

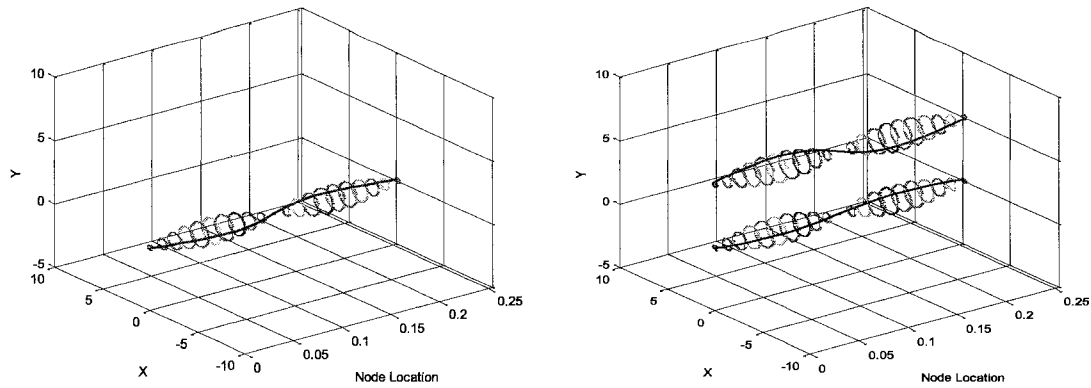


Figure 5.24: Comparison of 2nd Lateral Mode Shape - 2,858 Hz

Additionally, coupled lateral-torsional mode shapes arise from the gear-mesh. Figures 5.25 through 5.29 present the first five coupled mode shapes at a rotational speed of 1,000 rpm.

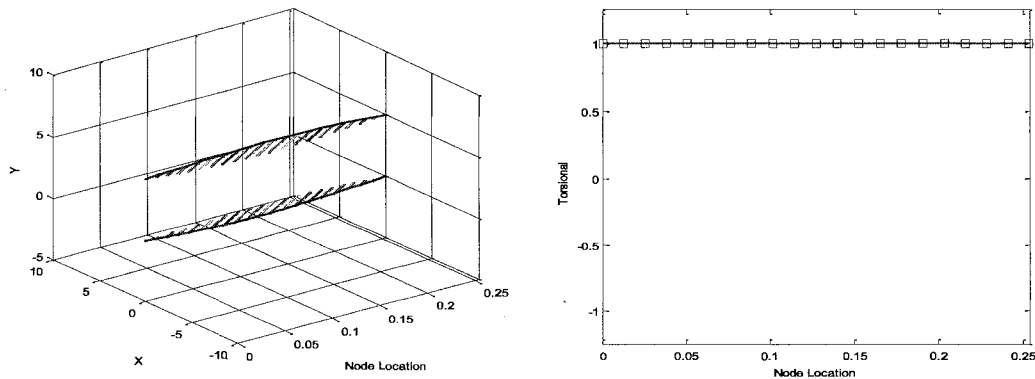


Figure 5.25: Coupled, 1st Lateral-Torsional Mode Shape - 505 Hz

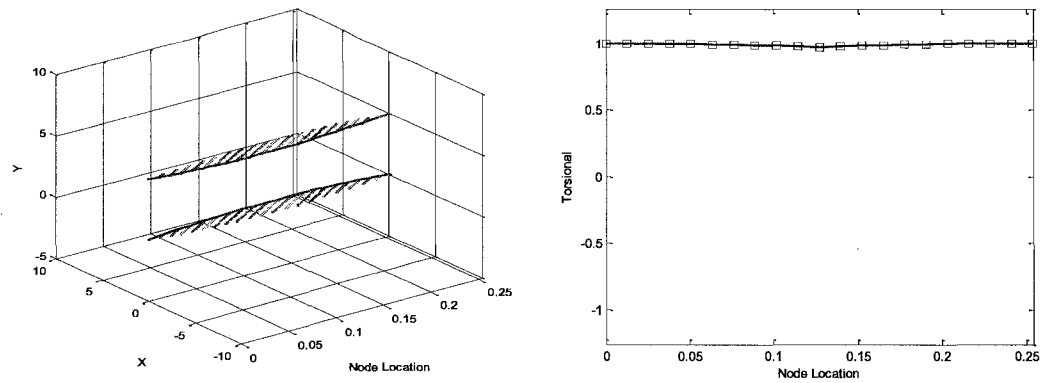


Figure 5.26: Coupled, 1st Lateral-Torsional Mode Shape - 2,090 Hz

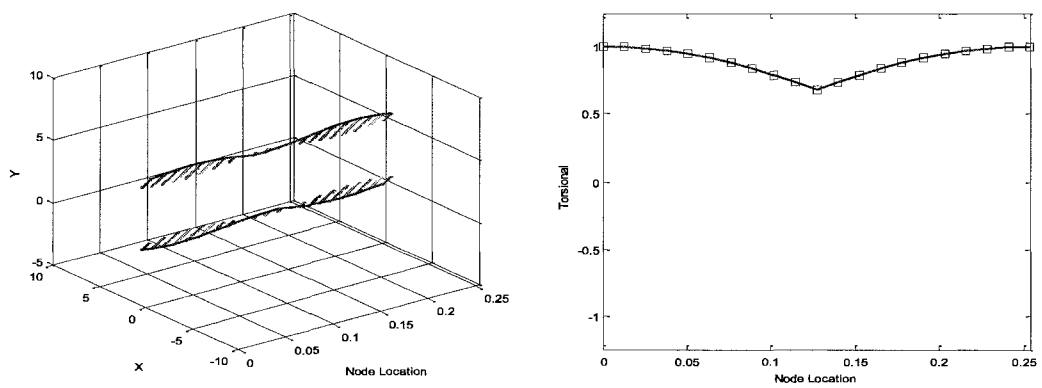


Figure 5.27: Coupled, 3rd Lateral-Torsional Mode Shape - 6,706 Hz

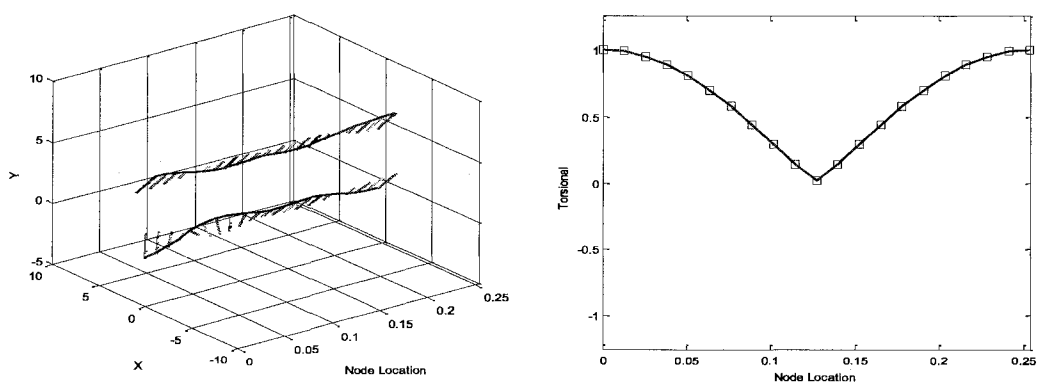


Figure 5.28: Coupled, 5th Lateral-Torsional Mode Shape - 12,798 Hz

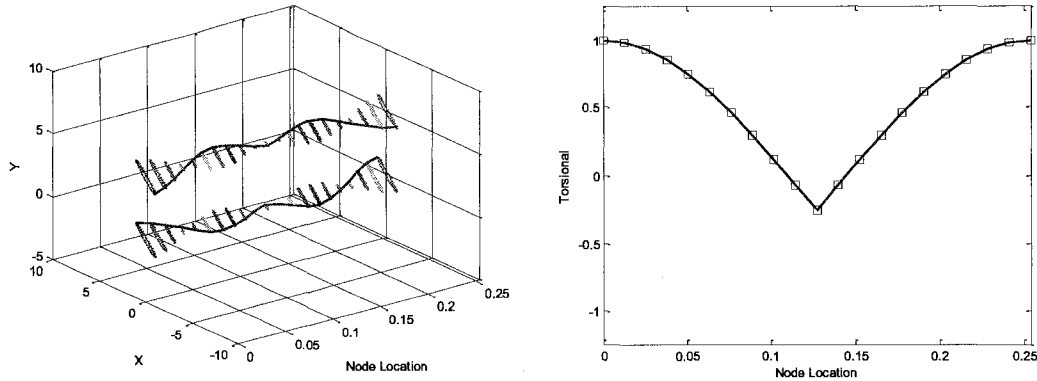


Figure 5.29: Coupled, 5th Lateral-Torsional Mode Shape -14,720 Hz

As with the single rotor system, the dual shafts are insensitive to the gyroscopic effects from 0-10,000 *rpm*. However, if the shaft rotating speed is examined from 0-10,000 Hz (0-600,000 *rpm*), the gyroscopics effects become significant. This is further discussed in a later section.

5.4.4 Observations

Comparing the two systems, the following observations are noted.

1. The uncoupled lateral, torsional, and axial frequencies of the dual shaft system are the same as the single shaft system.
2. The geared shaft system exhibits frequencies not appearing in the single shaft system. These frequencies identify coupled modes of vibration between the lateral and torsional DOFs.
3. The coupled mode shapes have distinct characteristics. First, the motions of the two shafts oppose each other, and are 180 degrees out of phase. Second, the shapes of the coupled modes are similar to those characterizing odd-numbered lateral modes of vibration.

4. The magnitude of the torsional coupling increases as the frequency increases. At the lowest frequency of 505 Hz, the torsional motion is rigid as depicted in Figure 5.25. As the frequency increases, the gear pair's effect on the torsional component of the shaft becomes more pronounced, as seen in Figure 5.25 through 5.29.

Lastly, these observations are limited by the symmetry of the system. The spur gear rig of Figure 5.10 consists of identical rotating shafts with identical spur gears placed in the center of the shaft. Due to the symmetry of the gears, bearings, and shaft, the mode shapes will always be symmetric.

5.4.5 Validation

The results of Table 5.2 were then compared with two previous studies by Choy (1993) and Kahraman (1990), and are presented in Figure 5.30. The x-axis groups the frequencies by mode of vibration. The mesh results are consistent with published data.

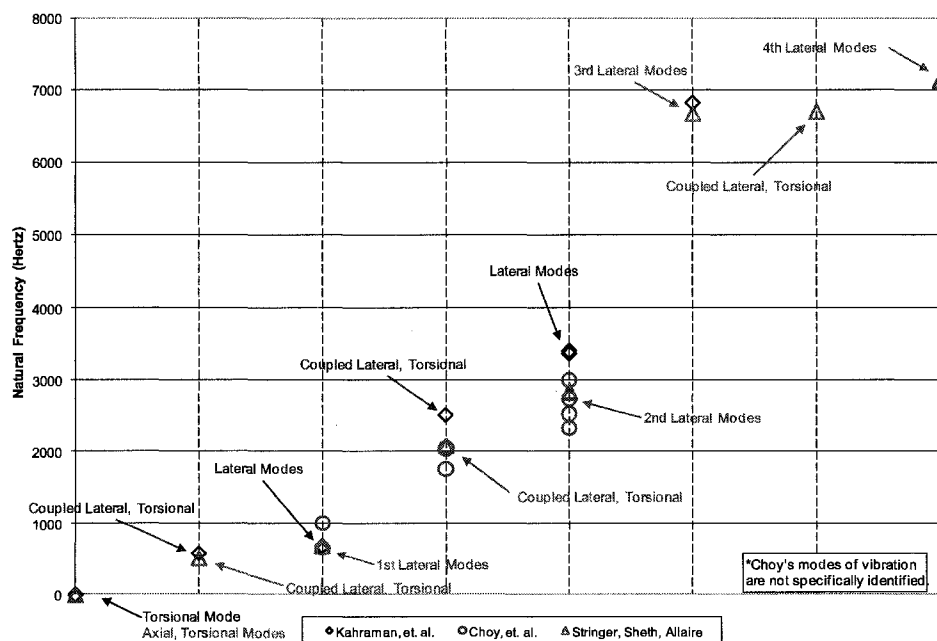


Figure 5.30: Validation with Previous Results (Choy, 1993, Kahraman, 1990)

5.5 Gear-Mesh Considerations

Energy dissipation in the gear mesh is a result of three sources. The first source of loss comes from the elastohydrodynamic effect: i.e., the presence of oil in between the meshing surfaces. Previous research has indicated that the damping coefficients due to EHD effects are very small, on the order of 0.0004-0.004 $N\cdot s/m$ (Choy, 1993). The second source includes the sliding friction action of the meshing gear teeth through the mesh cycle. The final source of energy loss is due to manufacturing errors. In this application, such losses were neglected.

5.6 Forced Response: Gear-Mesh Frequency

An excitation present in all geared systems is the gear-mesh frequency, also called the gear-tooth or tooth-pass frequency. Gear-mesh frequencies originate from the individual impacts of the gear teeth as they come into contact with one another (Houser, 2006, Bodre, 2008). The gear-mesh frequency is the number of gear teeth, (N_i) , multiplied by the rotational speed of the shaft, (Ω_i) .

$$\omega_{mesh} = N_i \Omega_i \quad (5.34)$$

The forcing function takes the form:

$$f(t) = F_o e^{j\omega_{mesh}t} = F_o e^{jN_i\Omega_i t} \quad (5.35)$$

The amplitude, (F_o) , is a result of the nonlinear loading pattern of the gear-mesh, which is further discussed in Chapter 6. In the finite element analysis, it acts at the nodes corresponding to the gear pair locations.

Like any other forcing frequency, the gear-mesh frequency becomes important if it coincides with a natural frequency of the system, causing resonance. The significance of the gear-mesh frequency is that it excites higher modes of vibration at lower speeds. This is further illustrated through a Campbell Diagram and frequency response chart.

Returning to the discussion of gyroscopics, the shafts were relatively insensitive to gyroscopic effects at low operating speeds. However, the results are much different when comparing the frequencies in Hertz (Hz), as illustrated in Figure 5.31.

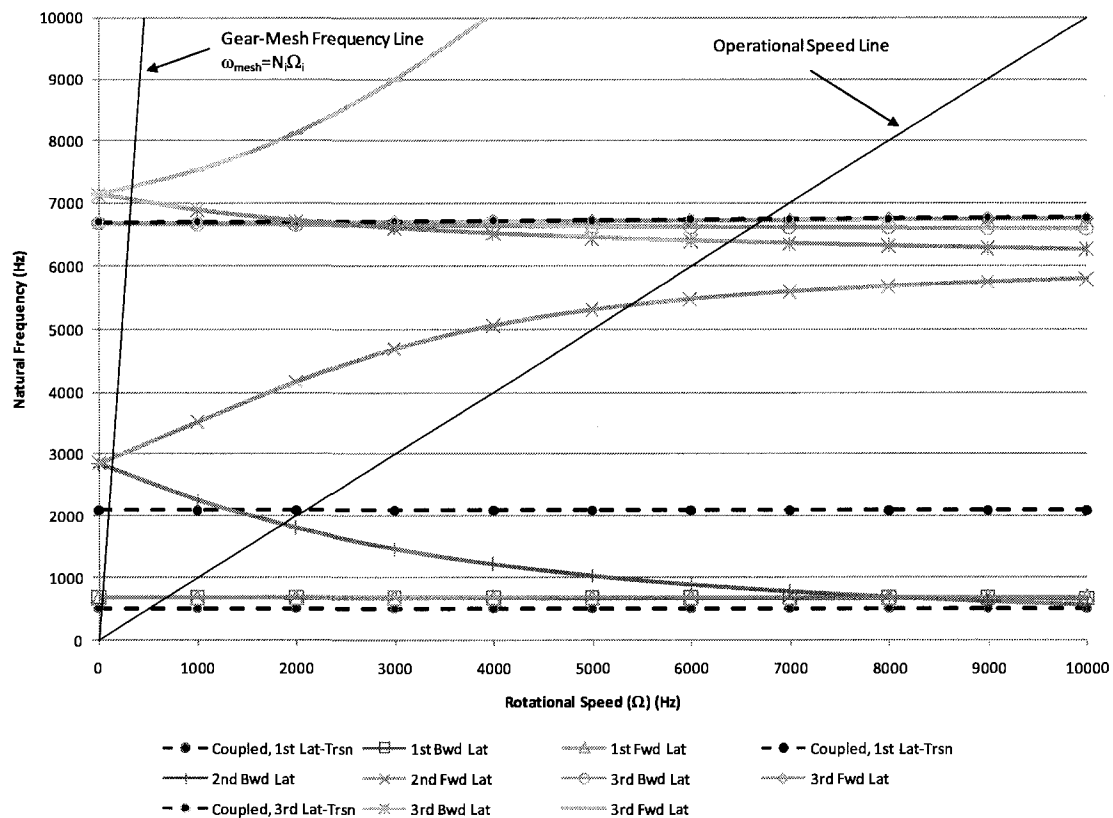


Figure 5.31: Campbell Diagram (0-5,000 Hz)

The gyroscopic effects are much more evident. Of more concern is the gear-mesh frequency line at the left of the chart. Every intersection between this line and a natural or coupled frequency mode indicates resonance at the corresponding rotational speed

on the x -axis. This is more clearly evident in the gear-mesh frequency response, depicted in Figure 5.32.

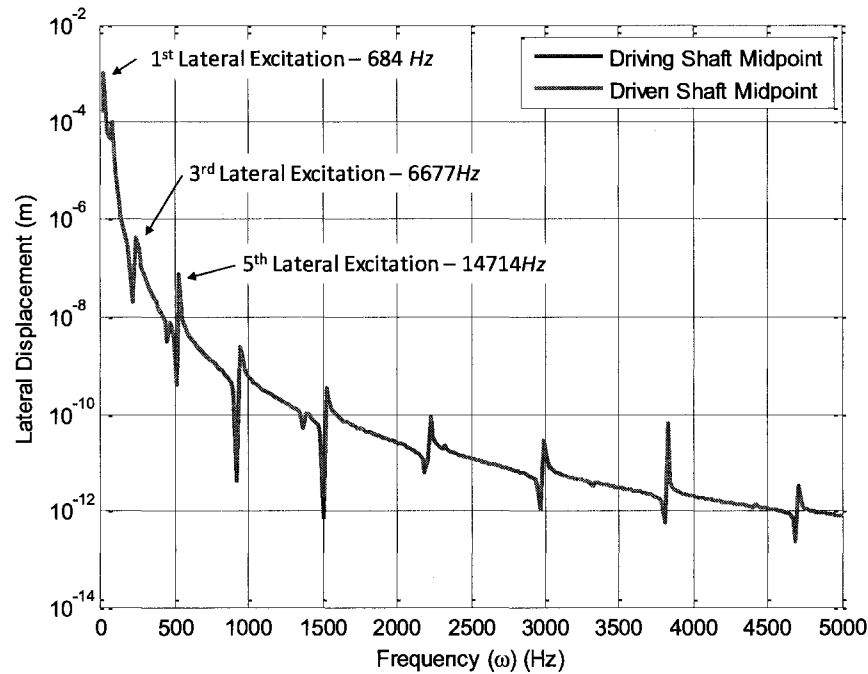


Figure 5.32: Gear-Mesh Frequency Response (0-5,000 Hz)

From Figure 5.32, certain observations are noted. First, each peak corresponds to a higher mode of vibration excited by the gear-mesh frequency. Second, resonant displacement is captured for odd-numbered modes only. This is due to excitation occurring at the midpoint of each shaft. Even-numbered mode shapes are characterized by minimum displacement at the shaft midpoints. The coupled lateral-torsional frequencies did not register on the frequency spectrum, due to their limited bandwidth.

5.7 Distributed Parameter Model Considerations

The mesh matrix in its present state applies to lumped parameter models only, where the coupling connects the center of two shafts, as first illustrated in Figure 5.1. In

this formulation, conventional rotor dynamic tools are used to analyze a discrete set of ordinary differential equations. A consequence of the lumped parameter model is that visibility on the overall structural dynamics of the system is lost.

If the dynamics of the gears themselves are important, such as in the context of a contact analysis, then the rigid disk assumption is no longer appropriate. The finite element model becomes a more complex distributed model, with approximate solutions to partial differential equations, rather than the closed-form solutions of the lumped parameter system (Meirovitch, 1997). The coupling between the two gears can be modeled as a force vector acting at the node of contact, as shown in Figure 5.33.

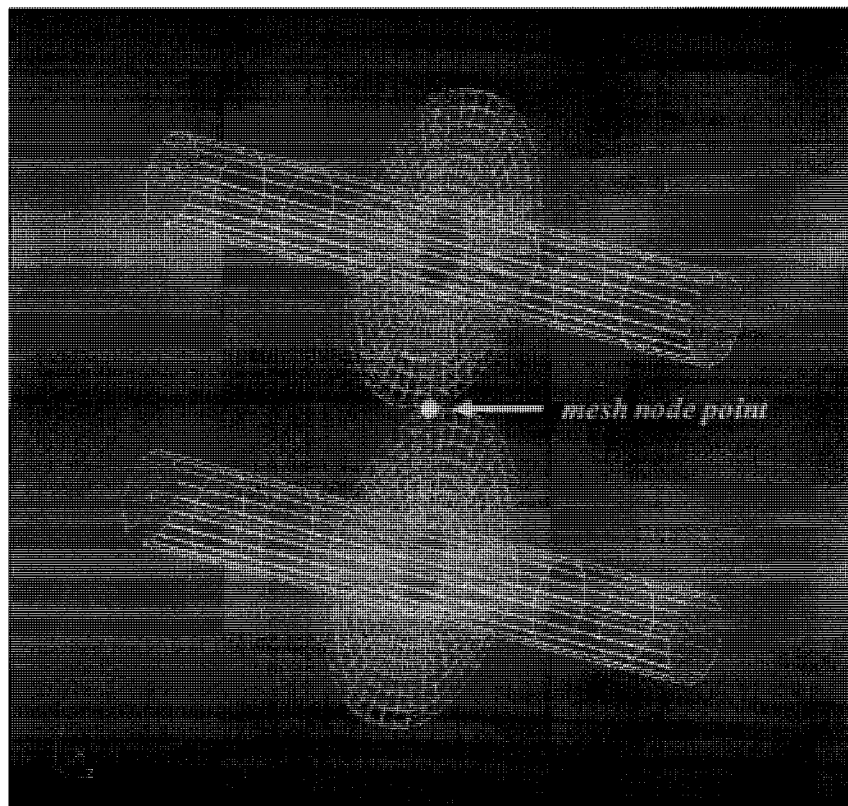


Figure 5.33: Distributed Parameter Model of Geared System

This reduces the mesh stiffness matrix from 12 DOFs to 6, with only the force relationships remaining in the matrix. The force components remain equal and

opposite. In the new mesh matrix, this equates to the first (3×3) elements of the four sub-matrices of the twelve DOF model of Equation 5.1, as shown below.

$$\begin{aligned}
 [\mathbf{K}]_{mesh} &= \mathbf{K}_g \begin{bmatrix} [\mathbf{K}_{ii}] & [\mathbf{K}_{ji}] \\ [\mathbf{K}_{ij}] & [\mathbf{K}_{jj}] \end{bmatrix}_{(12 \times 12)} = \mathbf{K}_g \begin{bmatrix} \begin{bmatrix} \mathbf{K}'_{ii(3 \times 3)} & & \\ & \ddots & \\ & & \mathbf{K}'_{ij(3 \times 3)} \end{bmatrix} & \begin{bmatrix} \mathbf{K}'_{ij(3 \times 3)} & & \\ & \ddots & \\ & & \mathbf{K}'_{jj(3 \times 3)} \end{bmatrix} \\ \begin{bmatrix} \mathbf{K}'_{ji(3 \times 3)} & & \\ & \ddots & \\ & & \mathbf{K}'_{jj(3 \times 3)} \end{bmatrix} & \begin{bmatrix} \mathbf{K}'_{jj(3 \times 3)} & & \\ & \ddots & \\ & & \mathbf{K}'_{jj(3 \times 3)} \end{bmatrix} \end{bmatrix} \\
 [\mathbf{K}']_{mesh} &= \begin{bmatrix} \mathbf{K}'_{ii} & \mathbf{K}'_{ij} \\ \mathbf{K}'_{ji} & \mathbf{K}'_{jj} \end{bmatrix}_{(6 \times 6)}
 \end{aligned} \tag{5.36}$$

Thus, the mesh matrix model can be tailored to both lumped and distributed parameter systems.

5.8 Summary

This chapter introduced a finite element methodology for studying a rotating shaft system containing helical or spur gear pairs. Previous mesh models from the literature were discussed regarding their loss of generality. A rigorous derivation resulted in a 12-DOF mesh stiffness matrix for a spur or helical gear pair. The complete matrix is presented in Appendix B.

Initial validation of the model was presented. A rotor dynamic analysis of the undamped system was conducted and compared to a non-geared rotating shaft having the same properties as the two geared shafts. The analysis identified coupled, lateral-torsional modes of vibration, consistent with geared systems. The mode shapes of the coupled and uncoupled frequencies were presented with noted observations. Gyroscopic effects were also discussed. The coupled, lateral-torsional frequencies were

insensitive to gyroscopic effects. These results compared well with other results from previous studies of the same rig.

The gear-mesh or tooth-pass frequency excitation was introduced. The gear-mesh frequency excites higher modes of vibration at lower rotational speeds. Its effects are further discussed in Chapter 6.

Lastly, a brief comparison of distributed and lumped parameter systems was presented. The differences in the mesh matrices between the two models were discussed. In the distributed parameter model, the mesh stiffness matrix simply reduced to a 6-DOF matrix, consisting of only the forces acting at the gear-mesh. This type of model would be useful in analyses in which the dynamics of the gears themselves were important.

Chapter 6

Nonlinear Mesh Analysis

In Chapter 5, the mesh stiffness parameter, (K_g) , was assumed to be a time-averaged constant, linearizing the equations of motion. In Chapter 3, the nonlinearity of the gear-mesh was discussed, based upon the total number of tooth pairs in contact. The objective of this chapter is to address these nonlinearities and perform a nonlinear analysis of the equations of motion using time and spatially-varying mesh stiffness values. The results are then compared to the linear results and discussed.

6.1 Methodology

The following steps outline the methodology for this nonlinear analysis.

1. The nonlinear mesh stiffness is assumed to be of the form given in Equation 6.1 (Luo, 1996).

$$K_g = K_{go} + \Delta K_g \tag{6.1}$$

where

K_{go} = average mesh stiffness

ΔK_g = periodic mesh stiffness

In Chapter 5, the periodic stiffness was neglected, leaving only the time-averaged component, $(K_g = K_{go})$.

2. The parameters of Equation 6.1 are determined using a finite element contact analysis program, CALYX®, developed by *Advanced Numerical Solutions (ANSOL), Inc.*
3. Once Equation 6.1 is assembled, it is substituted into the equations of motion. The equations of motion are then numerically integrated using the ODE solvers in MATLAB®.
4. The numerical integration results are then converted into the frequency domain using the Fast Fourier Transform (FFT).
5. The nonlinear results are then compared to the linear results of Chapter 5.

6.1.1 ANSOL CALYX® Results

The geometries and parameters of the spur gears from Table 5.1 were entered into the CALYX® program. A contact analysis of the gear teeth was conducted. In post-processing, the analysis included contact load and strain results as the teeth sequenced through mesh. Figures 6.1 through 6.3 illustrate the CALYX® output. Figure 6.1 provides the CALYX® graphic representation of the gears in mesh. Figure 6.2 and 6.3 show the contact load pattern of the teeth in mesh and the strain pattern, respectively.

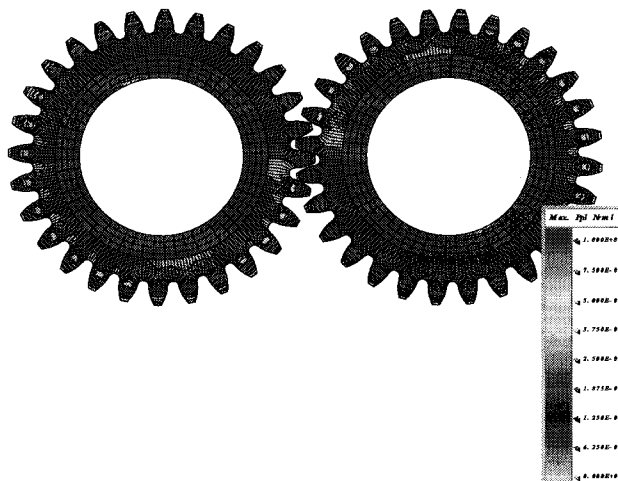


Figure 6.1: CALYX® Finite Element Representation

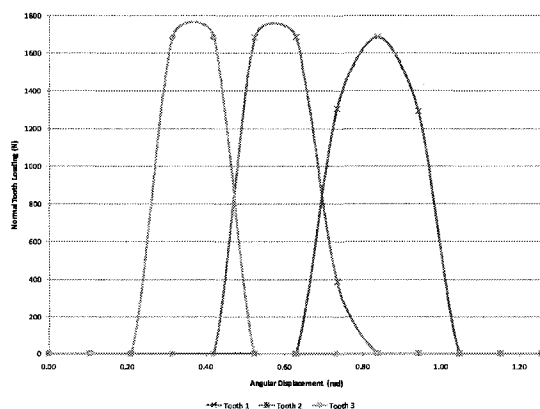


Figure 6.2: CALYX® Loading Pattern

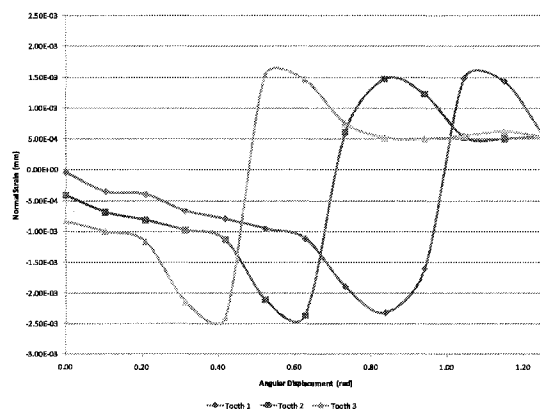


Figure 6.3: CALYX® Strain Pattern

6.1.2 Stiffness Approximation

From these results, the mesh stiffness was approximated. Tooth deformations were calculated using the strain results from Figure 6.3. A nonlinear stiffness profile was calculated using the loading and deformation calculations. Consequently, tooth deflection was the only contribution to the mesh stiffness. Elastohydrodynamic and disk contributions were not considered.

Using these calculations, a spatially-dependent mesh stiffness profile was generated. Time dependence was achieved by incorporating the shaft rotational speed, (Ω). The resulting mesh stiffness profile is presented in Figure 6.4.

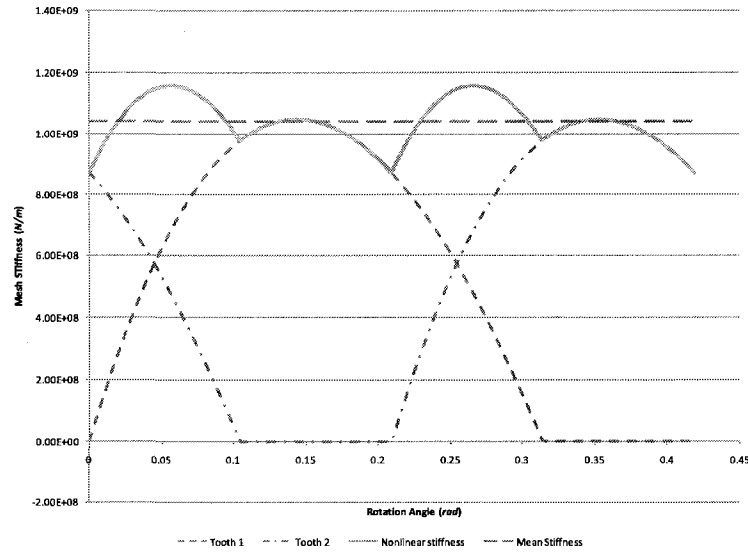


Figure 6.4: Periodic Mesh Stiffness Profile

This profile is very rudimentary. Nevertheless, it provides a periodic function centered about a mean stiffness value.

The next step was to apply a curve-fit to the profile and generate a nonlinear function. The Fourier Series of Equation 6.2 is appropriate.

$$K_g(t) = a_o + \sum_{n=1}^{\infty} a_n \cos(nw\Omega t) + \sum_{n=1}^{\infty} b_n \sin(nw\Omega t) \quad (6.2)$$

Using the MATLAB® curve-fitting tool, the Fourier coefficients were calculated. The results are depicted in Table 6.1. The series was truncated at $n = 3$.

Table 6.1: Nonlinear Mesh Stiffness Parameters

n	a_n	b_n
0	1.040E9	–
1	-4.145E7	6.126E7
2	-7.104E7	9.077E6
3	-5.579E6	2.559E6
w		29.81
(R^2)		0.9597

The term (w) is the scaling “frequency factor” that relates the spatially-dependent profile of Figure 6.4 to a time-dependent one. The value of this scaling factor corresponds to the number of teeth on the gear. The number of gear teeth is 28, very close to the approximate value determined by the curve-fit. Equation 6.3 illustrates this relationship, using the gear-mesh frequency presented in Equation 5.34.

$$nw_i\Omega_i = nN_i\Omega_i = nw_j\Omega_j = nN_j\Omega_j \quad (6.3)$$

These results are significant and reveal the nature of the mesh stiffness. The coefficient, (a_o) , is the average mesh stiffness, (K_{go}) . The frequencies of the Fourier series sine and cosine terms in Equation 6.2 represent the gear-mesh frequency and its harmonics. The periodic mesh stiffness, (ΔK_g) , is the source of the gear-mesh frequencies introduced in the previous chapter.

6.2 Results and Observations

The results of the linear and nonlinear responses due to imbalance for ten rotational speeds are shown in Figures 6.5 through 6.9. Both linear and nonlinear responses were calculated through numerical integration of the equations of motion at each operating speed. The time step, Δt , used for the numerical integration was 1.96E-5 sec. The frequency spectra were determined using the FFT algorithms in MATLAB®. Those results were then normalized to yield dimensionless responses for comparison.

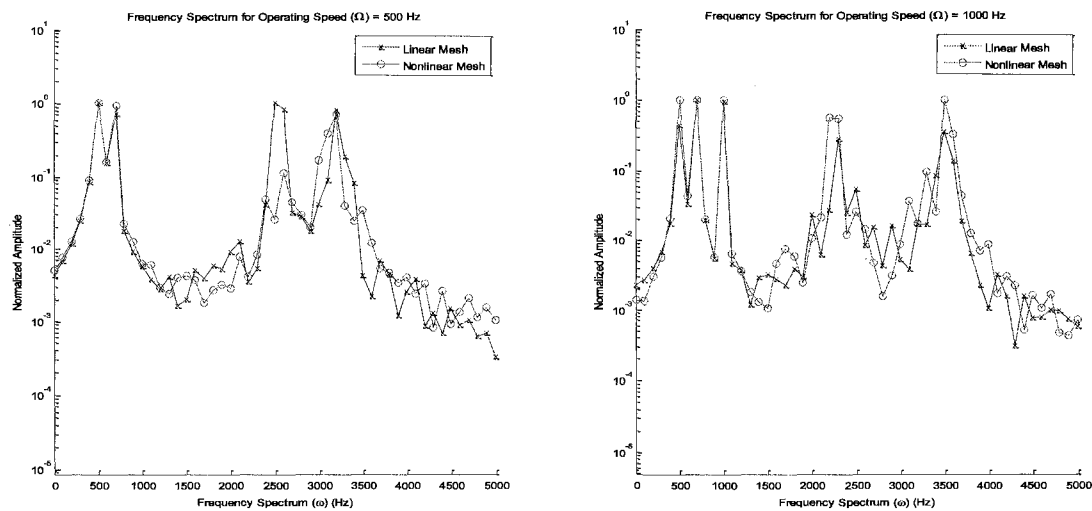


Figure 6.5: Linear versus Nonlinear Mesh Comparisons (500 and 1,000 Hz)

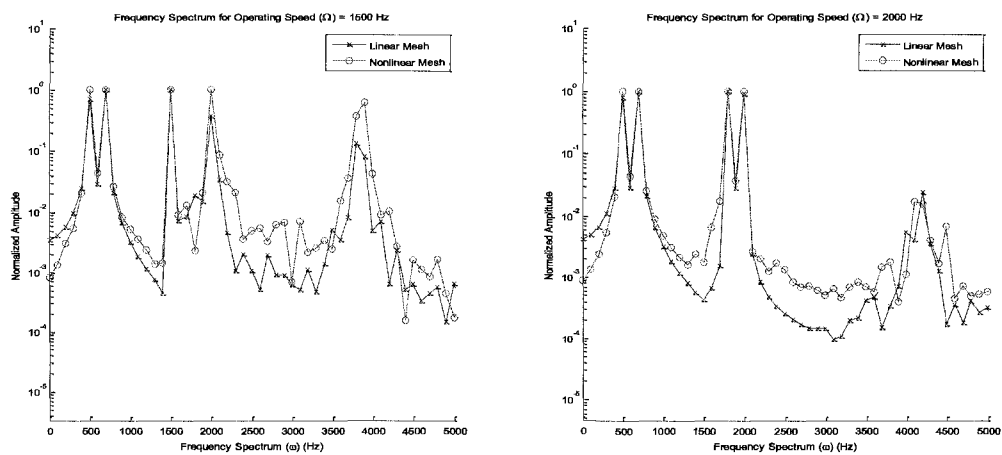


Figure 6.6: Linear versus Nonlinear Mesh Comparisons (1,500 and 2,000 Hz)

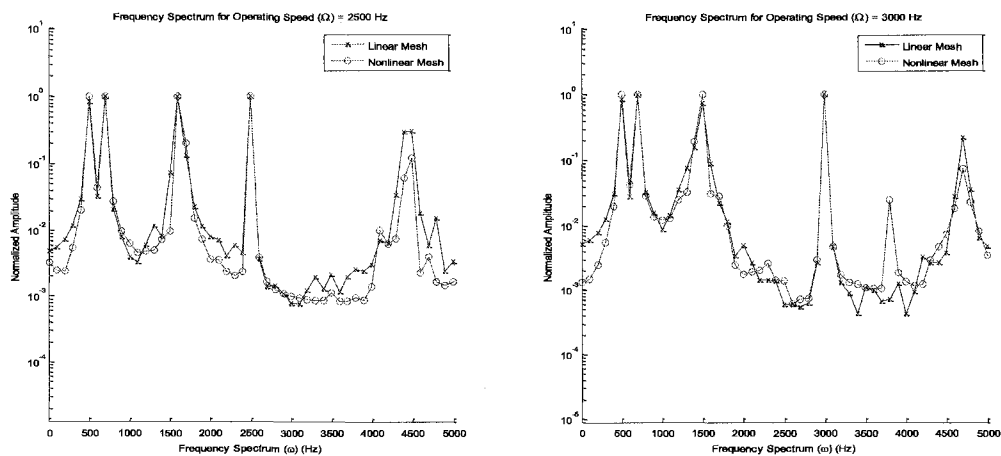


Figure 6.7: Linear versus Nonlinear Mesh Comparisons (2,500 and 3,000 Hz)

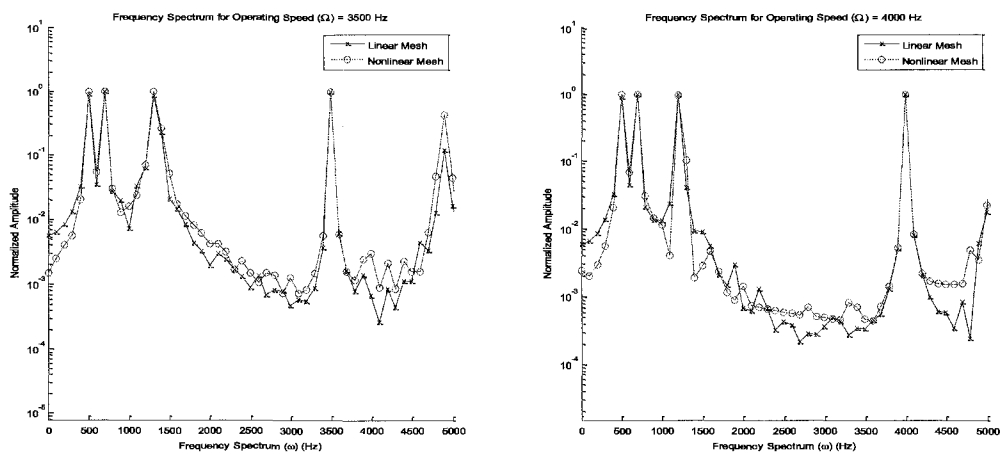


Figure 6.8: Linear versus Nonlinear Mesh Comparisons (3,500 and 4,000 Hz)

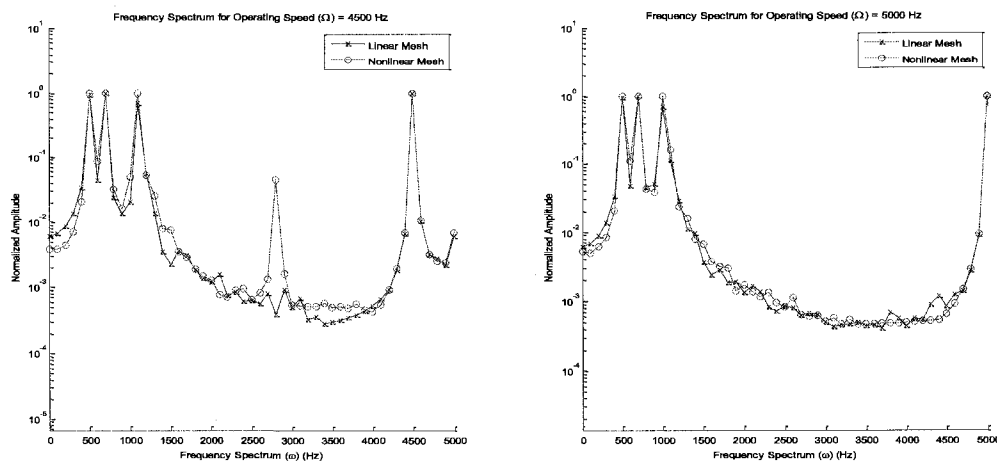


Figure 6.9: Linear versus Nonlinear Mesh Comparisons (4,500 and 5,000 Hz)

The results in Figures 6.5 through 6.9 show that the linear mesh results match closely with the nonlinear results. This is important for two reasons. First, it validates the linear treatment of the geared model for transmission modeling purposes. Second, it provides a very good approximation of the system dynamics at a fraction of the computing time.

Numerical integration of the nonlinear equations of motion required 3–4 hours on a personal computer with standard dual core processor and 2GB of memory, for each of the ten operating conditions. Integration of the linear equations of motion took approximately 4.25 minutes each on the same computer, or 1/48 of the processing time of the nonlinear system. Computing the same responses using traditional methods outlined in Chapter 4 took less than one minute.

In addition to being a more accurate and sophisticated model than the linear analysis, an additional advantage of the nonlinear treatment is the automatic inclusion of the gear-mesh frequency effects in the analysis. As introduced in Chapter 5, the gear-mesh frequency arises from the clatter between gear teeth as multiple teeth come in and

out of contact. Mathematically, this is modeled by the periodic mesh stiffness, ΔK_g . In the linear analysis, this is brought to the right-hand side of the equations of motion as a forcing function, using a load rather than a stiffness. From a linear standpoint, it is not treated as a natural occurrence of the system, but rather an external excitation.

6.3 Summary

The nonlinear methodology for calculating the system dynamics of the gear-mesh was presented. A nonlinear mesh stiffness model was constructed using the results of a spur gear contact analysis. The mesh stiffness was modeled as a periodic stiffness. Numerical integration of the equations of motion yielded time domain results, which were converted into the frequency domain using FFT methods. These results were then compared to the linear approximation.

The comparison showed that the linear treatment is a good approximation of the system dynamics. While the nonlinear results are more accurate, the linear results were obtained in only a fraction of the time required for the nonlinear analysis. However, the nonlinear analysis automatically included the effects of the gear-mesh frequency effects, whereas the linear analysis treated the gear-mesh frequency as an external excitation. Nevertheless, if one of the overarching goals of a comprehensive system model is efficiency, the time savings of the linear system are significant and cannot be dismissed.

Chapter 7

Modal Synthesis Methods via the State-Space

Analysis of a complex mechanical system can incur several hundred, thousand, or even million DOFs. Computing time increases with the number of DOFs in the system, in many cases exponentially. Consequently, computational engineers are always seeking to strike a balance between model accuracy and computational efficiency.

This chapter presents a method for conducting modal synthesis on a geared rotor dynamic system under the influences of non-proportional damping and gyroscopic effects. In addition to model simplification, this technique also exhibits further potential for use in optimization and system identification schemes.

7.1 Background

Hurty (1960) introduced the concept of modal synthesis, also called substructure or component mode synthesis. It has enjoyed popularity over four decades due to its straightforward methodology. The premise of modal synthesis is that the first few modes of vibration of a complex structure can be predicted from the lower modes of vibration of its substructures (Curnier, 1983). An extension of modal analysis, modal synthesis provides for a reduction in the overall number of DOFs. As such, the computational complexity decreases, but the reduced model still maintains the desired numerical precision.

The theory behind modal analysis is fairly simple. A coordinate transformation of the equations of motion yields an equivalent uncoupled system of equations, each of which is linearly independent of the others. The transformation yields the equations in terms of principal coordinates rather than physical coordinates. The physical motion of the system then is simply a linear combination of the motion of the principal coordinates such as in Equation 7.1.

$$\begin{aligned}\{q(z,t)\} &= [\Phi(z)][v(t)] \\ \{q(z,t)\} &= \Phi_1(z)v_1(t) + \Phi_2(z)v_2(t) + \dots + \Phi_N(z)v_N(t)\end{aligned}\tag{7.1}$$

where $v_i(t)$ is the i^{th} principal coordinate response of the diagonalized equations of motion, and $\{\Phi_i\}$ is its corresponding eigenvector.

A benefit of modal analysis is that it shows the contribution of the individual modes of vibration to the motion of the entire structure and thus explicitly captures the dynamics of the system. The lower modes extracted by modal analysis generally provide the most contribution to the motion of the system. Therefore, it provides a basis for DOF reduction and model simplification. Through modal analysis, many applications are available to include structural modification, sensitivity analysis, model reduction, correlation of model and finite element results, response prediction, substructure coupling, and structural damage detection (He, 2001).

One of the first rotor dynamic analyses to use modal synthesis resulted in the computer program Analysis of Rotor Dynamic Systems (ARDS). It could determine the frequency modes within 1-10% error (Nelson, 1982). However, gyroscopic effects were not explicitly defined within the program.

As introduced in Chapter 4, traditional modal analysis techniques have achieved varying degrees of success when a system undergoes damping and gyroscopic effects. Previous attempts have made approximations by different methods (Özgüven, 2002), the most common being the proportional damping assumption, which has not been particularly useful in rotor dynamics. Another approach ignores damping in the system and focuses only on the mass and stiffness contributions, resulting in the familiar eigenvalue problem: $(\mathbf{K} - \lambda^2 \mathbf{M}) = \mathbf{0}$.

A third method has been to transform the damping matrix via the traditional modal matrix and simply ignore the off-diagonal terms. Yang (1993) and Zehn (2003) used iterative procedures in transforming systems consisting of symmetric damping matrices only. However, a modal transformation currently exists using Garvey's technique (2002a, 2002b) to uncouple the equations of motion for a system undergoing general damping that can also include gyroscopic effects. Garvey's transformation provides the theoretical basis for the proposed methodology.

7.2 Coordinate Transformation

7.2.1 Theory

As presented in Chapter 4, the general eigenvalue problem in the state-space formulation becomes

$$-\begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K} \end{bmatrix} \begin{Bmatrix} \dot{\mathbf{q}} \\ \mathbf{q} \end{Bmatrix} + \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{M} & \mathbf{D} \end{bmatrix} \begin{Bmatrix} \ddot{\mathbf{q}} \\ \dot{\mathbf{q}} \end{Bmatrix} = \begin{Bmatrix} \mathbf{0} \\ \mathbf{0} \end{Bmatrix} \quad (7.2)$$

or

$$-[\mathbf{A}_1]\{\mathbf{q}'\} + [\mathbf{A}_2]\{\dot{\mathbf{q}}'\} = \{\mathbf{0}\} \quad (7.3)$$

where

$$[\mathbf{A}_1] = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K} \end{bmatrix}, [\mathbf{A}_2] = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{M} & \mathbf{D} \end{bmatrix}, \{\mathbf{q}'\} = \begin{Bmatrix} \dot{\mathbf{q}} \\ \mathbf{q} \end{Bmatrix}, \{\dot{\mathbf{q}}'\} = \begin{Bmatrix} \ddot{\mathbf{q}} \\ \dot{\mathbf{q}} \end{Bmatrix} \quad (7.4)$$

The generalized eigenvalue problem for the state-space formulation becomes

$$([\mathbf{A}_1] - \lambda[\mathbf{A}_2])\{\boldsymbol{\phi}_r\} = \{\mathbf{0}\} \quad (7.5)$$

where (λ) is the complex eigenvalue, and $\{\boldsymbol{\phi}_r\}$ is the corresponding right eigenvector.

The state-space formulation results in $(2N)$ eigenvalues occurring in complex conjugate pairs, where (N) is the number of DOF of the system. The objective is to determine the equivalent matrices, $[\mathbf{M}_D]$, $[\mathbf{K}_D]$, $[\mathbf{D}_D]$, that are diagonal matrices of the transformed equations of motion. Two modal matrices are also defined, $[\boldsymbol{\Psi}_r]$ and $[\boldsymbol{\Psi}_l]$, whose columns contain the right and left eigenvectors, respectively.

Since the eigenvalues occur in conjugate pairs, the eigenvalues and corresponding eigenvectors can arrange so that the conjugate pairs are separate. From this, two diagonal matrices of dimension $(N \times N)$ can be defined containing the conjugate eigenvalues on the diagonal:

$$\boldsymbol{\Lambda}_1 = \begin{bmatrix} \alpha_1 + j\beta_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \alpha_N + j\beta_N \end{bmatrix} \quad (7.6)$$

$$\mathbf{\Lambda}_2 = \begin{bmatrix} \alpha_1 - j\beta_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \alpha_N - j\beta_N \end{bmatrix} \quad (7.7)$$

If any of the eigenvalues contain purely real roots, the transformation scheme changes somewhat (Garvey, 2002a). For brevity, all roots are assumed to be complex.

To ensure the desired results, the right eigenvector must be scaled or normalized. One appropriate scaling results in

$$\mathbf{\Psi}'_r = [\mathbf{\Psi}_r] \left[[\mathbf{\Psi}_l]^T [\mathbf{A}_2] [\mathbf{\Psi}_r] \right]^{-1} \quad (7.8)$$

such that

$$\begin{aligned} [\mathbf{\Psi}_l]^T [\mathbf{A}_1] [\mathbf{\Psi}'_r] &= \begin{bmatrix} \mathbf{\Lambda}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{\Lambda}_2 \end{bmatrix} = [\mathbf{\Lambda}] \\ [\mathbf{\Psi}_l]^T [\mathbf{A}_2] [\mathbf{\Psi}'_r] &= \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} = [\mathbf{I}] \end{aligned} \quad (7.9)$$

Both the left and right eigenvectors are complex, and any further transformations will continue to produce complex numbers. The elimination matrix, $[\mathbf{\Xi}]$, is therefore defined to eliminate all complex notation from the transformation.

$$\mathbf{\Xi} = \begin{bmatrix} \frac{1}{\sqrt{2}} \mathbf{I} & \frac{-j}{\sqrt{2}} \mathbf{I} \\ \frac{1}{\sqrt{2}} \mathbf{I} & \frac{j}{\sqrt{2}} \mathbf{I} \end{bmatrix} \quad (7.10)$$

Using Equation 7.10,

$$[\Xi]^T [\Psi_l]^T [A_1] [\Psi_r] [\Xi] = \begin{bmatrix} L_x & L_y \\ L_y & L_z \end{bmatrix} \quad (7.11)$$

$$[\Xi]^T [\Psi_l]^T [A_2] [\Psi_r] [\Xi] = \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} \quad (7.12)$$

The matrices, $[L_x]$, $[L_y]$, and $[L_z]$, are real and diagonal. They are equal to the following:

$$\begin{aligned} L_x &= \text{real}(\Lambda_1) \\ L_y &= \text{imag}(\Lambda_1) \\ L_z &= -\text{real}(\Lambda_1) = -L_x \end{aligned} \quad (7.13)$$

A diagonal matrix, $[\Gamma]$, is defined.

$$\Gamma(k, k) = \frac{1}{2} \sinh^{-1} \left(\frac{L_z(k, k) - L_x(k, k)}{2L_y(k, k)} \right) \quad (7.14)$$

Then the following relationship holds.

$$\begin{bmatrix} \cosh \Gamma & \sinh \Gamma \\ \sinh \Gamma & \cosh \Gamma \end{bmatrix} \begin{bmatrix} L_x & L_y \\ L_y & L_z \end{bmatrix} \begin{bmatrix} \cosh \Gamma & \sinh \Gamma \\ \sinh \Gamma & \cosh \Gamma \end{bmatrix} = \begin{bmatrix} 0 & [\omega_n] \\ [\omega_n] & [2\zeta\omega_n] \end{bmatrix} \quad (7.15)$$

where the matrices, $[\omega_n]$ and $[2\zeta\omega_n]$, are diagonal, containing the natural and damped natural frequencies of the system, respectively.

The left and right transformation matrices can now be determined. These matrices will diagonalize the state-space matrices, $[A_1]$ and $[A_2]$.

$$\mathbf{T}_L = [\Psi_l][\Xi] \begin{bmatrix} \cosh \Gamma & \sinh \Gamma \\ \sinh \Gamma & \cosh \Gamma \end{bmatrix} \begin{bmatrix} [\omega_n] & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \quad (7.16)$$

$$\mathbf{T}_R = [\Psi_r][\Xi] \begin{bmatrix} \cosh \Gamma & \sinh \Gamma \\ \sinh \Gamma & \cosh \Gamma \end{bmatrix} \begin{bmatrix} [\omega_n] & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \quad (7.17)$$

Transforming, the diagonalized results become

$$[\mathbf{T}_L]^T [\mathbf{A}_1] [\mathbf{T}_R] = [\mathbf{B}_1] = \begin{bmatrix} \mathbf{0} & \mathbf{K}_D \\ \mathbf{K}_D & \mathbf{D}_D \end{bmatrix} \quad (7.18)$$

$$[\mathbf{T}_L]^T [\mathbf{A}_2] [\mathbf{T}_R] = [\mathbf{B}_2] = \begin{bmatrix} \mathbf{K}_D & \mathbf{0} \\ \mathbf{0} & -\mathbf{M}_D \end{bmatrix} \quad (7.19)$$

This transformation methodology results in $(2N)$ de-coupled linearly independent equations. The usual relationships achieved through modal analysis apply here as well, namely

$$\begin{aligned} \mathbf{M}_D &= \mathbf{I} \\ \mathbf{D}_D &= [2\zeta\omega_n] \\ \mathbf{K}_D &= [\omega_n^2] \end{aligned} \quad (7.20)$$

If carried through from start to finish, the process produces results consistent with the following simplification (Houlston, 2006).

$$\begin{aligned}
\lambda_{i1} &= \alpha + j\beta \\
\lambda_{i2} &= \alpha - j\beta \\
d_i &= -(\lambda_{i1} + \lambda_{i2}) = -2\alpha_i \\
k_i &= \frac{(\lambda_{i2} + \lambda_{i1})^2 - (\lambda_{i2} - \lambda_{i1})^2}{4} = \alpha_i^2 + \beta_i^2 \\
m_i &= 1
\end{aligned} \tag{7.21}$$

where $i = 1, 2, \dots, N$

From modal analysis, the transformed coordinates, $\{\mathbf{v}\}$, are principal coordinates.

The transformation to the physical coordinate system is

$$\begin{Bmatrix} \dot{\mathbf{q}} \\ \mathbf{q} \end{Bmatrix} = [\mathbf{T}_R] \begin{Bmatrix} \mathbf{v} \\ \mathbf{u} \end{Bmatrix} \tag{7.22}$$

In Equation 7.22, the term, $\{\mathbf{u}\}$, is not necessarily the time derivative of the general coordinate, $\{\mathbf{v}\}$, except when the system is in free vibration (Garvey, 2002a). Also, the vector order of the principal coordinate and its “derivative” are reversed compared to the physical coordinates.

7.2.2 Transformation Simplification

The full procedure outlined in the previous section is rigorous and lengthy. However, a simplified transformation is available using Equation 7.21 and similarity transformation relationships. From the definition of a similarity transformation,

$$[\Psi_l]^T [\mathbf{A}_1] [\Psi_r'] = [\Lambda] = [\Phi_l]^T [\mathbf{B}_1] [\Phi_r'] \tag{7.23}$$

$$[\Psi_l]^T [\mathbf{A}_2] [\Psi_r'] = [\mathbf{I}] = [\Phi_l]^T [\mathbf{B}_2] [\Phi_r'] \tag{7.24}$$

where $[\Phi_r]$ and $[\Phi_l]$ are the modal matrices of the reduced system. Using linear algebra, the transformation matrices, $[\mathbf{T}_L]$ and $[\mathbf{T}_R]$, can be written as (Houlston, 2006)

$$[\mathbf{T}_L] = [\Psi_l][\Phi_l]^{-1} \quad (7.25)$$

$$[\mathbf{T}_R] = [\Psi_r][\Phi_r]^{-1} \quad (7.26)$$

Using these relationships, the simplified transformation procedure is as follows:

1. Given matrices $[\mathbf{A}_1]$ and $[\mathbf{A}_2]$, solve the eigenvalue problem. Arrange the eigenvalues in matrices $[\Lambda_1]$ and $[\Lambda_2]$. Assemble the left, $[\Psi_l]$, and right normalized, $[\Psi_r]$, eigenvector matrices, corresponding to $[\Lambda_1]$ and $[\Lambda_2]$.

2. Using Equation 7.21, calculate the diagonal values of $[\mathbf{M}_D]$, $[\mathbf{K}_D]$, $[\mathbf{D}_D]$. Substitute into equations 7.18 and 7.19 for $[\mathbf{B}_1]$ and $[\mathbf{B}_2]$, respectively.

3. Determine the left, $[\Phi_l]$, and right, $[\Phi_r]$, normalized eigenvector matrices of the modal system, $[\mathbf{B}_1]$ and $[\mathbf{B}_2]$.

4. Substitute into equations 7.25 and 7.26 to solve for the left and right transformation matrices, $[\mathbf{T}_L]$ and $[\mathbf{T}_R]$, respectively.

To ensure more precise results, it is possible to further refine the transformation by calculating the left eigenvector matrix, $[\Psi_l]$, using either Equation 7.18 or 7.19, rather than using the original left eigenvector matrix. The results are then substituted into equations 7.25 and 7.26.

7.3 Modal Synthesis

7.3.1 Theory

In modal synthesis, analysis begins with the substructures. The equations of motion for the i^{th} substructure are

$$[\mathbf{M}_i]\{\ddot{\mathbf{q}}_i\} + [\mathbf{D}_i]\{\dot{\mathbf{q}}_i\} + [\mathbf{K}_i]\{\mathbf{q}_i\} = \{\mathbf{F}(t)_i\} \quad (7.27)$$

The substructure eigenvalue problem solution follows the state-space methodology introduced in Equation 7.2.

$$-\begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K} \end{bmatrix}_i \begin{Bmatrix} \dot{\mathbf{q}} \\ \mathbf{q} \end{Bmatrix} + \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{M} & \mathbf{D} \end{bmatrix}_i \begin{Bmatrix} \ddot{\mathbf{q}} \\ \dot{\mathbf{q}} \end{Bmatrix} = \begin{Bmatrix} \mathbf{0} \\ \mathbf{F}(t) \end{Bmatrix}_i \quad (7.28)$$

Once the substructure analysis is complete, the global, uncoupled system is constructed. However, unlike some other modal synthesis works (Luo, 1996), the entire state-space formulation of Equation 7.28 becomes part of the global model. The dimensions of the global state-space formulation including all (m) substructures are $[(2N_1 + \dots + 2N_m) \times (2N_1 + \dots + 2N_m)]$.

$$-\begin{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K} \end{bmatrix}_1 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K} \end{bmatrix}_m \end{bmatrix} \begin{Bmatrix} \begin{Bmatrix} \dot{\mathbf{q}} \\ \mathbf{q} \end{Bmatrix}_1 \\ \vdots \\ \begin{Bmatrix} \dot{\mathbf{q}} \\ \mathbf{q} \end{Bmatrix}_m \end{Bmatrix} + \begin{bmatrix} \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{M} & \mathbf{D} \end{bmatrix}_1 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{M} & \mathbf{D} \end{bmatrix}_m \end{bmatrix} \begin{Bmatrix} \begin{Bmatrix} \ddot{\mathbf{q}} \\ \dot{\mathbf{q}} \end{Bmatrix}_1 \\ \vdots \\ \begin{Bmatrix} \ddot{\mathbf{q}} \\ \dot{\mathbf{q}} \end{Bmatrix}_m \end{Bmatrix} = \begin{Bmatrix} \begin{Bmatrix} \mathbf{0} \\ \mathbf{F}(t) \end{Bmatrix}_1 \\ \vdots \\ \begin{Bmatrix} \mathbf{0} \\ \mathbf{F}(t) \end{Bmatrix}_m \end{Bmatrix} \quad (7.29)$$

or in simpler notation

$$\begin{bmatrix} [\mathbf{A}_1]_1 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & [\mathbf{A}_1]_m \end{bmatrix} \begin{Bmatrix} \{\dot{\mathbf{q}}\}_1 \\ \vdots \\ \{\dot{\mathbf{q}}\}_m \end{Bmatrix} + \begin{bmatrix} [\mathbf{A}_2]_1 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & [\mathbf{A}_2]_m \end{bmatrix} \begin{Bmatrix} \{\dot{\mathbf{q}}\}_1 \\ \vdots \\ \{\dot{\mathbf{q}}\}_m \end{Bmatrix} = \begin{Bmatrix} \{\mathbf{F}(\mathbf{t})\}_1 \\ \vdots \\ \{\mathbf{F}(\mathbf{t})\}_m \end{Bmatrix} \quad (7.30)$$

For simplicity, Equation 7.30 can be rewritten as

$$[\mathbf{A}_1^*] \{\dot{\mathbf{q}}^*\} + [\mathbf{A}_2^*] \{\dot{\mathbf{q}}^*\} = \{\mathbf{F}(\mathbf{t})^*\} \quad (7.31)$$

where the asterisk symbolizes the global system of individual substructures.

Substructure coupling has not been introduced yet.

Returning to the fundamental concept of modal analysis, the system equations of motion are transformed into principal coordinates. The global transformation matrices are composed of the substructure transformation matrices of Equations 7.25 and 7.26, respectively.

$$[\mathbf{T}_L^*]^T [\mathbf{A}_1^*] [\mathbf{T}_R^*] \{\dot{\mathbf{v}}^*\} + [\mathbf{T}_L^*]^T [\mathbf{A}_2^*] [\mathbf{T}_R^*] \{\dot{\mathbf{v}}^*\} = [\mathbf{T}_L^*]^T \{\mathbf{F}(\mathbf{t})^*\} \quad (7.32)$$

where

$$[\mathbf{T}_L^*] = \begin{bmatrix} [\mathbf{T}_L]_1 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & [\mathbf{T}_L]_m \end{bmatrix}, [\mathbf{T}_R^*] = \begin{bmatrix} [\mathbf{T}_R]_1 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & [\mathbf{T}_R]_m \end{bmatrix} \quad (7.33)$$

Equation 7.32 simplifies to

$$[\mathbf{B}_1^*] \{\dot{\mathbf{v}}^*\} - [\mathbf{B}_2^*] \{\dot{\mathbf{v}}^*\} = [\mathbf{T}_L^*]^T \{\mathbf{F}(\mathbf{t})^*\} \quad (7.34)$$

or in long form,

$$\begin{bmatrix} \begin{bmatrix} \mathbf{0} & \mathbf{K}_D \\ \mathbf{K}_D & \mathbf{D}_D \end{bmatrix}_1 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \begin{bmatrix} \mathbf{0} & \mathbf{K}_D \\ \mathbf{K}_D & \mathbf{D}_D \end{bmatrix}_m \end{bmatrix} \begin{Bmatrix} \begin{Bmatrix} \mathbf{u} \\ \mathbf{v} \end{Bmatrix}_1 \\ \vdots \\ \begin{Bmatrix} \mathbf{u} \\ \mathbf{v} \end{Bmatrix}_m \end{Bmatrix} - \begin{bmatrix} \begin{bmatrix} \mathbf{K}_D & \mathbf{0} \\ \mathbf{0} & -\mathbf{M} \end{bmatrix}_1 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \begin{bmatrix} \mathbf{K}_D & \mathbf{0} \\ \mathbf{0} & -\mathbf{M} \end{bmatrix}_m \end{bmatrix} \begin{Bmatrix} \begin{Bmatrix} \dot{\mathbf{u}} \\ \dot{\mathbf{v}} \end{Bmatrix}_1 \\ \vdots \\ \begin{Bmatrix} \dot{\mathbf{u}} \\ \dot{\mathbf{v}} \end{Bmatrix}_m \end{Bmatrix} = [\mathbf{T}_L^*]^T \begin{Bmatrix} \begin{Bmatrix} \mathbf{0} \\ \mathbf{F(t)} \end{Bmatrix}_1 \\ \vdots \\ \begin{Bmatrix} \mathbf{0} \\ \mathbf{F(t)} \end{Bmatrix}_m \end{Bmatrix} \quad (7.35)$$

Coupling occurs as a result of the gear-mesh. The mesh stiffness is the coupling agent. However, this treatment is general enough to apply to other types of couplings with only minor changes to the methodology. The mesh matrix, $[\mathbf{K}]_{mesh}$, breaks into four quadrants of dimension, (6×6) . For a two-shaft system, the coupling becomes

$$[\mathbf{A}_1^*]_{sys} = \begin{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K} \end{bmatrix}_1 & \mathbf{0} \\ \mathbf{0} & \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K} \end{bmatrix}_2 \end{bmatrix} + \begin{bmatrix} \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K}_{mesh(i,i)} \end{bmatrix} & \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K}_{mesh(i,j)} \end{bmatrix} \\ \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K}_{mesh(j,i)} \end{bmatrix} & \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K}_{mesh(j,j)} \end{bmatrix} \end{bmatrix} \quad (7.36)$$

or

$$[\mathbf{A}_1^*]_{sys} = [\mathbf{A}_1^*] + [\mathbf{A}_1^*]_{mesh} \quad (7.37)$$

Using the same transformation on the second term in Equation 7.36 results in the transformed coupling matrix, $[\mathbf{K}_{Dm}]$.

$$[\mathbf{B}_1^*]_{mesh} = [\mathbf{T}_L^*]^T \begin{bmatrix} \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K}_{mesh(i,i)} \end{bmatrix} & \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K}_{mesh(i,j)} \end{bmatrix} \\ \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K}_{mesh(j,i)} \end{bmatrix} & \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K}_{mesh(j,j)} \end{bmatrix} \end{bmatrix} [\mathbf{T}_R^*] = \begin{bmatrix} \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K}_{Dm(i,i)} \end{bmatrix} & \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K}_{Dm(i,j)} \end{bmatrix} \\ \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K}_{Dm(j,i)} \end{bmatrix} & \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K}_{Dm(j,j)} \end{bmatrix} \end{bmatrix} \quad (7.38)$$

Since this is a coupling matrix, it is not necessarily diagonal. The coupled system modal matrix, $[\mathbf{B}_1^*]_{sys}$, is

$$[\mathbf{B}_1^*]_{sys} = [\mathbf{B}_1^*] + [\mathbf{B}_1^*]_{mesh} \quad (7.39)$$

In this formulation, it is assumed that no damping occurs in the gear-mesh. Consequently, the gear-mesh does not affect the modal system matrix, $[\mathbf{B}_2^*]$ from Equation 7.34. It requires no modification:

$$[\mathbf{B}_2^*]_{sys} = [\mathbf{B}_2^*] \quad (7.40)$$

Equation 7.34 then becomes

$$[\mathbf{B}_1^*]_{sys} \{\dot{\mathbf{v}}^*\}_{sys} - [\mathbf{B}_2^*]_{sys} \{\dot{\mathbf{v}}^*\}_{sys} = [\mathbf{T}_L^*]^T \{\mathbf{F}(\mathbf{t})^*\}_{sys} \quad (7.41)$$

The general eigenvalue problem of Equation 7.41 yields the same results as the original problem of Equation 7.2. As with the transformation sequence, the steps for reduction are as follows:

1. Define a variable, (P) , that represents the number of DOFs of the reduced system model. The only caveats are that the selection of (P) must be divisible by 2 and less than or equal to the number of DOFs of the smallest substructure. The resulting state-space matrices, $[\mathbf{B}_1^*]_{sys}$ and $[\mathbf{B}_2^*]_{sys}$, will have dimensions $(2P \times 2P)$. It is possible to select individual reductions for each substructure, (P_i) , requiring only slight

modification to this procedure. However, the presentation here assumes uniform reduction across all substructures, (P) .

2. Reduce the size of modal matrices, $[\mathbf{T}_L^*]$ and $[\mathbf{T}_R^*]$, to dimensions $[(2N_1 + \dots + 2N_m) \times (2P)]$ by discarding the unnecessary columns of the modal matrices. Being a state-space formulation, the columns associated with each of the $(2m)$ vector components (i.e., $\{\dot{\mathbf{q}}_1\}_{N_1 \times 1}, \{\mathbf{q}_1\}_{N_1 \times 1}, \{\dot{\mathbf{q}}_2\}_{N_2 \times 1}, \{\mathbf{q}_2\}_{N_2 \times 1} \dots \{\dot{\mathbf{q}}_m\}_{N_m \times 1}, \{\mathbf{q}_m\}_{N_m \times 1}$) must undergo an equal reduction. The size of the reduced system is $(2P \times 2P)$. The selection matrices from $[\mathbf{T}_L^*]$ and $[\mathbf{T}_R^*]$ will result in an equal number of modes selected from the original eigen matrices associated with each vector component. The reduced transformation matrices, $[\mathbf{T}_L^*]$ and $[\mathbf{T}_R^*]$, will thus have dimensions $[(2N_1 + \dots + 2N_m) \times (2P)]$.

3. Transform the system matrices, $[\mathbf{A}_1^*]_{sys}$ and $[\mathbf{A}_2^*]_{sys}$, using the reduced transformation matrices, $[\mathbf{T}_L^*]$ and $[\mathbf{T}_R^*]$. This results in the reduced diagonal matrices of Equation 7.42.

$$\begin{aligned} [\mathbf{T}_L^*]_{(2P \times 2N_1 + \dots + 2N_m)}^T [\mathbf{A}_1^*]_{sys(2N_1 + \dots + 2N_m \times 2N_1 + \dots + 2N_m)} [\mathbf{T}_R^*]_{(2N_1 + \dots + 2N_m \times 2P)} &= [\mathbf{B}_1^*]_{sys(2P \times 2P)} \\ [\mathbf{T}_L^*]_{(2P \times 2N_1 + \dots + 2N_m)}^T [\mathbf{A}_2^*]_{sys(2N_1 + \dots + 2N_m \times 2N_1 + \dots + 2N_m)} [\mathbf{T}_R^*]_{(2N_1 + \dots + 2N_m \times 2P)} &= [\mathbf{B}_2^*]_{sys(2P \times 2P)} \end{aligned} \quad (7.42)$$

4. Solve the eigenvalue problem of the reduced state-space matrices, $[\mathbf{B}_1^*]_{sys(2P \times 2P)}$ and $[\mathbf{B}_2^*]_{sys(2P \times 2P)}$. The eigenvalues will result in $(2P)$ complex conjugate pairs, which

represent the first (P) frequencies of the system. They will have the same or very near the same value as the corresponding frequencies of the full system model.

7.3.2 Vibration Response

The determination of the vibration response is possible using the principal coordinates. Returning to Equation 7.41 and assuming a harmonic response to the forcing function, $\{\mathbf{F}(\mathbf{t})^*\}_{sys}$:

$$\{\mathbf{V}^*\}_{sys} = \{\mathbf{V}^*\} e^{j\omega t} \quad (7.43)$$

The equations of motion become

$$[\mathbf{B}_1^*]_{sys} \{\mathbf{V}^*\}_{sys} - j\omega [\mathbf{B}_2^*]_{sys} \{\mathbf{V}^*\}_{sys} = [\mathbf{T}_L^*]^T \{\mathbf{F}_0^*\}_{sys} \quad (7.44)$$

The displacements in modal coordinates are

$$\{\mathbf{V}^*\}_{sys} = \left[[\mathbf{B}_1^*]_{sys} - j\omega [\mathbf{B}_2^*]_{sys} \right]^{-1} [\mathbf{T}_L^*]^T \{\mathbf{F}_0^*\}_{sys} \quad (7.45)$$

Converting back to physical coordinates yields the following expression:

$$\{\mathbf{Q}^*\}_{sys} = [\mathbf{T}_R^*] \left[[\mathbf{B}_1^*]_{sys} - j\omega [\mathbf{B}_2^*]_{sys} \right]^{-1} [\mathbf{T}_L^*]^T \{\mathbf{F}_0^*\}_{sys} \quad (7.46)$$

where $\{\mathbf{Q}^*\}_{sys}$ is the vector containing the amplitudes of the response. Substituting the rotational speed, (Ω) , into Equation 7.46 will yield speed-sweep vibration responses, as shown later in Figures 7.1 and 7.2.

Gyroscopic effects play a unique role in a speed-sweep response. The modal transformation matrices are determined by calculating the eigenvectors of the system. The matrix, $[\mathbf{D}]$, contains the system gyroscopics, which is a function of rotational speed. Hence, the eigenvectors themselves are a function of rotational speed, and the transformation matrices themselves change with rotational speed. For a full, modal speed-sweep analysis, the transformation matrices must be calculated at each speed interval within the desired range. The traditional frequency response can be calculated by fixing the operational speed in the damping/gyroscopic matrix, $[\mathbf{D}]$, and varying the frequency, (ω) , in Equation 7.46.

7.3.3 Time Domain Response

If desired, it is also possible to determine the time domain forced response using Equation 7.46. The time domain response acts at a single frequency. The shaft operating speed is substituted into Equation 7.46 to solve for the amplitude vector, $\{\mathbf{Q}^*\}_{sys}$. The time domain response is the periodic function:

$$\{\mathbf{q}^*\}_{sys} = \{\mathbf{Q}^*\}_{sys} e^{j\omega t} \quad (7.47)$$

7.4 Application and Validation

The application for testing this methodology is the spur gear system of Chapter 5. Since the two substructures are the same, reduction in the complexity of Equation 7.28 is significant. All substructure matrix components are identical for each substructure.

The full finite element system model calculates the eigen solution of the system and the frequency response to a harmonic forcing function. Each shaft consists of 126 DOFs,

or 252 DOFs for the entire system. The state-space formulation doubles the size of the model to 504 DOFs.

7.4.1 Modal Reduction Comparisons

Since modal analysis is an approximation, it requires further validation through comparison with the full system model results. The first validation compares the eigenvalues of the system model with those of the reduced model. Tables 7.1 and 7.2 provide the results of reductions to 30 and 20 DOFs, respectively. The rotational speed for this analysis is 1,000 *rpm* (16.67 *Hz*). For brevity, only the first 10 frequencies are provided.

Table 7.1: Modal Analysis Eigenvalue Comparison (P=30).

	Modal Synthesis (Hz)	Full FEA (Hz)	Difference (%)
1	504.647	504.635	0.00%
2	684.261	684.260	0.00%
3	684.279	684.279	0.00%
4	684.296	684.297	0.00%
5	2091.562	2090.630	0.04%
6	2837.572	2837.560	0.00%
7	2837.572	2837.560	0.00%
8	2858.979	2858.969	0.00%
9	2858.979	2858.969	0.00%
10	6677.140	6676.739	0.01%

Table 7.2: Modal Synthesis Eigenvalue Comparison (P=20).

	Modal Synthesis (Hz)	Full FEA (Hz)	Difference (%)
1	504.652	504.635	0.00%
2	684.261	684.260	0.00%
3	684.279	684.279	0.00%
4	684.296	684.297	0.00%
5	2091.950	2090.630	0.06%
6	2837.572	2837.560	0.00%
7	2837.572	2837.560	0.00%
8	2858.979	2858.969	0.00%
9	2858.979	2858.969	0.00%
10	6677.140	6676.739	0.01%

The results clearly show the accuracy attainable by this methodology. Furthermore, the modal simplifications also produce substantial reductions in the amount of computing time required compared to the full system model: 62% and 63%, respectively.

The second validation method compares the reduced system's vibration responses to the unreduced model. The responses of two harmonic forcing functions are presented. The first is the gear-mesh frequency. The second models a mass imbalance in one of the gears. A finite imbalance exists in all rotating systems and is a function of the rotational speed squared, $(\Omega)^2$. Figures 7.1 and 7.2 provide the speed-sweep responses for the gear-mesh frequency and mass imbalance excitations, respectively, using a speed step, $(\Delta\Omega)$, of 50 Hz.

The response curves represent the lateral displacement of each shaft at its midpoint. Three curves are plotted for each shaft, providing the results of (1) the full finite element model – "Full FEA," (2) the full modal reduction – "Modal Synthesis," and (3) the modal reduction using a fixed operating speed – "Modal Fixed Gyro." The Modal Synthesis scenario recalculates the transformation matrices for each rotational speed interval. The

Modal Fixed Gyro scenario calculates the matrices only once for a single value of (Ω) within the desired speed range in order to reduce the number of burdensome eigenvector calculations. It then uses these matrices for all responses within the operating range, 0-5,000 Hz. The Modal Fixed Gyro curve is effectively the frequency response at a rotational speed of 5,000 Hz.

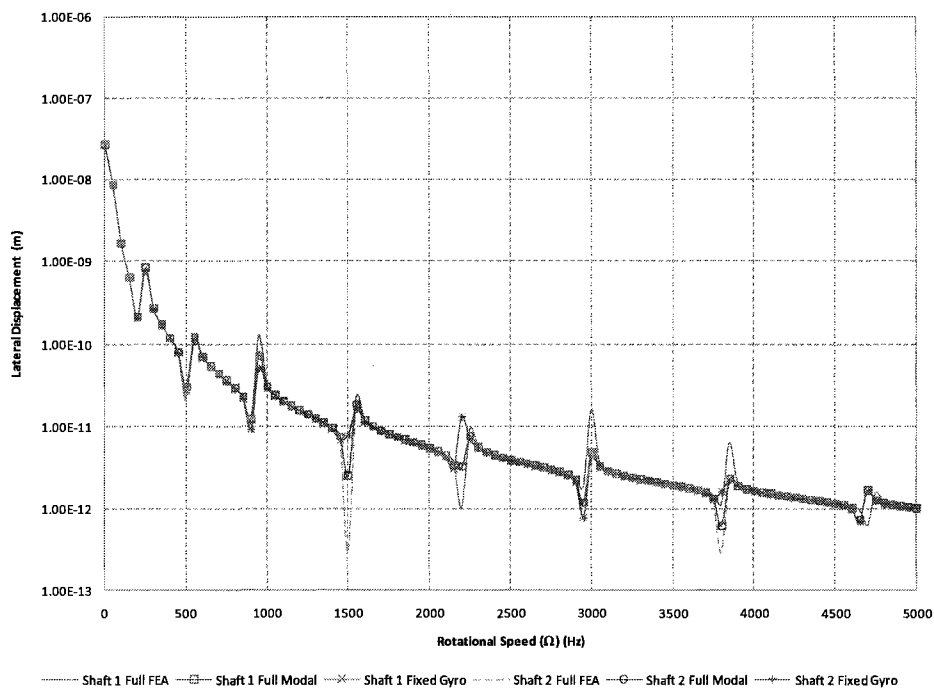


Figure 7.1: Gear-Mesh Response Comparison (0-5,000 Hz).

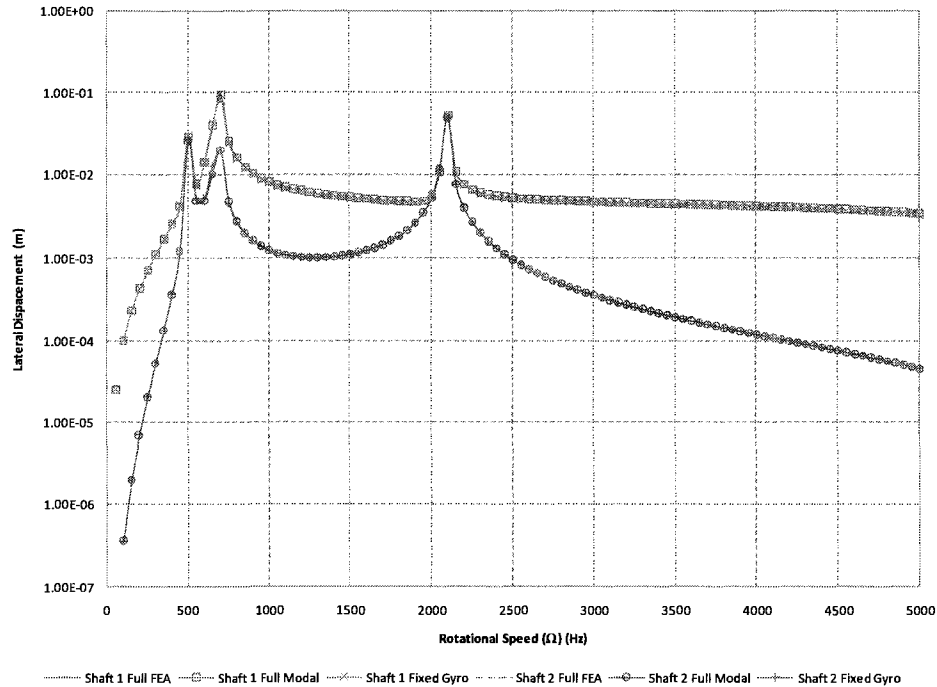


Figure 7.2: Imbalance Response Comparison (0-5,000 Hz).

7.5 Practical Limitations

For all its accuracy, this experiment does raise some questions regarding the utility of such procedure in terms of practicality and efficiency. These questions fall into two categories: limitations on modal reduction and the impact of the gyroscopic effect.

Once again, the modal synthesis procedure can produce a significant modal reduction without loss of accuracy. However, in the vibration responses, the gear-mesh frequency will limit the amount of reduction available. In spur and helical gears, the gear-mesh frequency will excite higher modes of vibration at lower speeds. Consequently, for the modal synthesis transformation to capture these higher modes, they must be included in the modal reduction. An example is presented in Figure 7.3.

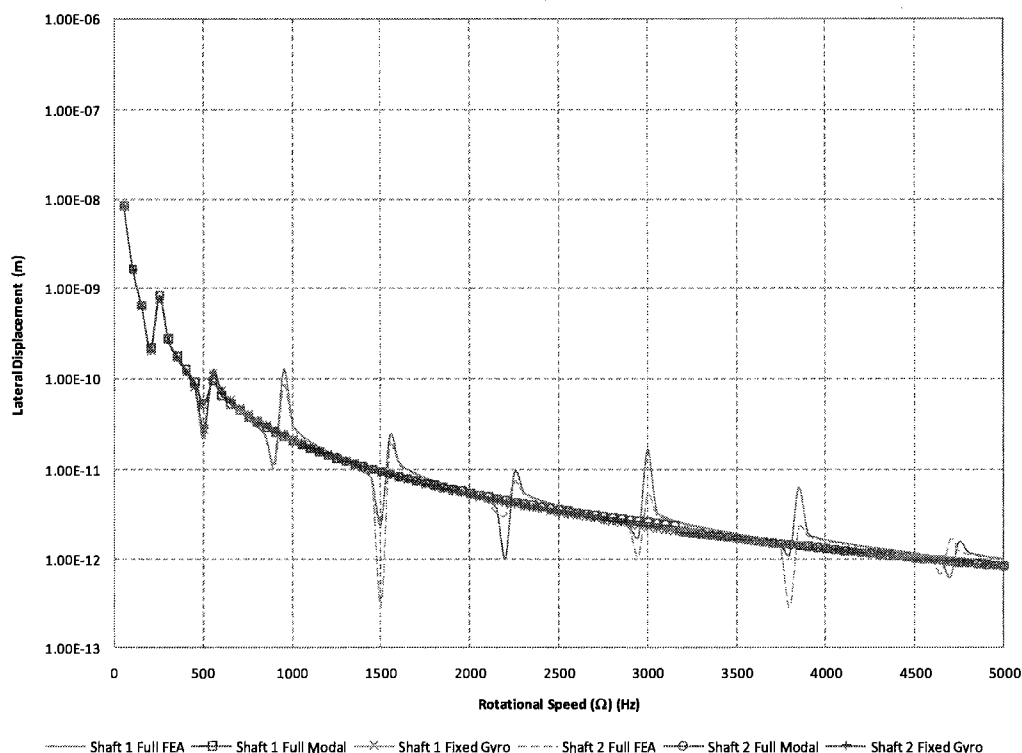


Figure 7.3. Gear-mesh frequency limitations on modal reduction.

The total number of DOF in the geared, two-shaft model is 252. This increases to 504 DOFs in the state-space. Figure 6 illustrates a modal reduction to 30 DOFs (60 DOFs in the state-space) in terms of the gear-mesh forcing function response. The reduced model has failed to accurately portray all resonance peaks within the operating range, 0-5,000 Hz. To capture the remaining peaks, the modal DOFs must increase to account for excitation of the higher modes. For the application system, the required number of DOFs to produce an accurate response is 168, or in the state-space, 336. Thus, an intended 88% DOF reduction will result in a reduction of only 33% if the gear-mesh frequency excitation is important to the analysis. The implication for geared system analysis is that there must be some *a priori* knowledge as to the number of higher modes excited within the desired operating speed range.

The presence of gyroscopics complicates the analysis of any system, and modal synthesis is no different. As previously discussed, a full modal speed-sweep response analysis requires the transformation matrices to be calculated at each speed interval, which greatly adds to computer processing time.

These two factors may adversely affect the utility of the methodology by increasing the computer processing time of a procedure originally designed to reduce it. A comparison of computing time is presented in table 4, illustrating the gyroscopic effects on processing time. The full 252-DOF system model is the comparison standard with a computer processing ratio of one. In the Full Modal Synthesis (speed-sweep response) models, where the transformation matrices are recalculated for each operating speed within the spectrum, the 168 and 30-DOF models take 8.7 and 5.9 times longer to process, respectively, than the full finite element model. The Modal Fixed Gyro (frequency response) scenarios reduce those times substantially. In the case of the 30-DOF model, the processing time is 20% of the full system model. However, as pointed out, these two reductions do not produce an accurate gear-mesh frequency response.

Table 7.3: Response Processing Time Ratio Comparison

DOF	252	168	30
Full Modal Synthesis Scenario	1	8.7	5.9
Modal Fixed Gyro Scenario	1	1.3	0.2

Consequently, while this experiment validates the modal synthesis procedure and results in computer processing time savings during eigen analysis, the cutoff frequency or limit of modal reduction is determined by the gear-mesh frequency. Therefore, it may or may not be particularly practical for use in vibration response calculations.

7.6 Summary

This chapter has presented a method for performing modal synthesis on a geared rotor dynamic system undergoing general damping and gyroscopic effects. The modal analysis produces the eigen solutions, frequency response, and time domain response of the system. Validation of the reduced model occurred by comparing the eigen-solutions and frequency responses of the reduced system with the original model. The results, in all instances, matched exactly.

In the speed-sweep response, the modal synthesis procedure resulted in limitations. The gear-mesh frequency excited higher modes of vibration, which required a higher DOF model to maintain accuracy. The introduction of gyroscopic effects increased processing time, substantially in some cases. These two limitations questioned the applicability and practicality of using the modal synthesis method for geared, rotating systems. However, the methodology's success in accurately determining system frequencies may instead provide a technique for the quick calculation of Campbell Diagrams.

Lastly, this methodology was applied to one system for analysis. Other systems may behave differently. It also may be that in other systems, the ratio of computer processing time is substantially lower for response calculations, making it an attractive alternative to the full finite element model, especially when the full model contains several thousand or millions of DOFs.

Chapter 8

Bevel and Planetary Gear Considerations

Chapter 5 presented the 12-DOF mesh model for spur and helical gears. This chapter extends that application to straight and spiral-bevel gears. Epicyclic gear trains are also discussed. A robust epicyclic model from the literature is presented.

8.1 Bevel Gears

Bevel gears are used for connecting non-parallel, intersecting rotating shafts as depicted in Figure 8.1. The teeth of straight-bevel gears are cut straight across the gear face. Spiral-bevel gears have teeth that are curved for increased energy transfer. A proper analogy compares straight-bevel gears to spur gears and spiral-bevel gears to helical gears.

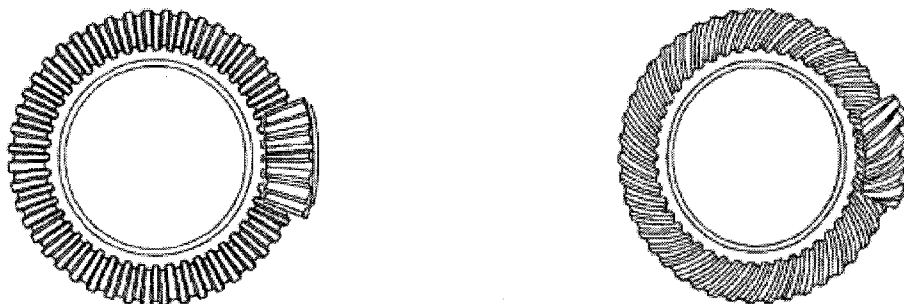


Figure 8.1: Straight-Bevel (left) and Spiral-Bevel (right) Gears (Coy, 1985).

The geometry of straight and spiral-bevel gears adds complexity to gear-mesh analysis, as depicted in Figure 8.2.

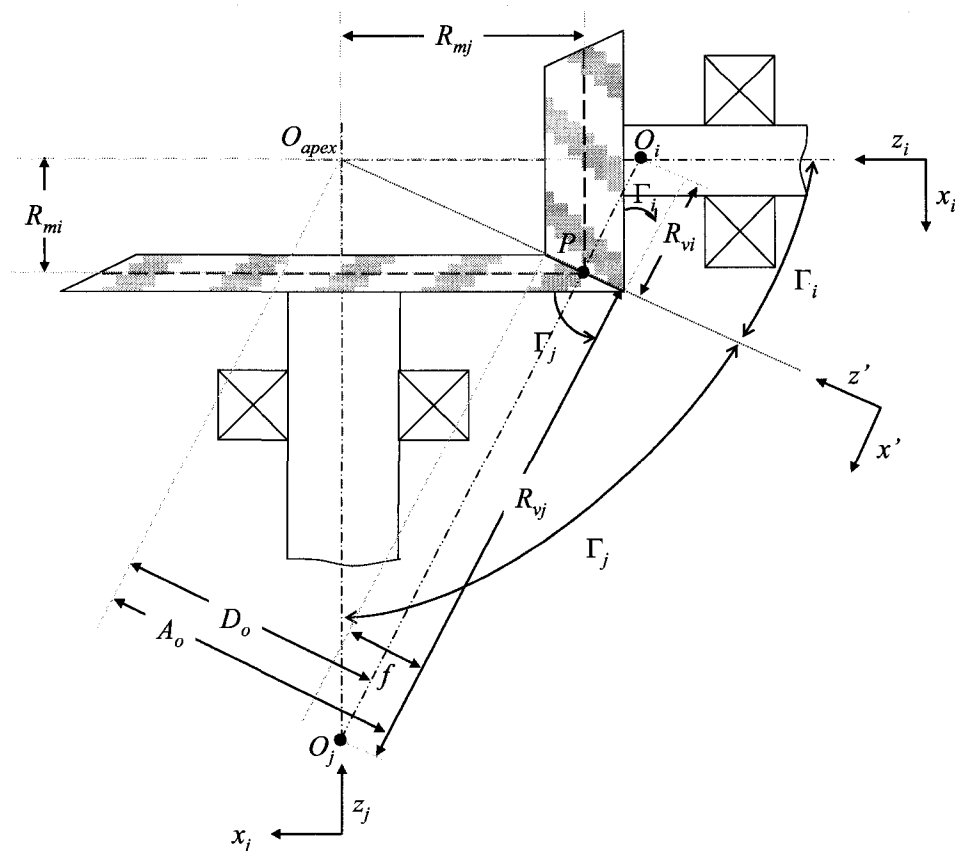


Figure 8.2: Straight-Bevel Gear Layout

where

A_o = outer cone distance

D_o = distance from apex to midpoint width of gear

f = facewidth of gears

P = pitch point

R_{mi} = mean radius of gear (i) at the pitch point

R_{vi} = equivalent spur gear radius of gear (i)

Γ_i = pitch angle of gear (i)

Three coordinate systems are also shown. For the two shafts, the convention introduced in Chapter 4 is used. The third, denoted with a prime in Figure 7.2, differs from the other two by a rotation about the y -axis: (Γ_i) for shaft (i) , and (Γ_j) for shaft (j) . The sum of the pitch angles equals the angle between the two intersecting shafts. The shaft angle is most commonly 90° .

In bevel gears, the shafts intersect, and the x - z planes of the three coordinate systems are co-planar. Consequently, the pitch point, (P) , always lies in the x - z plane. Thus, the orientation angle, (φ) , from the mesh model of Chapter 5 is unnecessary for bevel gear applications, since it is always zero.

The analysis of bevel gears and spiral-bevel gears is similar to that of spur gears and helical gears, respectively (Yoon, 1989). Tregold's Approximation (Buckingham, 1949) can be used to yield an equivalent spur gear analysis for two bevel gears in contact. The radii of the equivalent spur gears are represented in Figure 7.2 by the distances, (R_{vi}) and (R_{vj}) . The relationships associated with Tregold's approximation are presented below. The number of teeth of the equivalent spur gear is given by Equation 8.1.

$$N_{vi} = \frac{N_i}{\cos \Gamma_i} \quad (8.1)$$

For a straight-bevel gear, the radius of the equivalent spur gear is given by Yoon (1989).

$$R_{vi} = \frac{R_{mi}}{\cos \Gamma_i} = \frac{D_o \sin \Gamma_i}{\cos \Gamma_i} = D_o \tan \Gamma_i \quad (8.2)$$

with

$$\begin{aligned}
 D_o &= A_o - f/2 \\
 A_o &= \frac{d_i}{2 \sin \Gamma_i} \\
 d_i &= \frac{N_i}{P_i}
 \end{aligned}
 \tag{8.3}$$

where

d_i = diameter of gear (i) at the outer cone distance, (A_o).

P_i = diametral pitch of gear (i)

For a spiral-bevel gear, the analysis is somewhat different due to the presence of the spiral angle, (ψ), as depicted in Figure 8.3.

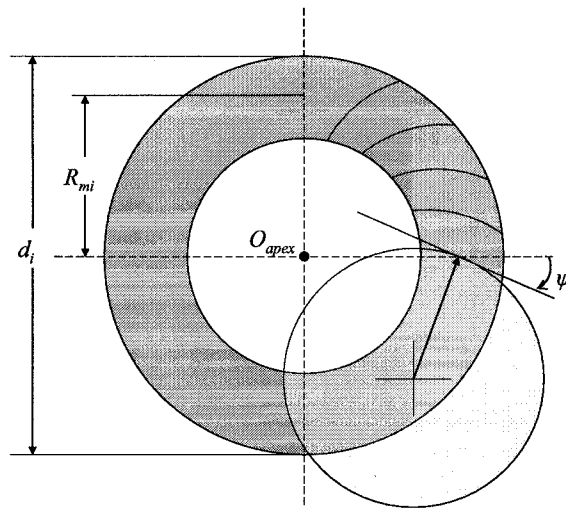


Figure 8.3: Spiral-Bevel Gear Geometry (Coy, 1985).

Tregold's Approximation requires a slight modification to account for the spiral angle, (ψ_i), as illustrated in Figure 8.4. The modification lengthens the equivalent spur gear radius, (R_{vi}), as a function of the spiral angle.

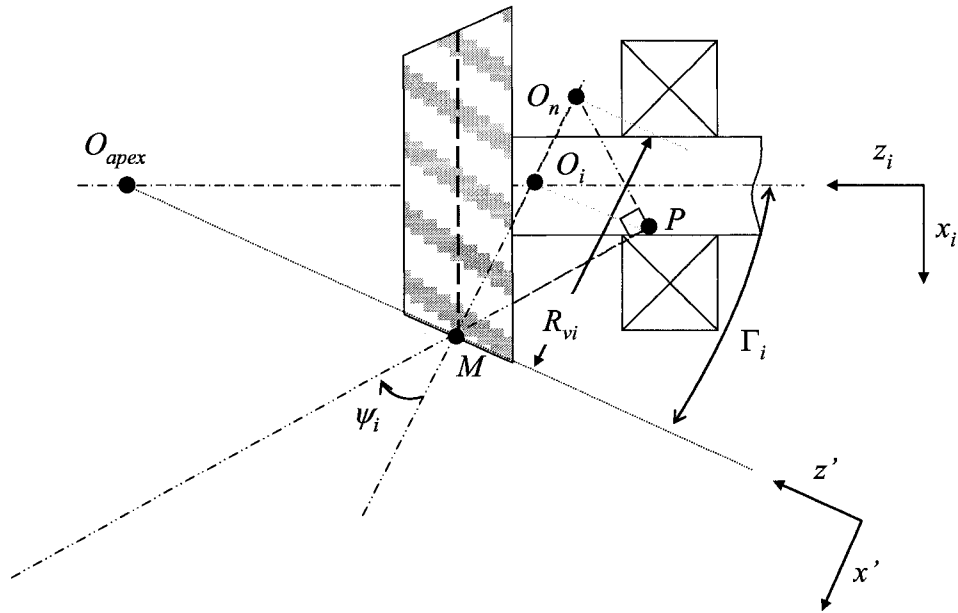


Figure 8.4: Tregold's Approximation for a Spiral-Bevel Gear (Yoon, 1989).

Tregold's Approximation for a spiral-bevel gear is as follows. In Figure 8.4, the distance, $\overline{O_i M}$, was the equivalent radius of the straight-bevel gear in Figure 8.2. The equivalent radius of the spiral-bevel gear takes into account the spiral angle, (ψ) .

Extending Equation 8.2:

$$\begin{aligned} O_i M &= D_o \tan \Gamma_i \\ O_i M &= PM \cos \psi_i \end{aligned} \quad (8.4)$$

$$\begin{aligned} PM &= \frac{D_o \tan \Gamma_i}{\cos \psi_i} \\ PM &= O_n M \cos \psi_i \end{aligned} \quad (8.5)$$

Therefore, since the equivalent spur gear radius equals the distance, $\overline{O_n M}$.

$$R_{vi} = \frac{D_o \tan \Gamma_i}{\cos^2 \psi_i} \quad (8.6)$$

When the spiral angle goes to zero, Equation 7.6 becomes Equation 7.2 for the straight-bevel gear case.

The most common spiral angle, (ψ_i) , is 35° . When used in aircraft, the pressure angle of bevel gears is commonly 20° (Dudley, 1984).

8.2 Planetary Gears

8.2.1 Background

Planetary or epicyclic gear trains are used for a variety of reasons. Planetary systems can provide a large amount of speed reduction in a very confined space, and are ideal for aerospace applications. Epicyclic gear trains are also unique in that the gear ratios are characterized by the configuration of the drive itself. The epicyclic gear system contains three components: a central sun gear, an outer ring gear, and intermediate planetary gears, as illustrated in Figure 8.5. The planets are connected to a rotating carrier which carries the torque output to, in the case of a helicopter, the output shaft running to the main rotor system.

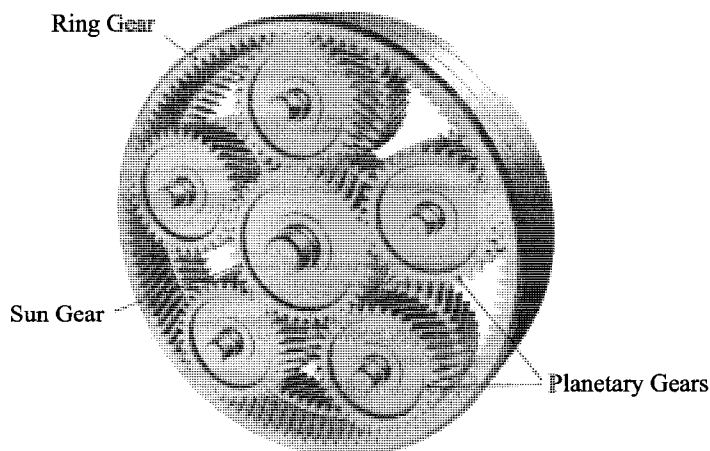


Figure 8.5: Epicyclic Gear Schematic (Litvin, 2004)

The uniqueness of planetary gears is that any one of the three components can provide the input with either of the other two providing the output, creating a variety of possible gear reduction ratios. A planetary gear system is one in which the ring gear is stationary. A star gear configuration uses a rotating ring system. Additionally, unlike the other gears discussed so far, the ring gear is an internal gear, with the gear teeth contained on the inside of the ring. Rotating ring gears in general are much more flexible than external gears (Tucker, 1977).

The planetary gear configuration is used exclusively in rotary-wing drive systems. The ring gear is stationary. Gear reduction ratios for the planetary configuration are given in Table 8.1.

Table 8.1: Planetary Gear Configuration Reduction Ratio

	Sun	Planet	Ring
Number of revolutions	$1 + \frac{N_r}{N_s}$	1	0

where (N_r) and (N_s) represent the number of teeth of the ring and sun gears, respectively.

Advantages of planetary gears include compactness, reduced loads on shaft bearing, and large torque-to-weight ratios (Parker, 2001). However, while enjoying reduced noise and vibration than other conventional gear configurations, the noise and vibration transmitted from the rotorcraft transmission to the passenger cabin is significant and can exceed 100 dB (Krantz, 1993).

Analytical considerations for modeling a planetary configuration are many to include multiple mesh points and intricate kinematics (Parker, 2001). From a rotor dynamic modeling standpoint, complexity increases due to the planets and carrier arms

rotating around, rather than on, the shaft centerline. Previous planetary gear configuration studies have largely been experimental or computational (Parker, 2001), such as August (1986) and Boyd (1989).

Due to the complexities of the epicyclic gear configuration, the derivation of a general, 12-DOF model is outside the scope of work of this study. Furthermore, the epicyclic trains of primary interest here use spur or herringbone gear teeth, such that all forces remain in the xy -plane. Lin (1999), Parker (2000, 2001), and Kiracofe (2007) have developed a lumped parameter planetary gear model that is well documented and validated against existing experimental data. It is used here for convenience.

8.2.2 Modeling Methodology

The components of the planetary gear configuration in Parker's model hinge on three DOFs each, two in translation and one in rotation. The model is represented in Figure 8.6.

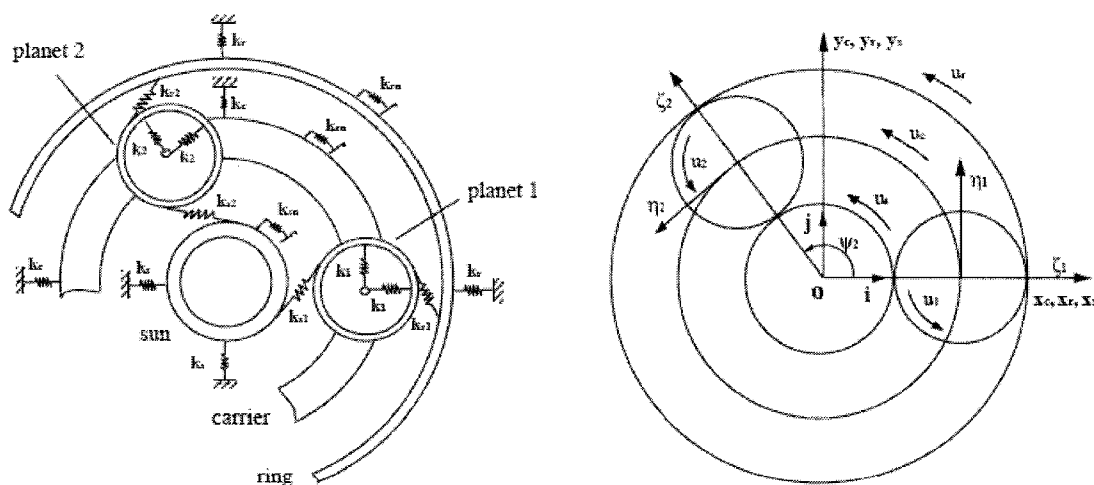


Figure 8.6: Planetary Gear Model with DOF (Parker, 2001).

Inspection of Figure 8.6 reveals the complexities of the planetary system in terms of the multiple stiffness components and DOFs of each subsystem. They are clarified in Table 8.2.

Table 8.2: Planetary Gear Model Definitions

Stiffness (k)	Subscript Definitions	DOF Coordinates	
pn	planet (n) bearing direct stiffness	u_h	rotational component
c	carrier bearing stiffness (direct)	x_h	lateral x component
cu	carrier bearing stiffness (rotational)	y_h	lateral y component
r	ring bearing stiffness (direct)	$h = c, r, s$	subscript indicating carrier, ring, or sun, respectively
rn	ring-planet (n) mesh stiffness	η_n	tangential displacement of planet (n)
ru	ring bearing stiffness (rotational)	ζ_n	radial displacement of planet (n)
s	sun bearing stiffness (direct)	u_n	rotational displacement of planet (n)
sn	sun-planet (n) mesh stiffness	ψ_n	circumferential angle of planet (n)
su	sun bearing stiffness (rotational)		

The rotational components, $(u_h, h = c, r, s, 1, \dots, n)$, are the familiar functions of angular displacement, (θ_h) , and radial arm, (r_h) .

$$u_h = r_h \theta_h \quad (8.7)$$

where the radii are those of the carrier, ring, sun, and planets, respectively. The carrier radius is the distance from the central axis of the gear train to the planet centers. For the

ring, sun, and planets, (r_h) is simply the base circle radius of each. Additionally, the circumferential angle of the first planet is fixed at zero, $\psi_1 = 0$.

The DOFs are measured with respect to a rotating reference frame affixed to the carrier. The carrier itself operates at the rotational speed, (Ω_c) . The equations of motion for the planetary gear configuration follow in Equation 8.8 (Parker, 1999).

$$\mathbf{M}\ddot{\mathbf{q}} + \Omega_c \mathbf{G}\dot{\mathbf{q}} + [\mathbf{K}_b + \mathbf{K}_{mesh} - \Omega_c^2 \mathbf{K}_\Omega] \mathbf{q} = \{\mathbf{F}(t)\}$$

$$\mathbf{q} = \left\{ \underbrace{x_c, y_c, u_c}_{carrier}, \underbrace{x_r, y_r, u_r}_{ring}, \underbrace{x_s, y_s, u_s}_{sun}, \underbrace{\zeta_1, \eta_1, u_1}_{1^{st} \text{ planet}}, \dots, \underbrace{\zeta_n, \eta_n, u_n}_{n^{th} \text{ planet}} \right\}^T \quad (8.8)$$

where

$[\mathbf{K}_b]$ = bearing stiffness matrix

$[\mathbf{K}_{mesh}]$ = mesh stiffness matrix

$[\mathbf{K}_\Omega]$ = stiffness hardening matrix due to high-speed carrier rotation

The complete component listing of the matrices in Equation 8.8 is presented in Appendix C.

The total number of DOFs in the epicyclic gear model depends upon two conditions. The first is the number of planets, (n) . The second is the configuration of the system rotation. If all components freely rotate, there are $3(n+3)$ total DOF. If the sun, carrier, or ring gear is fixed, the total number of DOF reduces to $3(n+2)$.

8.2.3 Frequency and Vibration Properties

The vibration characteristics of epicyclic trains are different from other rotor dynamic systems, due to multiple rotating components with multiple meshes. The vibrations of planetary gear systems are classified into three categories (Parker, 2000):

1. Translational modes. The carrier, ring, and sun gears have no rotational motion, only pure translation.
2. Rotational modes. The opposite of translational modes, the rotational modes are characterized by pure rotation of the carrier, ring, and sun.
3. Planet modes. The carrier, ring, and sun exhibit neither translational nor rotational motion. Deflection occurs with the planets only.

8.3 Summary

In Chapter 8, the considerations associated with modeling more complex gear trains have been discussed. The 12-DOF mesh model presented in Chapter 5 was extended to bevel gears, using Tregold's Approximation.

The dynamic complexities of planetary gears required a more intricate model. The current state-of-the-art model developed by Parker (2000, 2001) was a convenient and logical choice. The lumped parameter epicyclic model lends itself well for insertion into a larger system model. The mesh models presented here and in Chapter 5 will be instrumental to assembling system rotor dynamic models of geared transmission systems.

Chapter 9

Rotorcraft Transmission Assembly and Fault Modeling

This chapter represents the culmination of the material discussed in the previous seven chapters. The methodologies introduced are integrated into a comprehensive rotor dynamic finite element model, consisting of multiple shafts connected by multiple gears. An overview of the system characteristics is discussed, and preliminary results are presented. In terms of prognostics application, this chapter also discusses potential health indicators through fault classifications of the selected system and some of the resulting model output.

9.1 OH-58A Helicopter Transmission and Test Stand

The OH-58A is the military version of the Bell 206 helicopter. It is a single-engine, light helicopter used for observation and reconnaissance. Its civilian counterparts enjoy extended usage as news and traffic helicopters over major metropolitan areas around the United States. Photos of both helicopters are presented in Figure 9.1.

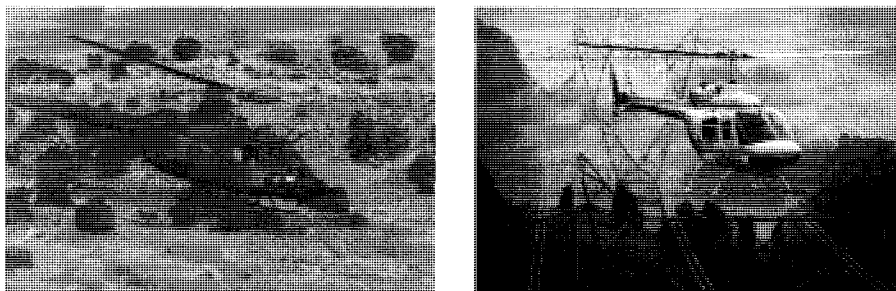


Figure 9.1: OH-58 (left) and Bell 206 (right) Helicopters

The OH-58A transmission provides speed reduction and torque transfer via two reduction stages. The first reduction occurs through a spiral-bevel gear set. The input pinion contains 19 gear teeth. The output gear has 71 teeth. The bevel-gear input shaft is mounted using three triplex ball bearings and one roller bearing. The bevel-gear output shaft uses duplex ball bearings and one roller bearing. The 71-tooth output gear also drives an accessory gear of 27 teeth. The accessory shaft operates the oil pump and thereby lubricates the system.

The second reduction occurs through a planetary gear set. The bevel-gear output shaft is splined to a sun-gear shaft. The sun gear contains 27 teeth, which drives three planetary gears of 35 teeth each. A later version of the transmission used four planets. A fixed ring gear of 99 teeth meshes with the planets at the top of the drive casing. The planets rest on double-row spherical bearings, which are attached to the carrier. The carrier is connected to the main rotor shaft. The shaft rests on a ball bearing and a roller bearing. The total reduction of the system is 17.44:1. A diagram of the transmission is presented in Figure 9.2. A more complete description of the transmission can be found in Lewicki (1987).

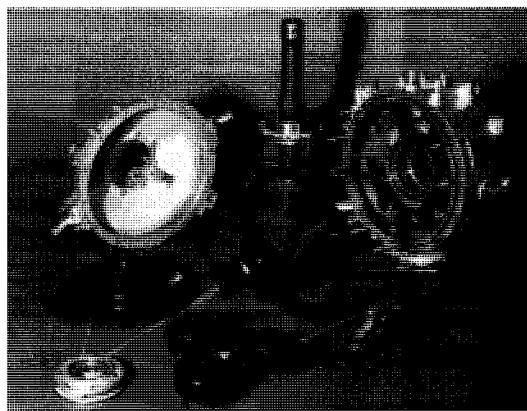
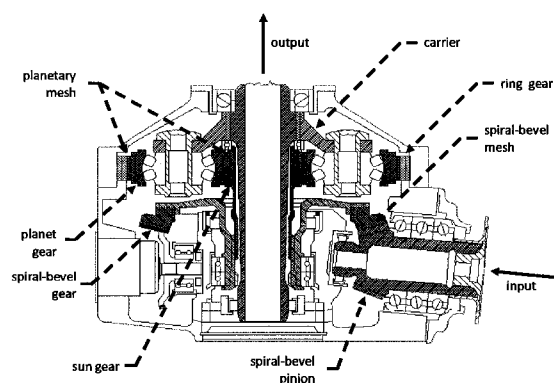


Figure 9.2: OH-58 Transmission Schematic (left) and Disassembled (right)

The OH-58A transmission has been extensively studied using the 500-hp Transmission Test Stand at the NASA Glenn Research Center (Figure 9.3). The test stand operates on the closed-loop principle. A 149-kW motor powers the stand and controls the speed. A second 11-kW motor provides the torque in the closed loop. Using this method, power circulates around the loop such that the 149-kW motor only replenishes friction losses. A shaft loading system acting on the mast shaft output can simulate loads from a spinning rotor. A fully detailed description is provided in Lewicki (1992). The test stand itself is pictured in Figure 9.3.

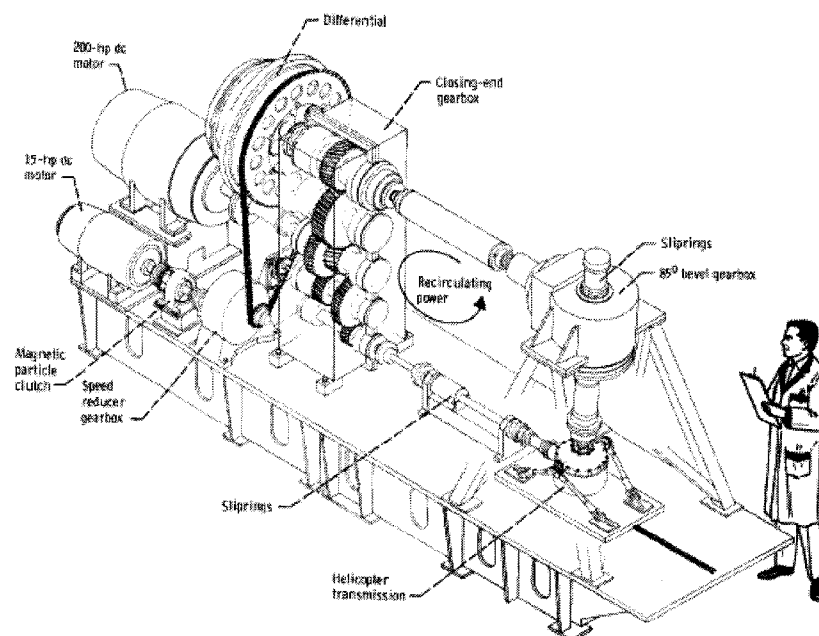


Figure 9.3: NASA Glenn 500-hp Transmission Test Stand (Lewicki, 1987).

9.2 Transmission Model

9.2.1 Assembly

The lumped parameter model consists of three shafts. Shaft A is the input shaft. Shaft B is the mated bevel and sun gear shaft. Shaft C is the mast output shaft.

Additional DOFs are incurred by the number of planets and the epicyclic ring gear. The system's geometric and material parameters for the model were obtained from technical data and manufacturer drawings of the transmission.

Each of the three shafts is divided into elements. The finite element nodes are placed at key locations as determined by shaft geometry, gear-mesh, and bearing locations. The resulting elements are each therefore unique. Furthermore, the three shafts are not uniformly distributed along their lengths. An equivalent (eq) shaft is subsequently generated for each of the three shafts to simplify the modeling process. Shaft A consists of eight nodes, with four bearing locations and one mesh location. Shaft B also has eight nodes with two mesh and four bearing locations. Shaft C has thirteen nodes with only two bearing locations. Shaft C also contains the lumped epicyclic carrier parameters, acting at a single node. Figures 9.4 through 9.6 depict the finite element formulation of the three shafts.

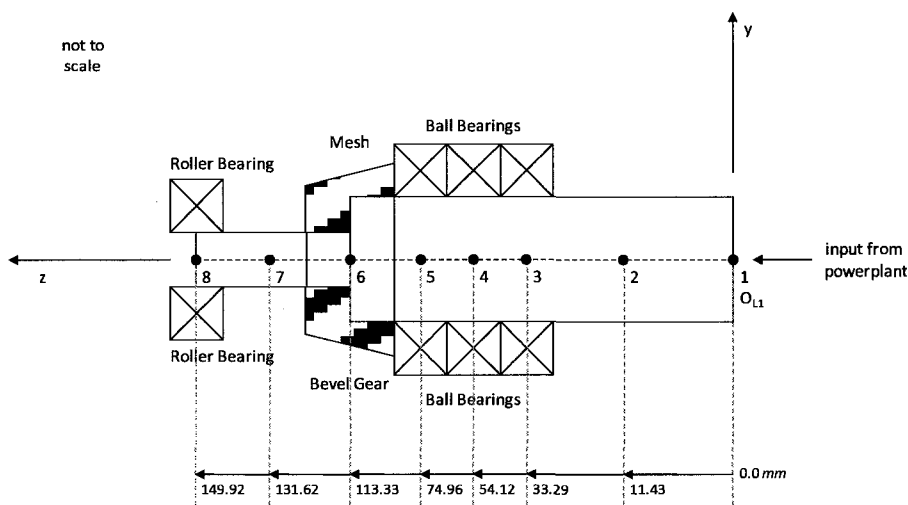


Figure 9.4: Finite Element Layout - Shaft A

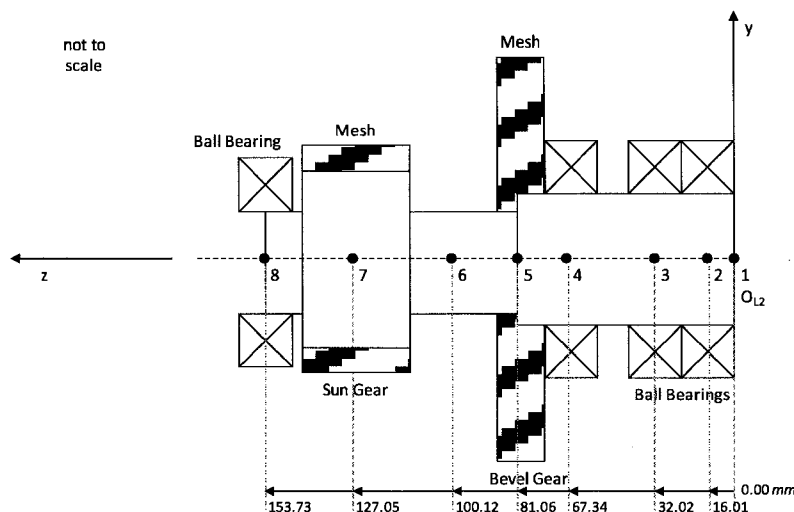


Figure 9.5: Finite Element Layout - Shaft B

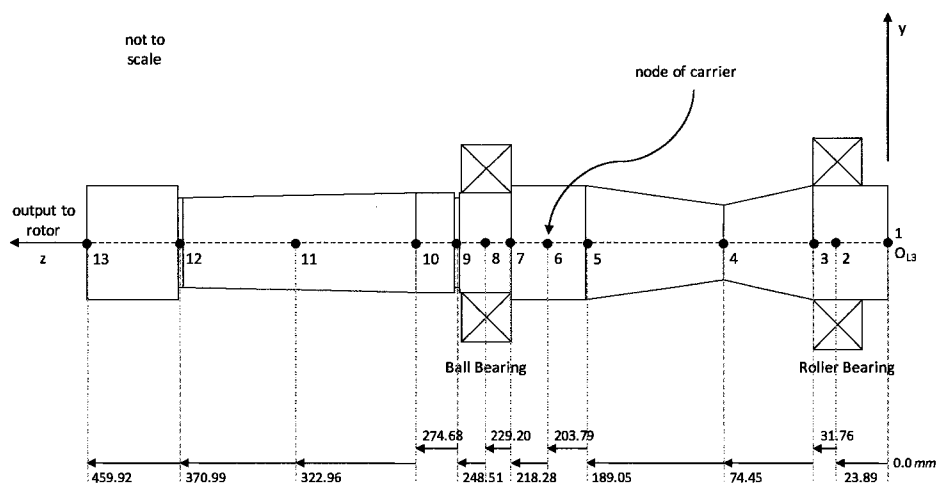


Figure 9.6: Finite Element Layout - Shaft C

The ring and planet gears are not depicted. The configuration of the *OH-58* transmission in this model uses three planetary gears. The ring and the planets each have three DOF. Each node along the three shafts has 6 DOFs, resulting in 186 DOFs for the model. The mast output shaft, Shaft C, is different than the actual shaft used in the aircraft. The shaft in this application has been modified for use in the transmission test

stand. The primary difference is in the lengths of the two shafts. The primary shaft is much longer than that of the test stand.

Shaft coupling occurs via the gear-meshes. Shafts A and B are coupled by the spiral-bevel mesh. The gear mesh is modeled as two equivalent spur gears using Tregold's Approximation, and the mesh model of Chapter 5. Shafts B and C are coupled using the epicyclic model discussed in Chapter 8.

A complete list of model parameters follows in Tables 9.1 and 9.2. The shaft parameters and spiral-bevel mesh parameters used are presented in Table 9.1. Table 9.2 lists the epicyclic train parameters. The model itself is a MATLAB® code containing several scripts and subroutines.

Table 9.1: Shaft and Spiral-Bevel Mesh System Parameters

Material Parameters	
Young's Modulus (E) (N/m^2)	2.05E11
Shear Modulus (G) (N/m^2)	8.0E10
Poission's Ratio (ν)	0.29
Density (ρ) (kg/m^3)	7850
Shaft A Parameters: Input Shaft	
Length (m)	0.150
Outer Diameter (eq) (d_{so}) (m)	0.050
Inner Diameter (eq) (d_{si}) (m)	0.033
Shaft B Parameters: Mated Bevel and Sun Gear Shaft	
Length (m)	0.154
Outer Diameter (eq) (d_{so}) (m)	0.073
Inner Diameter (eq) (d_{si}) (m)	0.058
Shaft C Parameters: Mast Output Shaft	
Length (m)	0.460
Outer Diameter (eq) (d_{so}) (m)	0.057
Inner Diameter (eq) (d_{si}) (m)	0.021
Input Pinion Parameters (Shaft A)	
Mass (kg)	0.47
Moment of Inertia (I_t) ($kg-m^2$)	1.4E-4
Polar Moment of Inertia (I_p) ($kg-m^2$)	2.8E-4
Diameter (d_g) (m)	0.062
Pitch Angle (Γ) (deg)	15.3
Pressure Angle (α_n) (deg)	20
Spiral Angle (ψ) (deg)	30.0
Orientation Angle (ϕ) (deg)	0.0
Average Mesh Stiffness (K_g) (N/m)	1.0E8
Number of Teeth	19
Input Gear Parameters (Shaft B)	
Mass (kg)	2.24
Moment of Inertia (I_t) ($kg-m^2$)	1.5E-4
Polar Moment of Inertia (I_p) ($kg-m^2$)	3.0E-4
Diameter (d_g) (m)	0.228
Pitch Angle (Γ) (deg)	78.7
Pressure Angle (α_n) (deg)	20
Spiral Angle (ψ) (deg)	30.0
Orientation Angle (ϕ) (deg)	0.0
Average Mesh Stiffness (K_g) (N/m)	1.0E8
Number of Teeth	71

Table 9.2: Epicyclic Component Parameters

Epicyclic Gear Parameters	
Carrier	
Mass (kg)	5.43
Moment of Inertia (I_i) (kg)	6.29
Diameter (d_g) (m)	0.1768
Pressure Angle (α_n) (deg)	20
Bearing Stiffness (k_{bc}) (N/m)	1.0E8
Bearing Stiffness (k_{cu}) ($N\cdot m/rad$)	100
Ring	
Mass (kg)	2.35
Moment of Inertia (I_i) (kg)	3.00
Diameter (d_g) (m)	0.275
Pressure Angle (α_r) (deg)	24.6
Bearing Stiffness (k_{br}) (N/m)	1.0E15
Bearing Stiffness (k_{ru}) ($N\cdot m/rad$)	1.0E15
Number of Teeth	99
Sun	
Mass (kg)	0.40
Moment of Inertia (I_i) (kg)	0.39
Diameter (d_g) (m)	0.0774
Pressure Angle (α_s) (deg)	24.6
Bearing Stiffness (k_{bs}) (N/m)	1.0E8
Bearing Stiffness (k_{su}) ($N\cdot m/rad$)	100
Number of Teeth	27
Planet	
Number of Planets	3
Mass (kg)	0.66
Moment of Inertia (I_i) (kg)	0.61
Diameter (d_g) (m)	0.1003
Pressure Angle (α_n) (deg)	20
Bearing Stiffness (k_{br}) (N/m)	1.0E8
Number of Teeth	35
Duplex, Triplex, and Roller Bearing Parameters	
k_{xx}, k_{yy} (N/m)	1.0E9

9.2.2 System Frequencies and Mode Shapes

The nominal input speed of the model is 6060 *rpm* (101 *Hz*) to match the transmission's normal operating speed. The first 20 natural frequencies are listed in

Table 9.3. Additionally, comparison data results extracted from Lewicki (1987) and Coy (1988) and the error are listed for model validation purposes.

Table 9.3: Natural Frequency Comparison – Model Results vs. Experimental Data

Model (Hz)	Experimental Data (Hz)	Error (%)	Model (Hz)	Experimental Data (Hz)	Error (%)
483	475	1.64	2627	2600 <i>m</i>	1.03
486	475	2.28	3198	3250	1.60
1035	1050	1.40	3286	3250	1.10
1400	1450	3.44	3417	3350	2.00
1405	1450	3.08	3635	3700	1.76
1754	1720 <i>m</i>	1.97	3656	3700	1.18
1754	1720 <i>m</i>	2.01	3977	4014 <i>m</i>	0.93
1797	1720 <i>m</i>	4.49	3993	4014 <i>m</i>	0.52
2150	2125	1.19	4010	4014 <i>m</i>	0.11
2166	2125	1.92	5108	5160 <i>m</i>	1.01

The lowercase *m* signifies a gear-mesh frequency harmonic.

Table 9.3 also illustrates the impact of the gear-mesh frequencies and their harmonics on the experimental data. Lewicki (1987) and Dempsey (2004a) discuss the presence of these frequencies and their dominant impact on analysis. Eight of the natural frequencies listed in Table 9.3 are within 5.0% of a gear-mesh frequency harmonic, as depicted with a lowercase (*m*). Experimentally, these peaks dominate the response. However, all other frequency values match closely with error values well under 10%. Graphically, Figure 9.7 captures this comparison of the first 10 modes of vibration between the actual system and the finite element model.

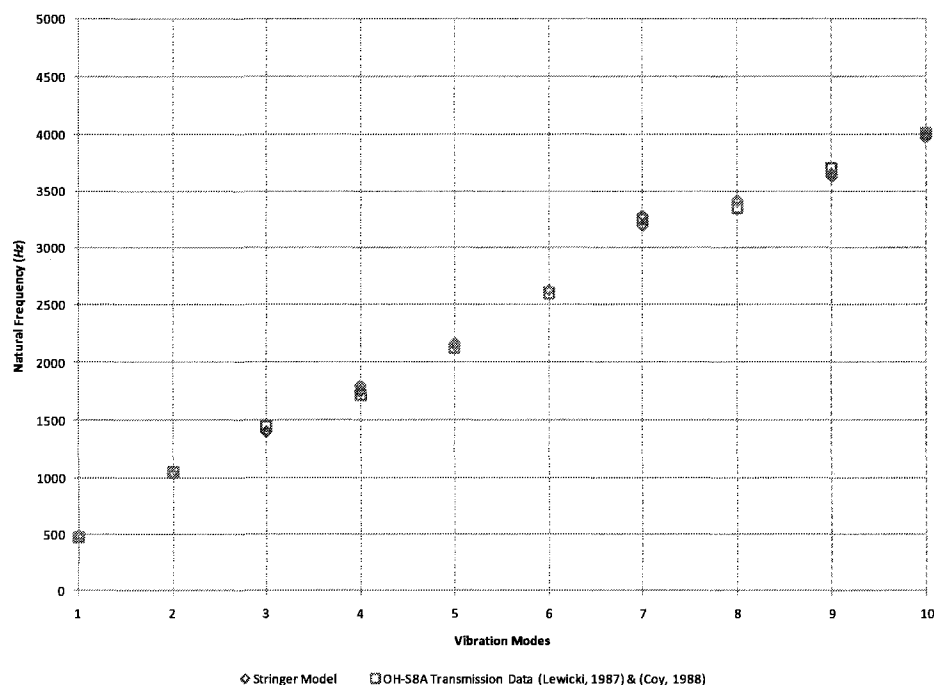


Figure 9.7: Modes of Vibration: Model vs. Experimental Comparison

The mode shapes of the system were also determined from the system eigenvectors. Unlike the spur gear application in Chapter 5, the mode shapes for the transmission model are more complex. All mode shapes with lateral motion have some degree of torsional coupling. Examples of this coupling can be seen in Figures 9.8 through 9.10. For each mode shape, there are three sets of charts. Each set consists of a lateral motion plot (top) and a torsional magnitude plot (bottom). The three columns correspond to Shafts A, B, and C, respectively. Modes corresponding to the planets and ring gear are not presented.

Additionally, each torsional plot contains one or two dotted vertical lines. These correspond to the gear-mesh or carrier locations for each shaft. It is at these vertical lines that the coupling between shafts occurs.

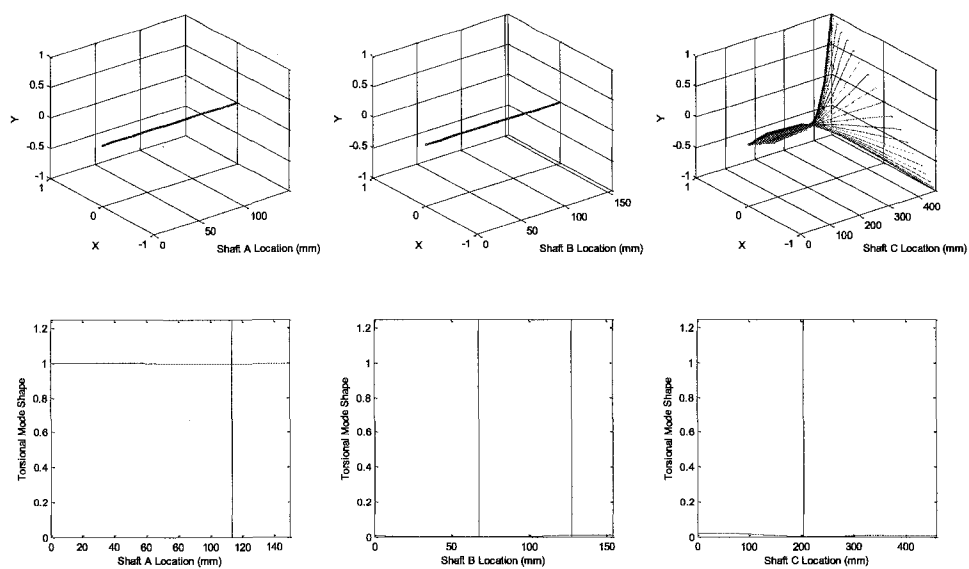


Figure 9.8: Dominant Shaft C Mode Shape (483 Hz)

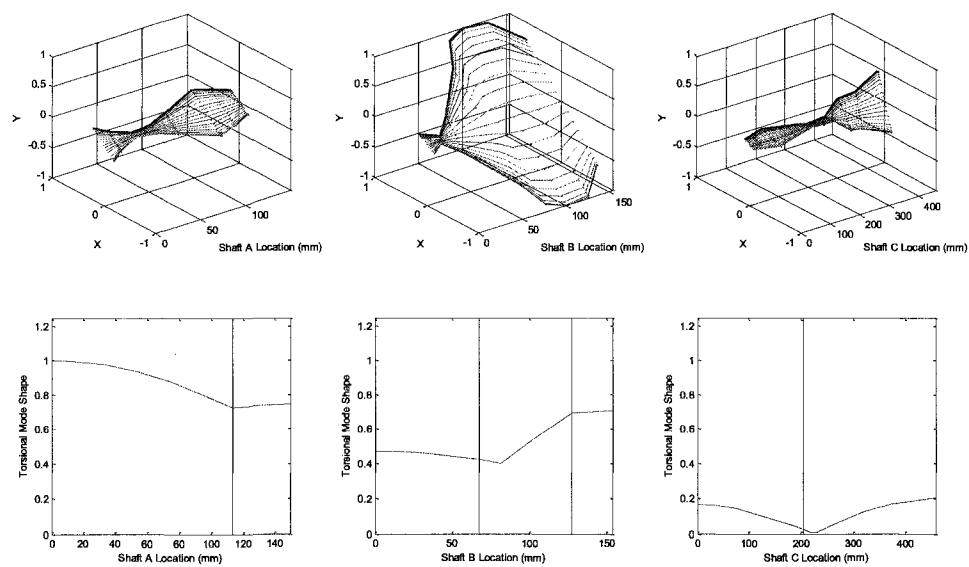


Figure 9.9: Dominant Shaft B Mode Shape (3,417 Hz)

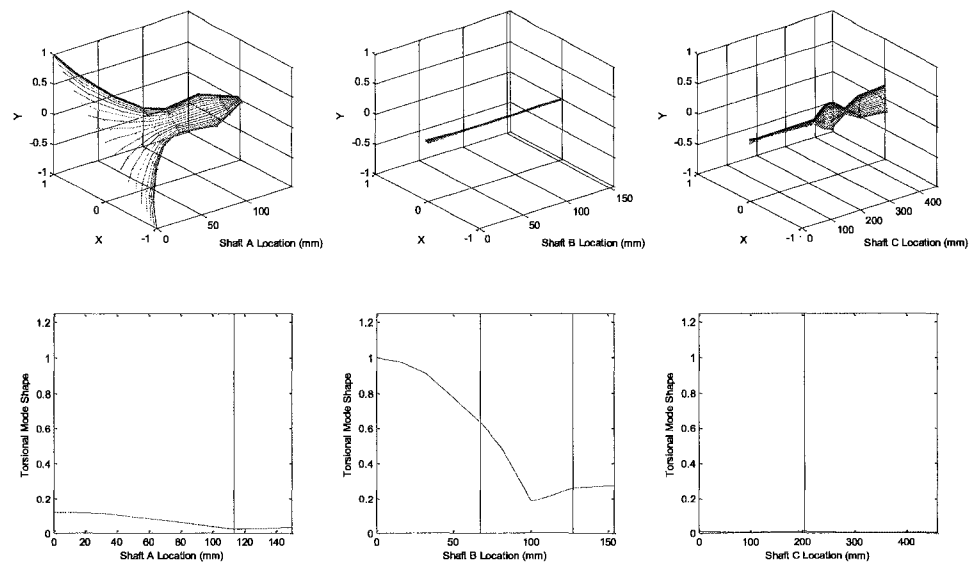


Figure 9.10: Dominant Shaft A Mode Shape: (6,669 Hz)

The three mode shapes in the previous charts were selected because they each show a different shaft to have the dominant motion. In Figure 9.8, the mode shape at 483 Hz shows Shaft C to be dominant with an almost rigid torsional mode shape in Shaft A. Figure 9.9 shows the mode shape at 3,417 Hz, characterized by Shaft B's dominance in lateral motion and a significant torsional coupling in all three shafts. Finally, Figure 9.10 shows Shaft A as the dominant shaft at 6,669 Hz, with significant coupling of the torsional modes of Shafts A and B.

The non-symmetry of the model coupled with multiple gear-meshes produce a complex set of mode shapes. Lin (1999) and Parker (2001) noted similar observations regarding the modes of their epicyclic gear train model. Further analysis and characterization of the total system mode shapes is necessary, to include the planet and ring contributions.

A brief discussion comparing and contrasting this transmission model to Parker's epicyclic model is appropriate. Parker's gear model, presented in Equation 8.12, is a lumped parameter model of the carrier, ring, sun, and planets. It is a convenient model for rotor dynamic analysis, because it simplifies the kinematic complexities of the gear train, such as the rotation of the planets about the central shaft axis, rather than along the axis itself. Parker's model also accounts for the multiple gear-mesh interactions of the epicyclic gear train.

However, Parker's model only addresses the epicyclic gear train, which is a sub-component of the helicopter transmission. The shafts and other meshes of the transmission are not included. Parker's model is therefore a convenient tool for analyzing the dynamics of the epicyclic gear train as a critical sub-component of the larger system. This is particularly true with the OH-58A transmission model, where the primary dynamics of concern are the sun gear's contribution to the intermediate shaft (Shaft B), and the carrier's contribution to the output shaft (Shaft C). The difference between this model and Parker's model is that the transmission model replicates the dynamics of the complete system, while Parker's model focuses solely on the epicyclic gear components themselves.

9.2.3 Modal Synthesis Results

An investigation into the feasibility of applying the modal synthesis method presented in Chapter 7 was conducted. Modal analysis was conducted independently on the three shafts. Coupling was then inserted into the modal model between the substructures. Several iterations of varying DOF reductions were carried out. The results of one such iteration are presented below. Table 9.4 provides the modal

reduction for each component. In Table 9.5, the natural frequencies obtained using the modal synthesis method are compared to the frequencies of the full finite element model. Figure 9.11 shows the frequency response comparison between the two.

Table 9.4: Modal Synthesis Reduction

	Original Modes	Reduced Modes
Shaft A	48	20
Shaft B	48	20
Shaft C	78	20
Epicyclic Gear Set	12	12

Table 9.5: Finite Element vs. Modal Synthesis Frequency Comparisons

FE Model (Hz)	Modal Synthesis (Hz)	Error (%)	FE Model (Hz)	Modal Synthesis (Hz)	Error (%)
482.78	483.19	0.09	2626.81	2667.87	1.56
485.83	483.32	0.52	3198.14	3199.51	0.04
1035.29	1034.38	0.09	3285.69	3294.88	0.28
1400.14	1402.80	0.19	3416.96	3432.21	0.45
1405.29	1406.26	0.07	3634.86	3655.38	0.56
1753.94	1754.63	0.04	3656.19	3656.29	0.00
1754.50	1754.67	0.01	3976.54	4010.05	0.84
1797.16	1798.10	0.05	3992.96	4010.97	0.45
2150.22	2170.74	0.95	4009.58	4020.85	0.28
2165.86	2173.75	0.36	5107.97	5107.53	0.01

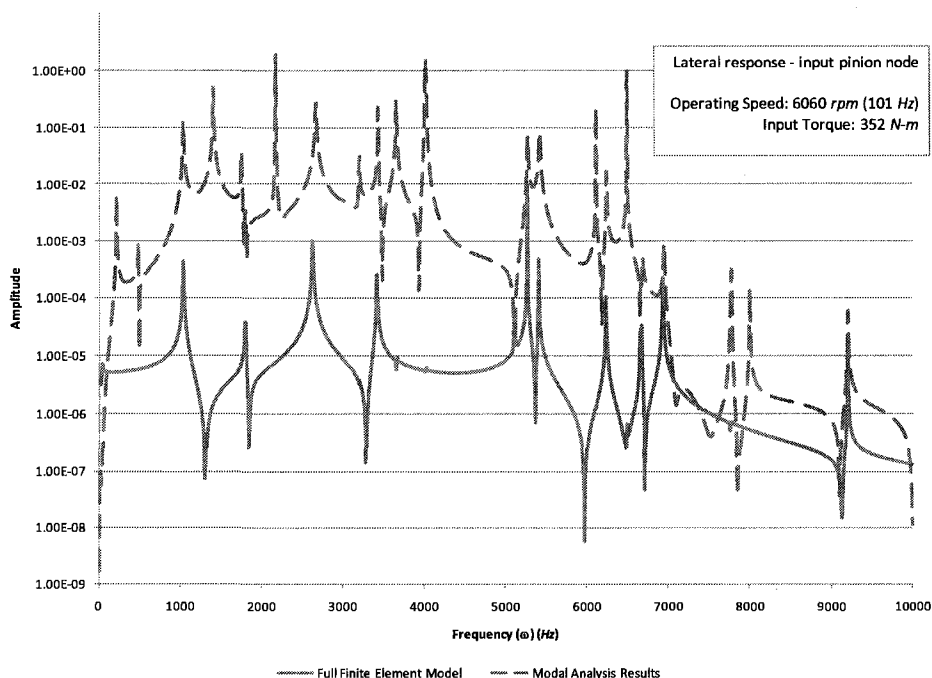


Figure 9.11: Frequency Response Comparison of Input Node Lateral Response

Based on these results, the modal synthesis method provides a good approximation of the system model. In Table 9.5, the maximum error between the natural frequencies was less than 2.0%.

However, from Figure 9.11, the responses do not match exactly as with the results of the spur gear application of Chapter 5. This is most likely due to the non-symmetry and added sophistication of the transmission model. Nevertheless, the modal synthesis response is still valid for two reasons. First, the modal synthesis results represent a conservative approximation. The modal synthesis results over-predict the response. Secondly, the approximation very clearly identifies the resonant peaks within the system. In some instances, the modal synthesis results more clearly identify the resonant peaks than the full finite element model.

While the modal synthesis results match the system results well, the rigor of the process in conjunction with its practical limitations as discussed in Chapter 7 still apply. If the goal of the modal synthesis process is to study the underlying dynamics of the geared system, then the modal synthesis procedure has benefit. If the goal is strictly model simplification through modal reduction, then the modal synthesis method does not necessarily provide the simplification in terms of processing time.

The modal synthesis method presented results in a valid approximation of the system dynamics. However, the results also show that the procedure warrants further investigation, especially for other potential applications such as system identification or design optimization. Investigations of this method to other more sophisticated models should be conducted to further determine its benefits. Lastly, other modal methods should be considered, especially those that do not require formulation in the state-space. One such possible method was introduced by Meirovitch (1997). Meirovitch proposed a perturbation method for systems with small damping, while his treatment of gyroscopic effects used the standard state-space formulation. If the gyroscopic effects are small enough as they are in these systems, then theoretically, the perturbation method is still applicable. This procedure has yet to be investigated thoroughly, but it may provide a more attractive alternative to the rigor of the modal synthesis method.

9.3 Health Monitoring: Faults, Fault-Seeding and Model Predictions

The previous two sections have detailed the construction, validation, and initial results of the OH-58A transmission model. This section focuses attention on certain faults common to rotorcraft drive systems. Faults indigenous to gears and bearings are discussed. Seeding these faults into the system model is discussed, and the methods for

detecting and analyzing them are presented. The model's utility and importance to health diagnostics and prognostics becomes readily apparent in the results.

9.3.1 Common Faults: The Nature of Transmission Failures

From previous experiments, the following common faults are identified.

Spall. A spall is a chip or a fragment that splinters off from a larger body due to some action. In mechanical engineering, it can occur when objects undergo large loading or pressure fluctuations. It most frequently occurs in ball bearings where the shear stress has exceeded the shear stress at a critical point, usually beneath the surface, causing the surface to chip. Figure 9.12 shows various images of spall damage to the OH-58A transmission bearings. Figure 9.12(a) and (b) show damage to the inner races of both duplex and triplex ball-bearings, respectively. Figure 9.12(c) shows spall damage to one of the balls in the bearing. Figure 9.12(d) shows fatigue spall damage to the inner race of one of the planet bearings.

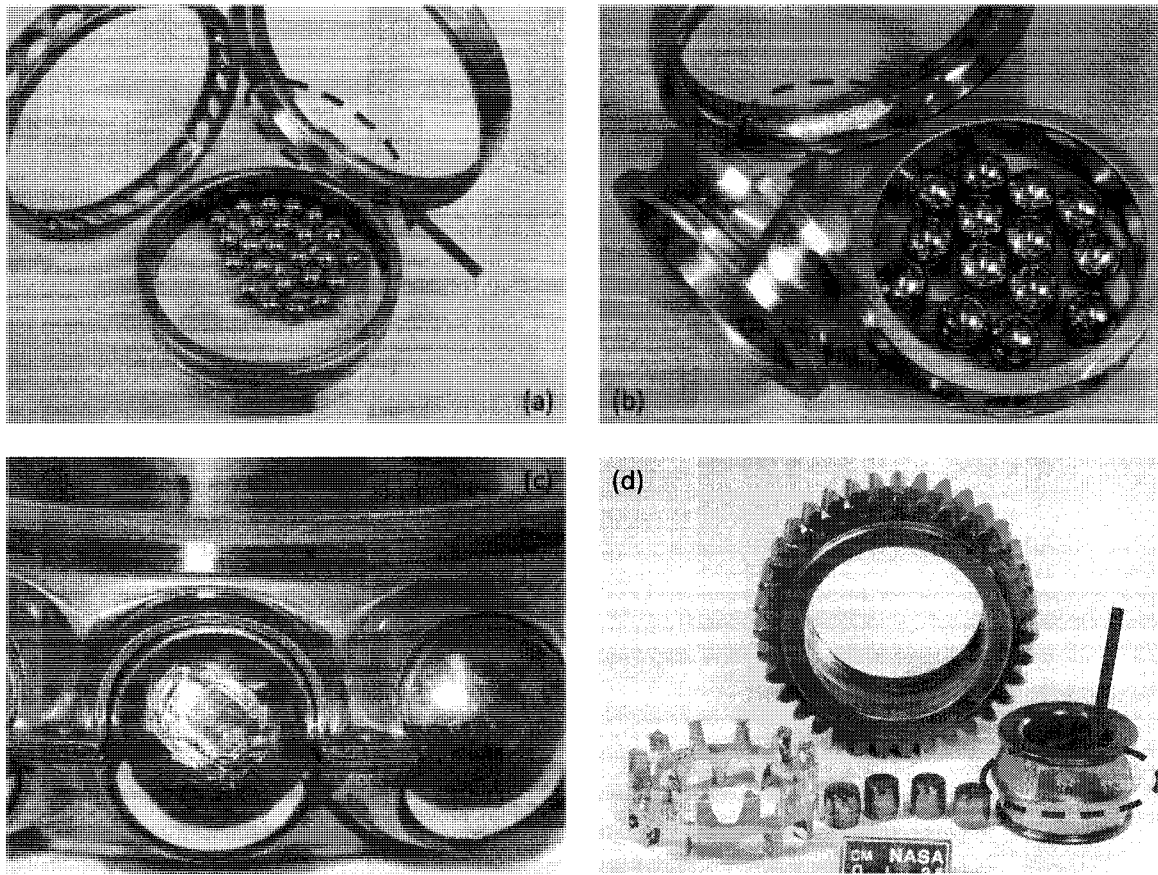


Figure 9.12: Spall Damage to Various OH-58A Transmission Bearings

Pitting. Pitting occurs when the gear material exceeds its surface fatigue limit. Pits form on the contacting surfaces of the gear teeth as pieces of the gear material break off (Dempsey, 2004a). Pitting is characterized as a fatigue failure. Fatigue type failures are significant due to the extreme number of cycles the gear teeth undergo. For example, the spiral-bevel input pinion undergoes approximately $36.4E6$ tooth loading cycles in 100 hours of operation at the nominal operating speed of 6060 rpm (101 Hz). The other gears in the transmission undergo fewer loading cycles, although on the same order of magnitude (Lewicki, 1992). Clearly, fatigue damage plays a significant role in transmission health. Figure 9.13 shows a spur gear tooth with pitting damage.

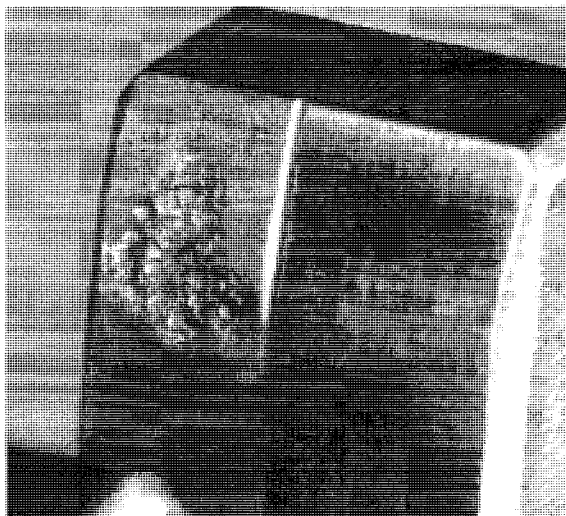


Figure 9.13: Pitting Damage to Gear Tooth

Cracking and Fracture. Gear cracks and resulting fracture are much less common than other forms of transmission damage, but they are potentially more catastrophic (Decker, 2003). A crack forms when the stress at a point on the gear exceeds its ultimate stress. This is normally located at the root of the gear tooth. During successive cycles, the crack continues to propagate as the material in the crack area weakens. Finally, the entire tooth or a portion breaks off, resulting in a fracture.

Unlike the other types of damage discussed here, cracks propagate quickly, such that the time from initial damage to complete failure is much less. Figure 9.14 shows three images of crack propagation leading to fracture. Figure 9.14(a) shows a finite element representation of two gear teeth coming into mesh, with the highest stress concentrations at the root of the tooth. Figure 9.14(b) shows a crack propagating from the root of the gear tooth. Lastly, Figure 9.14(c) shows the catastrophic fracture of a segment of the OH-58A spiral-bevel pinion.

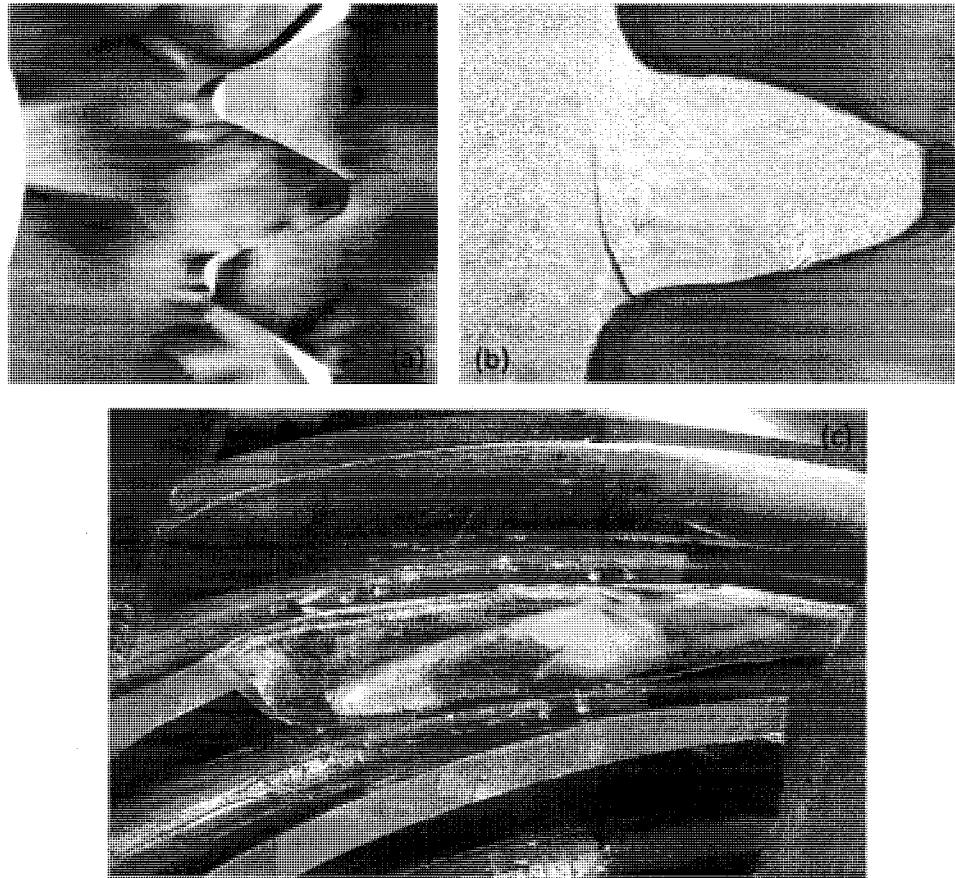


Figure 9.14: Gear Crack Propagation and Fracture

Surface Contact Wear. Gear surface contact wear has been labeled as “one of the most important failure modes” in geared systems (Krantz, 2007). While gear manufacturers generally concentrate on preventing pitting and scoring, they tend not to focus on the prevention of contact wear (Flodin, 1997). Contact wear is the mild deterioration of the mating surface of a gear tooth due to the sliding action between teeth. Wear is a gradual process, characterized by increasing dynamic contact loads, which can accelerate the likelihood of fatigue failures (Kuang, 2001). The debris can also negatively impact bearings or other components of the drive system (Krantz, 2007). Figure 9.15 depicts a gear tooth with contact surface wear.

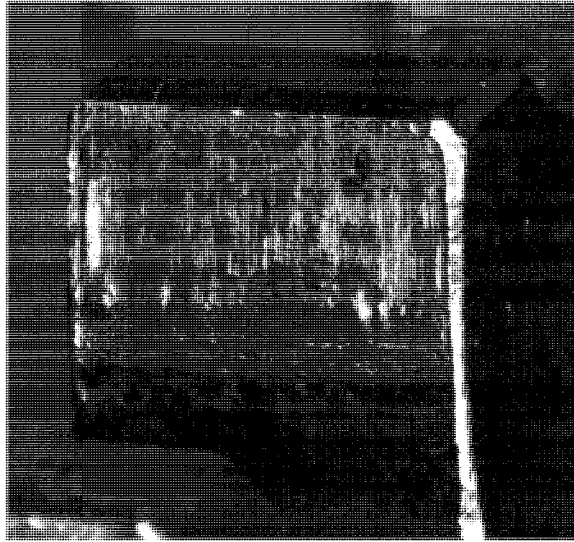


Figure 9.15: Gear Tooth with Surface Contact Wear

9.3.2 Fault Detection Methodology

Currently, two major methods are used for fault detection: oil-debris monitoring and vibration monitoring. After repeated use, the gears and bearings in the drive train begin to deteriorate. This damage manifests itself either by shedding transmission material into the transmission oil (Dempsey, 2004a) or by an increase in machinery vibration levels (Lebold, 2000).

Oil-debris monitoring uses magnetic sensors to measure debris buildup and particle size as an indicator of transmission health. However, this method has limitations. The oil contains relatively large particles of debris. Debris indicating failure may be smaller than non-failure debris, resulting in false alarms. The current average rate of false alarms is approximately 1 per 100 flight hours (Dempsey, 2004a). Furthermore, the transmission components have the same material composition. This makes it difficult to determine which component is failing without disassembling the transmission.

The second method of fault detection involves vibration monitoring. Current vibration patterns are compared to previous patterns to determine if the component is functioning normally. This procedure is complex. Comparisons of the signals themselves are not necessarily effective. However, the extraction and calculation of diagnostic parameters from these vibration signatures can be effective (Lebold, 2000), although failure detection rates using this method average approximately 70% (Larder, 1999). Effective vibration monitoring is also dependent upon sensor location. Placing sensors at every location on the transmission system is costly and unfeasible. Identifying the optimal sensor locations is critical to monitoring meaningful vibration patterns.

While both methods are necessary, the missing component for improving health monitoring rates is a vibration signature database (Choy, 1996). With a comprehensive database, alarms due to oil-debris indicators could be double-checked and verified, thereby reducing unnecessary downtime for aircraft maintenance.

9.3.3 Fault Modeling Scenarios and Methodology

From the list of common transmission faults, two scenarios were selected for demonstration. The first simulates contact surface wear on the spiral-bevel input pinion of the input shaft (Shaft A) using a surface wear loading profile similar to one presented in Flodin (1997). The second fault scenario examines the effects of damage to one of the triplex ball bearings on the input shaft. The methodology of modeling these faults is described below.

1. The equations of motion are integrated to calculate the steady-state responses for the "normal" operating cases. Two nodal locations are selected: the input node

(Node 1) of the input shaft and the carrier node (Node 6) of the output shaft (Shaft C). A curve fit is applied to the lateral- x response of these two nodes, generating a truncated Fourier series. This allows the response to be calculated over a large time period. The procedure is repeated for the “damaged” operating case.

2. The two responses are placed together in three phases: normal operation phase, a transition-to-damage phase, followed by a damaged operation phase. This simulates a period of normal operation, followed by onset of damage, and ending with the nominal failure condition.

3. Vibration-monitoring diagnostic parameters are determined for the time duration of the combined response. These parameters are defined and discussed in the next section.

4. The frequency responses of the damaged and normal simulations are determined using the FFT algorithm in MATLAB®. They are used to compare the differences in the frequency responses of the normal and abnormal operating conditions.

9.3.4 Vibration Diagnostic Parameters

Vibration diagnostic parameters are traditionally based upon a statistical measurement of the vibration signal (Decker, 2003). Several methods exist and are documented in a variety of sources to include Lebold (2000) and Decker (2003). These parameters are normally used with vibration signal data and require some sort of pre-processing to condition the signal for meaningful interpretation. Only four parameters are discussed here.

Root Mean Square. The root mean square (RMS) is the square root of the average of the sum of the squares of an infinite number of samples of the signal, as per Equation

9.1. It is also sometimes referred to as the standard deviation of the signal average. It can be effective in tracking system noise, but it does not provide meaningful information on which component is failing (Lebold, 2000).

$$\text{RMS} = \sqrt{\frac{1}{N} \left[\sum_{i=1}^N (S_i)^2 \right]} \quad (9.1)$$

where

N = number of data points

S = signal

Crest Factor (CF). The Crest Factor (Swansson, 1980) is calculated by dividing the maximum positive peak signal by the RMS value. This results in a normalized metric of the signal amplitude. A signal with few high-amplitude peaks produces a higher crest factor, such as the case with a damaged component (Decker, 2003). The formula for the crest factor is

$$\text{CF} = \frac{\text{Peak Level}}{\text{RMS}} \quad (9.2)$$

Kurtosis. The kurtosis is the normalized fourth moment of the vibration signal (Miller, 1985), given by Equation 9.3. The moment is normalized to the square of the variance of the signal. The kurtosis metric is a statistical measure of the number and amplitude of peaks in a signal. A signal with a high number of large amplitude peaks will have a higher kurtosis.

$$\text{Kurtosis} = \frac{N \sum_{i=1}^N (S_i - \bar{S})^4}{\left[\sum_{i=1}^N (S_i - \bar{S})^2 \right]^2} \quad (9.3)$$

where

N = number of data points

S = signal

$\bar{S}$ = mean signal value

M6A. The last diagnostic parameter presented is the M6A parameter. Introduced by Martin (1989), to detect surface damage, the M6A is an extension of the kurtosis (Decker, 2003). Rather than using the raw signal (S), the M6A uses the conditioned difference signal (Lebold, 2000). The M6A is calculated using EQN(9.4), which is very similar to the kurtosis expression in EQN (9.3). The only difference is in the magnitude of the exponents. The M6A parameter is expected to be more sensitive to peaks in the difference signal.

$$\text{M6A} = \frac{N^2 \sum_{i=1}^N (d_i - \bar{d})^6}{\left[\sum_{i=1}^N (d_i - \bar{d})^2 \right]^3} \quad (9.4)$$

where

d = difference signal

$\bar{d}$ = mean difference signal

For illustrative purposes, the following figure is provided. Figure 9.16 shows sample results of the four diagnostic parameters calculated from OH-58A transmission vibration

data: (a) RMS, (b) Crest Factor, (c) Kurtosis, and (d) M6A. The x -axis of each chart shows the time of operation in hours. The y -axis shows the parameter value. In each of the four cases, the point of detectable damage is indicated by a clear change in the diagnostic parameter behavior and marked by the vertical line labeled "inspection." The five data sets in each chart correspond to vibration data recorded at different sensor locations.

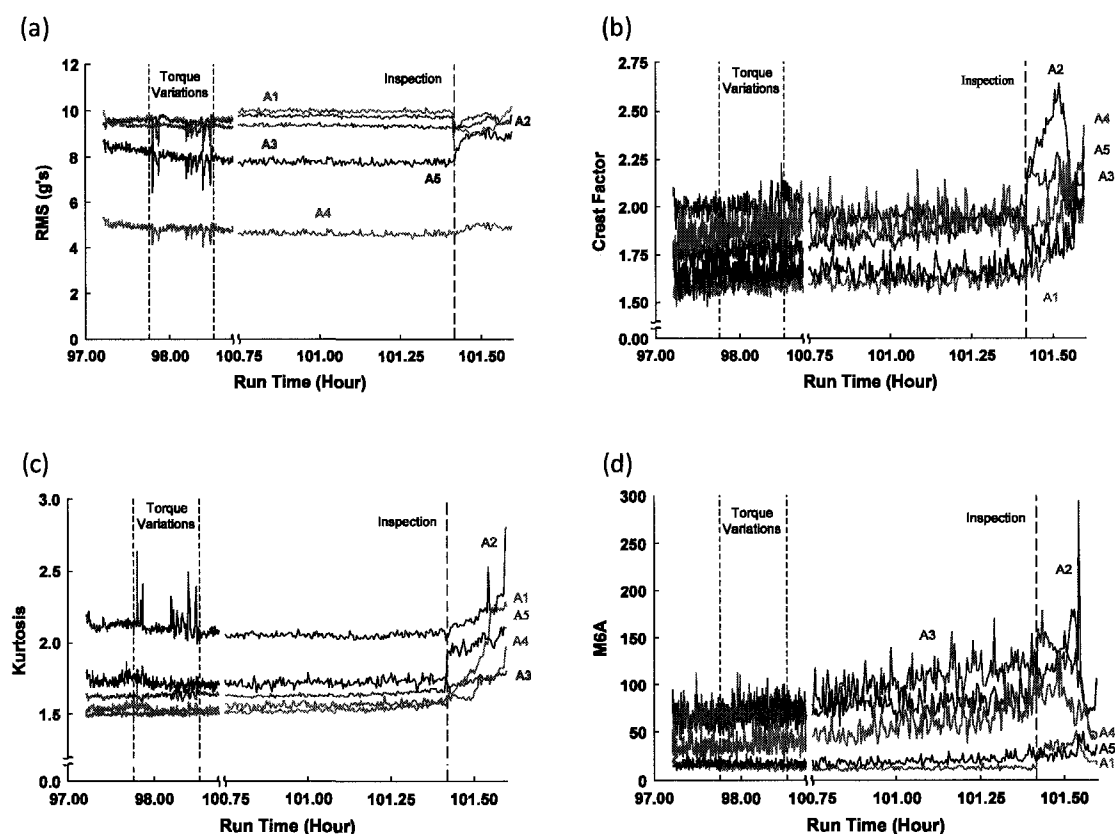


Figure 9.16: OH-58A Experimental Data Diagnostic Parameters (Decker, 2003)

9.3.5 Surface Contact Wear Simulation and Results

Gear surface contact wear of the input pinion was simulated by changing the gear-mesh loading profile as presented in Flodin (1997). Figure 9.17 depicts the two gear loading profiles representing normal and wear conditions. The normal loading

condition is represented by the solid line in Figure 9.17. It is an assumed loading pattern based upon the geometry of the input pinion, similar to the treatment of the nonlinear mesh stiffness formulation presented in Chapter 6. In the damage condition, the profile changes as illustrated by the dotted line in Figure 9.17. This represents the variation in loading brought on by the wear condition.

In a normal gear-mesh sequence, the load is shared equally between the number of teeth in contact. In systems exhibiting contact wear, the variations in force sharing are not as evenly distributed, illustrated by the dotted curve in Figure 9.17. This is due to the loss of tooth material such that the tooth contact surface area decreases. The loading therefore decreases as the tooth surface contact area is lost, changing the loading profile of the gear teeth. The curve in Figure 9.17 is a generic representation. Actual loading patterns follow the same trends, but must be determined experimentally or computationally for each gear pair.

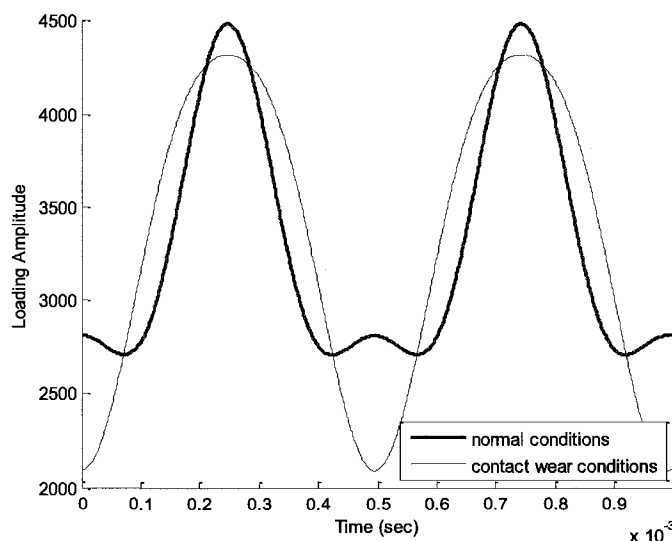


Figure 9.17: Gear Loading Profiles - Normal and Contact Wear Conditions

These forcing functions took the form of a second-order Fourier series as given in EQN (6.2). The coefficients used for the two conditions are presented in Table 9.6.

Table 9.6: Fourier Coefficients for Normal & Wear Gear Loading Profiles

n	normal conditions		wear conditions	
	a_n	b_n	a_n	b_n
0	0	-	0	
1	-842.5	6.364	-1055	-55.64
2	345.9	-5.226	-141.6	-14.97
w	20		20	

Simulations were run for both normal and wear conditions. The equations of motion for both systems were numerically integrated to obtain time domain responses. From these, a curve-fit was generated for the lateral-x motion of the transmission input node and the lateral-x motion of the carrier node of the output shaft. From the curve-fits, a representative vibration response was determined. For this simulation, the onset of damage was assumed to occur at 95 hours of operation, followed by full damage at 100 hours. The resulting time responses for the input and carrier nodes are shown in Figure 9.18. The vertical line marks the point of the damage condition. The input node experiences a significant change in the response. The carrier node is little affected by the contact wear of the input pinion.

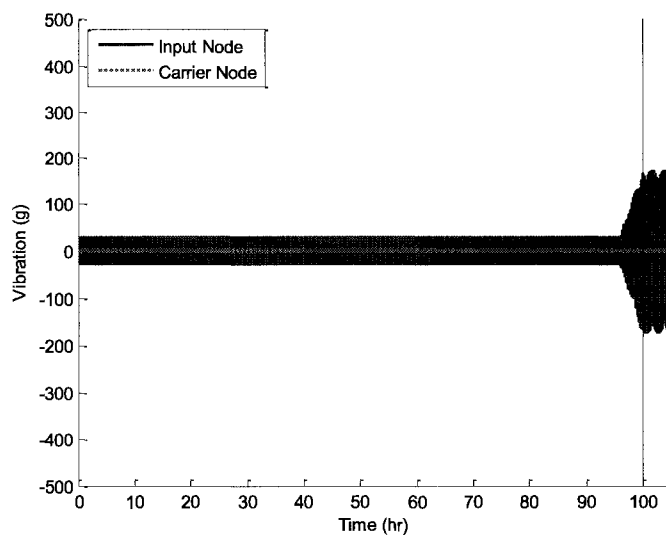


Figure 9.18: Time Response Showing Onset of Contact Wear Damage

The RMS, Crest Factor, Kurtosis, and M6A, were calculated for the responses of Figure 9.18. The parameter results are presented in Figure 9.19: (a) RMS, (b) Crest Factor, (c) Kurtosis, and (d) M6A. Just as in Figure 9.16, the vertical black dotted line indicates the point of full damage conditions.

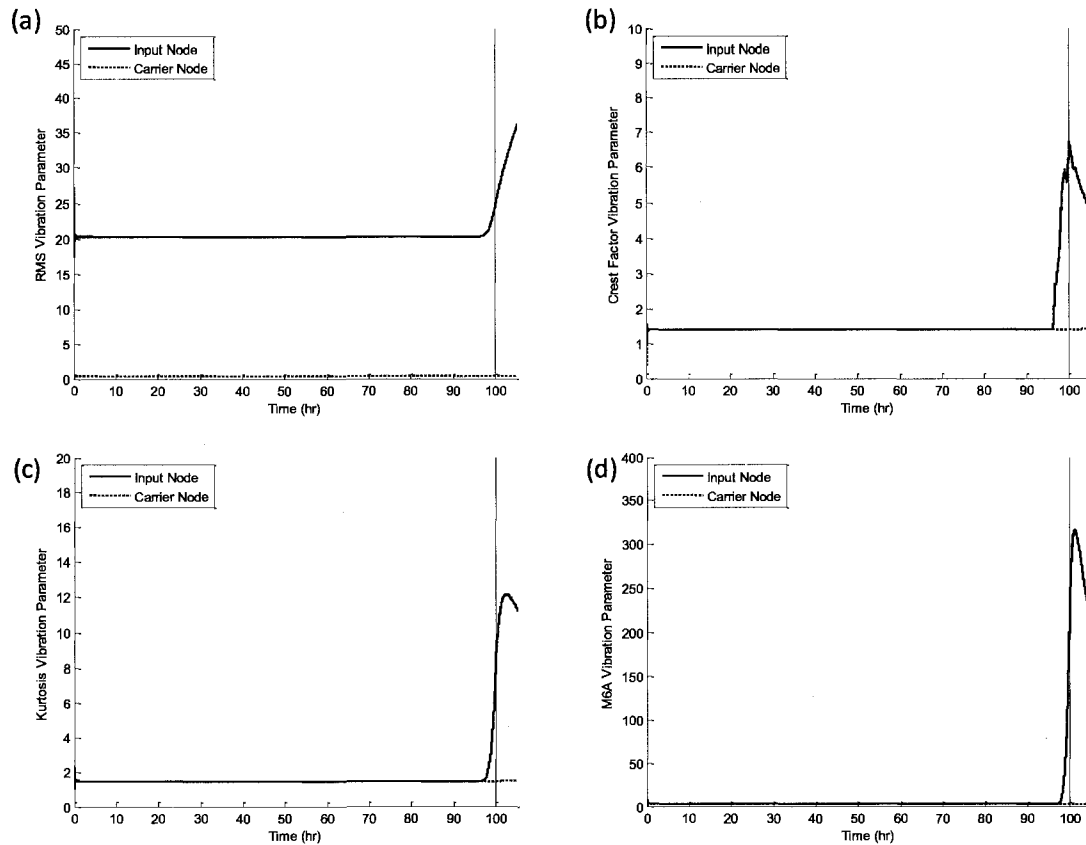


Figure 9.19: Diagnostic Parameters – Surface Contact Wear Results

The input node detects changes in operating conditions for all parameters. The carrier node response does not detect damage with any parameter, illustrating the sensitivity of the diagnostic parameters to nodal or “sensor” location. These results are consistent with Decker (2003). Additionally, the frequency responses at the two nodal locations were calculated through FFT of the time data and are depicted in Figure 9.20. The differences in the input node responses for the two conditions are evident. The carrier node responses match closely for the normal and damaged conditions, and at lower frequencies, they are nearly identical.

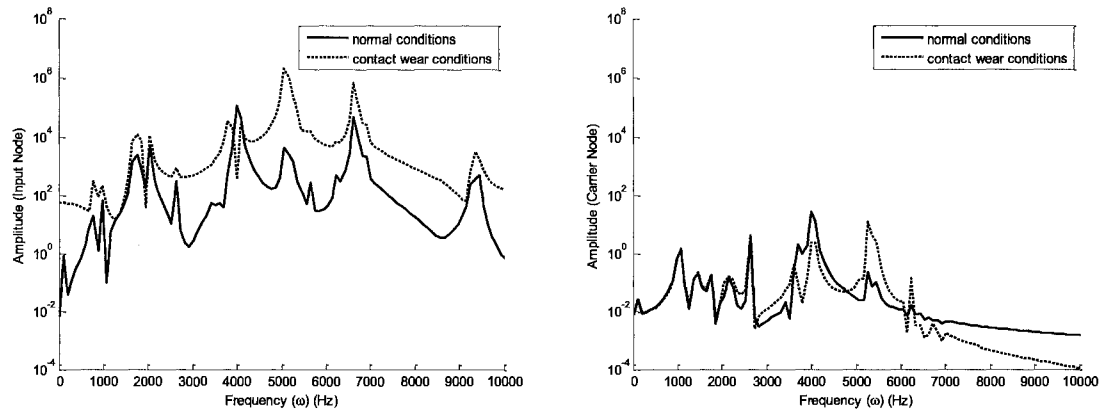


Figure 9.20: Contact Wear Frequency Response – Input (left) and Carrier (right) Nodes

Therefore, a metric was defined to better measure the difference between the frequency response patterns to detect when a component crosses the threshold from normal operation to damage. In this case, the square of the percentage error at each data point was used (Frequency Error²). This increases the visibility of a damaged condition by reducing the magnitude for small errors (<1), while increasing the magnitude of larger errors (>1). The Frequency Error² metric is illustrated in Figure 9.21.

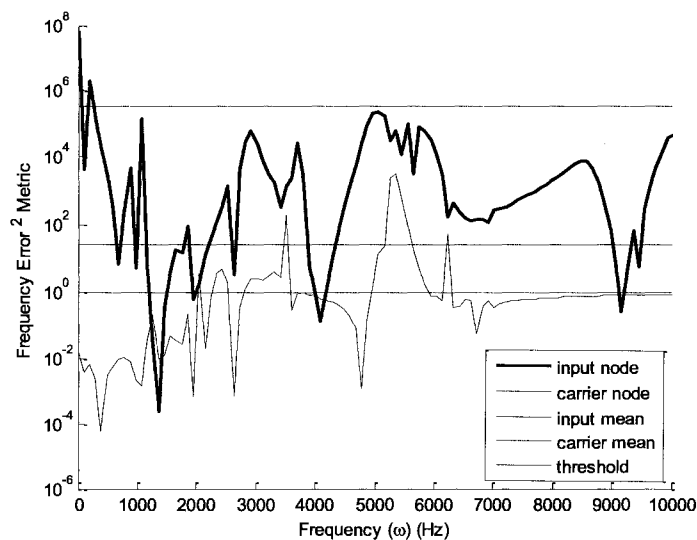


Figure 9.21: Contact Wear Frequency Error² Metric

The mean Frequency Error² is also shown in Figure 9.21 for the two nodal locations and compared to a threshold value of 1.0. The means representing both nodal locations are indeed above the threshold. This is a key observation regarding the Frequency Error² metric. In Figure 9.19, the four vibration diagnostic parameters did not register damage conditions at the carrier node in the time domain. The Frequency Error² metric does detect the damage. This suggests that analysis of the frequency responses may provide an equal or better indicator of damage than the traditional vibration patterns of the time domain under certain circumstances, and thus, both are significant. Figure 9.21 also illustrates the importance of sensor placement to adequately detecting damage. While both nodal locations have means above the threshold value, the input node is more indicative of damage than the carrier node.

9.3.6 Ball Bearing Damage Simulation and Results

The second fault condition is the failure of the triplex ball bearing acting at Node 5 of Shaft A. A bearing fault indication differs from that of a gear. When a bearing begins to fatigue, fault frequencies are generated, known as the fundamental defect frequencies (Dempsey, 2004b). There are four fault frequencies: the ball pass frequency outer race (BPFO), ball pass frequency inner race (BPFI), ball spin frequency (BSF), and the fundamental train frequency (FTF). The equations for each are given in EQN (9.5).

$$BPFO = \frac{N_b}{2} \left(1 - \frac{B_d}{P_d} \cos \theta \right) \frac{RPM}{60}$$

$$BPFI = \frac{N_b}{2} \left(1 + \frac{B_d}{P_d} \cos \theta \right) \frac{RPM}{60}$$

$$BSF = \frac{P_d}{2B_d} \left(1 - \left(\frac{B_d}{P_d} \right)^2 \cos^2 \theta \right) \frac{RPM}{60}$$

$$FTF = \frac{1}{2} \left(1 - \frac{B_d}{P_d} \cos \theta \right) \frac{RPM}{60}$$

(9.5)

where

B_d = ball or roller diameter

P_d = bearing pitch diameter

N_b = number of balls or rollers

θ = contact angle

The parameters and fundamental defect frequencies of the triplex bearing used in the simulation are given in Tables 9.7 and 9.8, respectively (Dempsey, 2004b).

Table 9.7: Triplex Bearing Parameters

Roller Diameter (mm)	Bearing Pitch Diameter (mm)	Number of Rolling Elements	Contact Angle (deg)
14.306	77.526	14	35

Table 9.8: Triplex Bearing Fundamental Defect Frequencies (Hz) at 6060 rpm

Ball Pass Frequency Inner Race (BPFI)	600.2
Ball Pass Frequency Outer Race (BPFO)	813.8
Ball Spin Frequency (BSF)	267.7
Fundamental Train Frequency (FTF)	42.9

Two conditions, representing a slight and moderate damage to the bearing, were simulated. The vibration response for each was generated using the same techniques as in the surface contact wear simulation. In these two simulations, the onset of damage occurred at 40 hours, with the full damage condition occurring at 45 hours. No distinguishable difference existed between the vibration responses of the two operating conditions, as it did in the contact wear simulation of Figure 9.18. This is illustrated in Figure 9.22 for the mild case of bearing damage. The moderate case is not depicted.

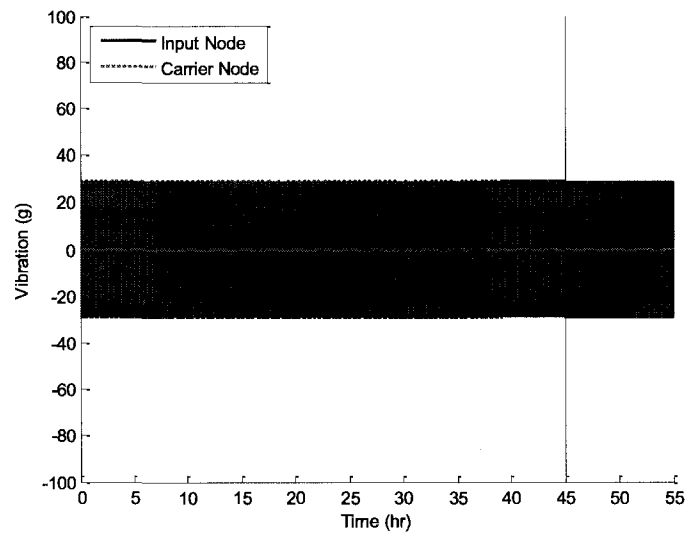


Figure 9.22: Time Response Showing Onset of Mild Bearing Damage

The four diagnostic parameters were calculated for the mild and moderate damage simulations. Figure 9.19 depicts the results of the moderate damage simulation. None of the parameters indicated any damage whatsoever.

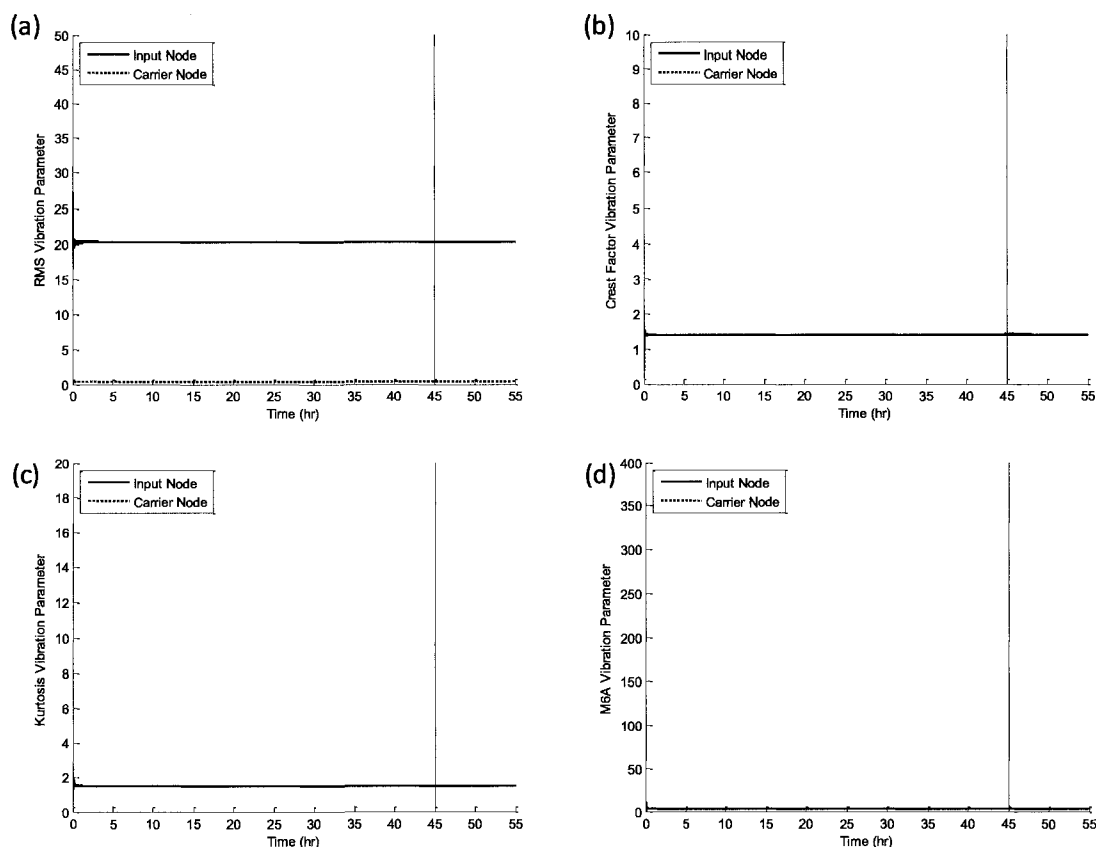


Figure 9.23: Diagnostic Parameters – Moderate Bearing Damage

In the frequency domain, however, the differences are noticeable due to the fundamental defect frequencies. The frequency responses of the input and carrier nodes are depicted in Figures 9.24 and 9.25 for the slight and moderate damage conditions, respectively. The vertical lines in each figure mark the location of the defect frequencies in the spectrum. Between the two figures, the only noticeable difference is with the amplitude of the defect frequencies. Moderate damage shows higher amplitudes, which is expected. The carrier node provides no indication of damage for either the slight or moderate condition. Furthermore, there is no apparent difference between the carrier responses for the two conditions.

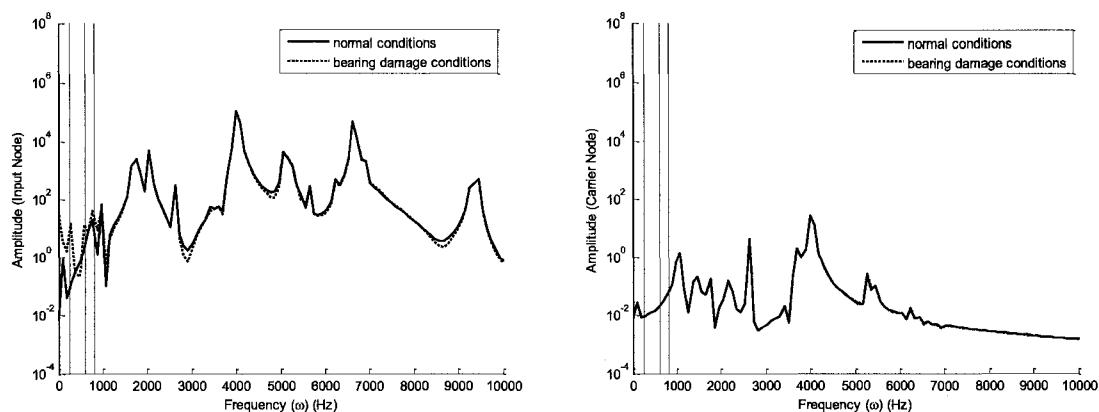


Figure 9.24: Slight Bearing Damage Frequency Response – Input (left) and Carrier (right) Nodes

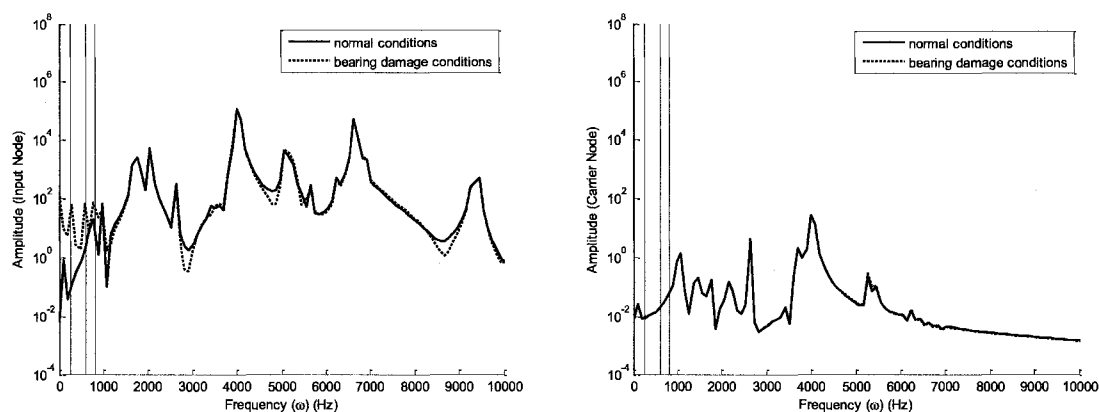


Figure 9.25: Moderate Bearing Damage Frequency Response – Input (left) and Carrier (right)

Finally, the Frequency Error² metric was calculated for both damage conditions and is presented in Figure 9.26. These results are informative. First, as in Figures 9.24 and 9.25, the severity of damage is indicated by the amplitudes of the input responses. The moderate damage response has a higher magnitude at the input node, resulting in a higher input mean value. The carrier responses for the two cases, however, are identical. The carrier node provides no indication of damage.

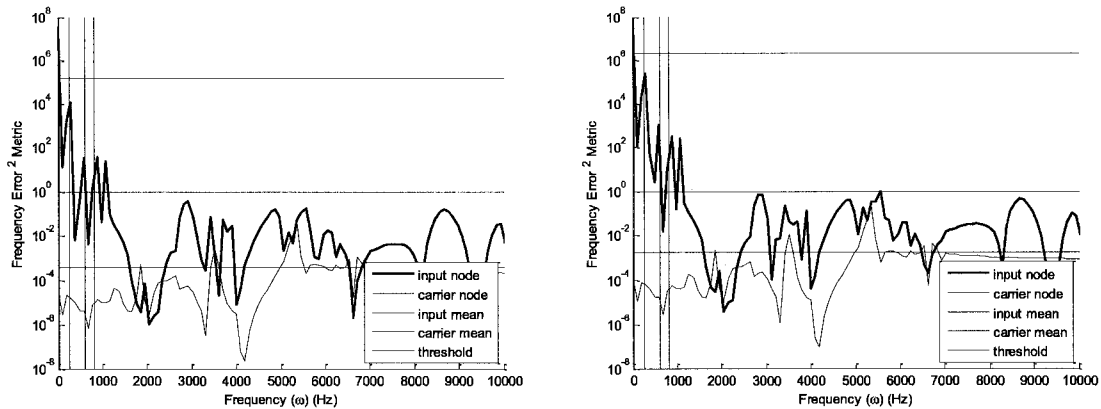


Figure 9.26: Slight (left) and Moderate (right) Bearing Damage Frequency Error² Metrics

The results of the bearing simulations support two conclusions. First, conventional diagnostic parameters, such as RMS, Crest Factor, Kurtosis, M6A, etc., are not effective in detecting bearing faults. Rather, quantifying the differences in the frequency responses shows more consistent results as suggested in Dempsey (2004b). The second conclusion has been mentioned repeatedly. Optimal sensor placement is critical to accurate fault detection.

9.4 Summary

This chapter is the culmination of the methodologies presented in the previous chapters. The transmission system, consisting of multiple geared shafts, and the finite element formulation have been introduced and discussed. The natural frequencies and mode shapes of the system have been presented, along with preliminary model validation. Results using the modal synthesis technique have also been presented.

Application to diagnostics and prognostics has also been discussed. First, common transmission failures are addressed as to the cause and nature of each failure. Four vibration diagnostic parameters currently used in health monitoring are introduced. A

new frequency response error metric, Frequency Error², has been introduced. The simulation results were consistent with previously published results.

Chapter 10

Conclusions and Recommendations

This research has presented a comprehensive set of methodologies for conducting geared rotor dynamic analysis of a complex system using physics-based modeling techniques. Furthermore, the importance of physics-based modeling to health monitoring and prognostics has been demonstrated. In closing, this chapter details some important conclusions from this research and recommendations for furthering the development of prognostic technology through the synergy of physics-based and probabilistic modeling.

10.1 Conclusions

The primary conclusions from this research include the following:

1. Gear-Mesh Model.

Through a rigorous derivation, a 12-DOF, lumped parameter, mesh stiffness matrix was presented for coupling two nodes of a finite element analysis. The matrix can incorporate both linear and nonlinear mesh stiffness coefficient values into the model. Application to a distributed parameter, 6-DOF model was discussed.

The lumped parameter model was validated using a simple gear application. The natural frequencies and mode shapes were characterized. Of particular interest was the emergence of coupled mode shapes due to the presence of the gear-mesh.

Linear and nonlinear mesh stiffness analyses were conducted to determine the significance of any differences between the two approaches. The analyses showed similar results.

The model was extended to bevel gears using Tregold's Approximation, producing excellent results. An existing lumped parameter, epicyclic gear model was presented and used for convenience.

The gear-mesh model has several capabilities. Additionally, it has some limitations in its present form. The capabilities of the 12-DOF mesh stiffness matrix are as follows.

- a. The model is a fully general formulation of a spur or helical gear pair.
- b. The model provides the contributions of all DOFs of a gear pair to a global coordinate system.
- c. The mesh stiffness coefficient, (K_g) , can accommodate constant or time-varying values for use in either linear or nonlinear analyses, respectively.
- d. The model can also incorporate mesh coefficients generated by more sophisticated analyses acting at the contact point to include elastohydrodynamic considerations.
- e. The formulation can be extended to a full bevel gear pair configuration, although in this research, Tregold's Approximation was utilized to demonstrate applicability to a bevel gear pair.
- f. The mesh matrix can be viewed as a reduced model of the full geometric contact model of the gear-mesh—i.e., the full geometric contact model of two teeth in mesh does not need to be directly incorporated in a rotor dynamic model. The full

contact model can be reduced to a 12-DOF matrix which can then be included in the rotor dynamic model.

Some of its limitations include

a. It does not include the effects of sliding action or friction, since all forces are assumed to operate at the pitch point along the line-of-action. As the gears experience sliding action, the point of contact moves up and down the tooth face. This results in a continuously changing point of contact during the mesh cycle, translating to a continuous shift in the geometric parameters of the mesh matrix. These instantaneous parameters are not captured in the mesh model.

b. In its present form, the model does not extend to systems where the axes of coupled shafts are in relative motion to one another, for instance, in epicyclic gear trains where the local axes of the planetary gears revolve around the central shaft axis.

2. Modal Synthesis.

A modal synthesis method was presented for geared, gyroscopic systems undergoing general damping via the state-space. The procedure diagonalized the state-space formulation of the equations of motion, providing a technique for modal reduction and model simplification.

The technique was applied to two model applications. The modal reduction showed good accuracy with the full system models. This demonstrates the potential for application of such a modal reduction technique to a full geometric model. The full geometric model would necessarily incur several thousand DOFs. A modal reduction scheme such as the one presented might allow for a more sophisticated model to be used with negligible penalties in processing time and cost.

The effectiveness of the procedure was questionable when dealing with gyroscopic and geared effects.

3. System Transmission Model.

The OH-58A helicopter transmission system was modeled with preliminary validation.

Common transmission faults to include the causes and nature of the types of failure were presented and discussed.

Two of these faults were simulated to show the system responses. Conventional vibration diagnostic parameters were used to monitor the seeded faults.

Another health metric, the Frequency Error² was introduced as an effective means of monitoring component condition.

The model demonstrated the importance of sensor location to accurately detecting damage.

10.2 Recommendations

Many recommendations for follow-on research emerged during this effort, to include the following areas:

1. Gear-Mesh Model.

The examination of combined tooth stiffness and elasto-hydrodynamic (EHD) effects on the mesh stiffness. Both tooth stiffness and EHD determine the mesh stiffness. The incorporation of both can provide a truly robust and sophisticated model of the dynamics of the gear-mesh and its contribution to the overall dynamics of the system.

Further investigation of nonlinear analysis and any significant advantages versus the linear treatment. In Chapter 6, the differences between the linear and nonlinear analyses

were negligible. However, further nonlinear analyses of different systems may reveal system characteristics not easily visible in the linear analysis.

Full DOF treatment of epicyclic trains. The epicyclic model used here consisted of spur gears in the sun, ring, and planets. Many epicyclic configurations use spur or herringbone arrangements, for which the current model is acceptable. If configurations employ a helical arrangement, then a more robust model is needed.

Inclusion of backlash and transmission error. For a more representative model, these gear effects must be included.

2. Modal Synthesis.

Further investigation of modal synthesis techniques to simplify the modeling process. The procedure as presented is rigorous. The perturbation technique mentioned may have potential for reducing the rigor of this methodology.

Extension to design analysis and optimization. Modal analysis techniques are often used for system identification and optimization. This technique may provide a benefit in these areas as well.

3. System Transmission Model.

Further validation of transmission model. Further validation is required to enhance the model's robustness. This occurs through modeling all known faults and operating conditions, and comparing the results with experimental data. This also includes determining how different types of failures propagate and the ability to model that propagation accurately.

System identification. System identification allows the incorporation of experimental data into the model. Furthermore, system identification techniques can help determine how to best model a fault from a physics-based perspective.

Inclusion of transmission casing in model. The transmission model contains only the shafts, gears, and bearings within the transmission. The casing is isolated via the bearings. The casing should eventually be included to determine the effect on the system model. Furthermore, the casing provides the conduit for the vibration of the transmission to the rest of the aircraft.

Extension to entire drive train. Inclusion of the entire drive train to include tail shaft and tail rotor gearboxes would greatly increase the robustness of the model.

Incorporation of gear life predictions and uncertainty in model. Life prediction must be included to develop prognostic capabilities. This is perhaps one of the most critical pieces to maturing the predictive capability of health monitoring systems. Additionally, uncertainty methods must be employed to include design of experiment and Monte Carlo simulation techniques. This adds robustness to the model and increases the credibility of the information it provides.

Inclusion of signal noise into the model. When processing vibration data, many times it must be filtered to obtain a useful signal. The inclusion of noise would make the model signal more representative of actual data. By using signal conditioning techniques, the model results would be closer to the actual system, making it easier to monitor and predict component condition.

Characterization of mode shapes. As illustrated in Chapter 9, the system mode shapes are complex. Further investigation and characterization of the transmission mode shapes is important for understanding the dynamics of the system.

Incorporation of harmonic frequencies within the spectrum. For accuracy, the model must incorporate all harmonic frequencies within the designated spectrum.

Development of a Graphic User Interface (GUI) tool. A GUI tool provides a user-friendly environment for reducing the time necessary in constructing, seeding, and executing the model, and providing concise, neat results.

Capture of influencing effects on transmission. For increased capability, the effects of the main rotor and engine on transmission vibration must be included.

Development of sensor locator tool. The physics-based model shows promise for determining optimum sensor placement.

10.3 Summary

In summary, this research has presented a comprehensive investigation into the considerations behind building a rotor dynamic system model for health monitoring and prognostic purposes. Physics-based modeling has not been as popular as time-series or statistical methods in modeling approaches. However, this research has demonstrated the robustness of the physics-based approach in simulating multiple operating and fault conditions. This effort provides a means of building the vibration signature database called for over a decade ago. In concluding, the combined advantages of the probabilistic and physics-based approaches will continue to further the current state-of-the-art in drive system technology.

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Appendix A

Finite Element Matrix Equations

This appendix presents the matrices used in Equation 4.29, with the exception of the gear-mesh matrix, $[\mathbf{K}]_{mesh}$. These matrices are found from a variety of sources to include Younan (2004), ANSYS (2004), Chen (2005), and Yokoyama (1990, 1996).

A.1 12-DOF Matrices

In this finite element formulation, the 12-DOF matrices model only the rotating shaft and the gear-mesh. The mesh matrix is presented in Chapter 5 and Appendix B. From the assumptions in Chapter 4, the shaft itself contains no damping. Only the mass, stiffness, and gyroscopic matrices are presented.

Two preliminary parameters for both the shaft mass and stiffness matrices are the shape factor, (κ) , and the transverse shear parameter, (Φ) .

For a solid shaft:

$$\kappa = \frac{6(1+\nu)}{(7+6\nu)} \quad (\text{A.1})$$

For a thin-walled shaft:

$$\kappa = \frac{2(1+\nu)}{(4+3\nu)} \quad (\text{A.2})$$

where (ν) is Poisson's Ratio.

The transverse shear parameter, (Φ) , is given by the following:

$$\Phi = \frac{12EI}{\kappa AGl^2} \quad (\text{A.3})$$

where

A = Shaft cross-sectional area

E = Young's modulus

G = Shear modulus

I = Area moment of inertia

l = Element length

A.1.1 Shaft Consistent Mass Matrix

The following relationships from Yokoyama (1990, 1996) are used in determining the element matrices for the consistent mass matrix. Equation A.4 defines the radius of gyration, (r_g) .

$$r_g = \sqrt{\frac{I}{A}} \quad (\text{A.4})$$

$$a = \frac{\rho Al}{(1+\Phi)^2} \left(\frac{13}{35} + \frac{7}{10}\Phi + \frac{2}{3}\Phi^2 + \frac{6}{5} \left(\frac{r_g}{l} \right)^2 \right) \quad (\text{A.5})$$

$$b = \frac{\rho Al}{(1+\Phi)^2} \left(\frac{9}{70} + \frac{3}{10}\Phi + \frac{1}{6}\Phi^2 - \frac{6}{5} \left(\frac{r_g}{l} \right)^2 \right) \quad (\text{A.6})$$

$$c = \frac{\rho A l^2}{(1+\Phi)^2} \left(\frac{11}{210} + \frac{11}{120} \Phi + \frac{1}{24} \Phi^2 + \left(\frac{1}{10} - \frac{1}{2} \Phi \right) \left(\frac{r_g}{l} \right)^2 \right) \quad (\text{A.7})$$

$$d = \frac{\rho A l^2}{(1+\Phi)^2} \left(\frac{13}{420} + \frac{3}{40} \Phi + \frac{1}{24} \Phi^2 - \left(\frac{1}{10} - \frac{1}{2} \Phi \right) \left(\frac{r_g}{l} \right)^2 \right) \quad (\text{A.8})$$

$$f = -\frac{\rho A l^3}{(1+\Phi)^2} \left(\frac{1}{140} + \frac{1}{60} \Phi + \frac{1}{120} \Phi^2 + \left(\frac{1}{30} + \frac{1}{6} \Phi - \frac{1}{6} \Phi^2 \right) \left(\frac{r_g}{l} \right)^2 \right) \quad (\text{A.9})$$

$$g = \frac{\rho A l^3}{(1+\Phi)^2} \left(\frac{1}{105} + \frac{1}{60} \Phi + \frac{1}{120} \Phi^2 + \left(\frac{2}{15} + \frac{1}{6} \Phi + \frac{1}{3} \Phi^2 \right) \left(\frac{r_g}{l} \right)^2 \right) \quad (\text{A.10})$$

The element consistent mass matrix, superscripted (*e*) for the element number, follows as

$$[M]_s^{(e)} = \begin{bmatrix} a & & & & & & & & & & \\ 0 & a & & & & & & & & & \\ 0 & 0 & \frac{1}{3} \rho A l & & & & & & & & \\ 0 & -c & 0 & g & & & & & & & \\ c & 0 & 0 & 0 & g & & & & & & \\ 0 & 0 & 0 & 0 & 0 & \frac{J \rho A l}{3A} & & & & & \\ b & 0 & 0 & 0 & d & 0 & a & & & & \\ 0 & b & 0 & -d & 0 & 0 & 0 & a & & & \\ 0 & 0 & \frac{1}{6} \rho A l & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} \rho A l & & \\ 0 & d & 0 & f & 0 & 0 & 0 & c & 0 & g & \\ -d & 0 & 0 & 0 & f & 0 & -c & 0 & 0 & 0 & g \\ 0 & 0 & 0 & 0 & 0 & \frac{J \rho A l}{6A} & 0 & 0 & 0 & 0 & 0 & \frac{J \rho A l}{3A} \end{bmatrix} \quad (\text{A.11})$$

where

ρ = shaft density

J = polar area moment of inertia

A.1.2 Shaft Stiffness Matrix

Equations A.13 through A.16 develop the shaft stiffness matrix:

$$a = \frac{12EI}{l^3(1+\Phi)} \quad (\text{A.12})$$

$$b = \frac{6EI}{l^2(1+\Phi)} \quad (\text{A.13})$$

$$c = \frac{(4+\Phi)EI}{l(1+\Phi)} \quad (\text{A.14})$$

$$d = \frac{(2-\Phi)EI}{l(1+\Phi)} \quad (\text{A.15})$$

$$[K]_s^{(e)} = \begin{bmatrix} a & & & & & & & & & \\ 0 & a & & & & & & & & \\ 0 & 0 & \frac{AE}{l} & & & & & & & \\ 0 & -b & 0 & c & & & & & & \\ b & 0 & 0 & 0 & c & & & & & \\ 0 & 0 & 0 & 0 & 0 & \frac{GJ}{L} & & & & \\ -a & 0 & 0 & 0 & -b & 0 & a & & & \\ 0 & -a & 0 & b & 0 & 0 & 0 & a & & \\ 0 & 0 & -\frac{AE}{l} & 0 & 0 & 0 & 0 & 0 & \frac{AE}{l} & \\ 0 & -b & 0 & d & 0 & 0 & 0 & b & 0 & c \\ b & 0 & 0 & 0 & d & 0 & -b & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{GJ}{L} & 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} \\ \\ \\ sym \\ \\ \\ \\ \\ \\ \\ \end{matrix} \quad (\text{A.16})$$

A.1.3 Shaft Gyroscopic Matrix

The shaft gyroscopic matrix is skew symmetric and is multiplied by the shaft rotational speed, (Ω) .

$$a = \frac{\rho Al}{(1+\Phi)^2} \frac{6}{5} \left(\frac{r_g}{l} \right)^2 \quad (\text{A.17})$$

$$b = -\frac{\rho Al}{(1+\Phi)^2} \left(\frac{1}{10} - \frac{1}{2} \Phi \right) \left(\frac{r_g^2}{l} \right) \quad (\text{A.18})$$

$$c = \frac{\rho Al}{(1+\Phi)^2} \left(\frac{2}{15} + \frac{1}{6} \Phi - \frac{1}{3} \Phi^2 \right) r_g^2 \quad (\text{A.19})$$

$$d = -\frac{\rho Al}{(1+\Phi)^2} \left(\frac{1}{30} + \frac{1}{6} \Phi - \frac{1}{6} \Phi^2 \right) r_g^2 \quad (\text{A.20})$$

$$\Omega [G]_s^{(e)} = 2\Omega \begin{bmatrix} 0 & & & & & & & & & & \\ -a & 0 & & & & & & & & & \\ 0 & 0 & 0 & & & & & & & & \\ -b & 0 & 0 & 0 & & & & & & & \\ 0 & -b & 0 & -c & 0 & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & & & & & \\ 0 & -a & 0 & -b & 0 & 0 & 0 & & & & \\ a & 0 & 0 & 0 & -b & 0 & -a & 0 & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & & \\ -b & 0 & 0 & 0 & d & 0 & b & 0 & 0 & 0 & \\ 0 & -b & 0 & -d & 0 & 0 & 0 & b & 0 & -c & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (\text{A.21})$$

skew
sym

A.2 6-DOF Matrices

The following matrices represent mass, stiffness, gyroscopics, and damping effects acting at individual nodes. This incorporates gears and bearings acting at specific nodes into the model.

A.2.1 Rotor Disc (Gear) Matrices

From the assumptions in Chapter 4, the gears or discs are assumed to be rigid. Therefore, only the mass and gyroscopic matrices are necessary for the finite element analysis. Since the gears are also assumed to be point masses, only a 6-DOF model is necessary.

$$[M]_d = \begin{bmatrix} m_d & & & & & \\ 0 & m_d & & & & \\ 0 & 0 & m_d & & & \\ 0 & 0 & 0 & I_T & & \\ 0 & 0 & 0 & 0 & I_T & \\ 0 & 0 & 0 & 0 & 0 & I_P \end{bmatrix} \quad (A.22)$$

The gyroscopic matrix is

$$\Omega[G]_d = \Omega \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I_P & 0 \\ 0 & 0 & 0 & -I_P & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (A.23)$$

where

$$m_d = \text{mass of disk (gear)}$$

I_T = transverse mass moment of inertia

I_p = polar mass moment of inertia

A.2.2 Bearing Matrices

The bearings hold the spinning shaft in place and are assumed to be rigid. As such, they contribute no mass to the system, but they do store and dissipate energy from the shaft. Thus, the stiffness and damping matrices are presented. Both matrices are symmetric. The off-diagonal terms in either matrix may or may not equal zero, depending upon assumptions or the types of bearings used.

$$[K]_b = \begin{bmatrix} k_{xx} & & & & \\ & k_{yy} & & & \\ & & k_{zz} & & \\ & & & K_{\theta x \theta x} & \\ & & & & K_{\theta y \theta y} \\ k_{x \theta z} & & \dots & & & K_{\theta z \theta z} \end{bmatrix} \quad (\text{A.24})$$

Similarly,

$$[C]_b = \begin{bmatrix} c_{xx} & & & & \\ & c_{yy} & & & \\ & & c_{zz} & & \\ & & & C_{\theta x \theta x} & \\ & & & & C_{\theta y \theta y} \\ c_{x \theta z} & & \dots & & & C_{\theta z \theta z} \end{bmatrix} \quad (\text{A.25})$$

Appendix B

12-DOF Mesh Stiffness Matrix

This appendix presents the 12-DOF mesh stiffness matrix from the derivation in Chapter 5. For ease of writing, the following shorthand is used.

$$\begin{aligned}
 c\varphi &= \cos \varphi \\
 s\varphi &= \sin \varphi \\
 c\phi_x &= \cos \phi_x = \cos \beta \cos \alpha_n \\
 c\phi_y &= \cos \phi_y = \sin \alpha_n \\
 c\phi_z &= \cos \phi_z = \sin \beta \sin \alpha_n
 \end{aligned} \tag{B.1}$$

From Equation 5.1, the gear-mesh stiffness matrix can be written as

$$\begin{Bmatrix} \mathbf{F}_i \\ \mathbf{F}_j \end{Bmatrix} = [\mathbf{K}]_{mesh} \begin{Bmatrix} \mathbf{q}_i \\ \mathbf{q}_j \end{Bmatrix} = K_g \begin{bmatrix} [\mathbf{K}_{ii}] & [\mathbf{K}_{ji}] \\ [\mathbf{K}_{ij}] & [\mathbf{K}_{jj}] \end{bmatrix} \begin{Bmatrix} \mathbf{q}_i \\ \mathbf{q}_j \end{Bmatrix} \tag{B.2}$$

The four sub-matrices of Equation B.2 are presented in Equations B.3 through B.6.

Appendix C

Planetary Gear System Matrices

This appendix presents the matrix components for the planetary gear model of Equation 8.12, presented in Chapter 8 (Parker, 1999).

$$\begin{aligned}\mathbf{M} &= \text{diag}(\mathbf{M}_c, \mathbf{M}_r, \mathbf{M}_s, \mathbf{M}_1, \dots, \mathbf{M}_n) \\ \mathbf{M}_j &= \text{diag}(m_j, m_j, I_j/r_j^2), j = c, r, s, 1, \dots, n\end{aligned}\tag{C.1}$$

$$\begin{aligned}\mathbf{G} &= \text{diag}(\mathbf{G}_c, \mathbf{G}_r, \mathbf{G}_s, \mathbf{G}_1, \dots, \mathbf{G}_n) \\ \mathbf{G}_j &= \begin{bmatrix} 0 & -2m_j & 0 \\ 2m_j & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, j = c, r, s, 1, \dots, n\end{aligned}\tag{C.2}$$

$$\mathbf{K}_\Omega = \text{diag}(m_c, m_c, 0, m_r, m_r, 0, m_s, m_s, 0, m_1, m_1, 0, \dots, m_n, m_n, 0)\tag{C.3}$$

$$\begin{aligned}\mathbf{K}_b &= \text{diag}(\mathbf{K}_{cb}, \mathbf{K}_{rb}, \mathbf{K}_{sb}, \mathbf{0}, \dots, \mathbf{0}) \\ \mathbf{K}_{jb} &= \text{diag}(k_{jx}, k_{jy}, k_{ju}), j = c, r, s\end{aligned}\tag{C.4}$$

$$\mathbf{K}_m = \begin{bmatrix} \Sigma \mathbf{K}_{c1}^n & \mathbf{0} & \mathbf{0} & \mathbf{K}_{c2}^1 & \mathbf{K}_{c2}^2 & \dots & \mathbf{K}_{c2}^n \\ & \Sigma \mathbf{K}_{r1}^n & \mathbf{0} & \mathbf{K}_{r2}^1 & \mathbf{K}_{r2}^2 & \dots & \mathbf{K}_{r2}^n \\ & & \Sigma \mathbf{K}_{s1}^n & \mathbf{K}_{s2}^1 & \mathbf{K}_{s2}^2 & \dots & \mathbf{K}_{s2}^n \\ & & & \mathbf{K}_{pp}^1 & \mathbf{0} & \dots & \mathbf{0} \\ & sym & & & \ddots & \dots & \mathbf{0} \\ & & & & & \ddots & \vdots \\ & & & & & & \mathbf{K}_{pp}^n \end{bmatrix}\tag{C.5}$$

$$\mathbf{K}_{pp}^n = \mathbf{K}_{c3}^n + \mathbf{K}_{r3}^n + \mathbf{K}_{s3}^n \quad (\text{C.6})$$

$$\mathbf{K}_{c1}^n = k_{pn} \begin{bmatrix} 1 & 0 & -\sin \psi_n \\ & 1 & \cos \psi_n \\ \text{sym} & & 1 \end{bmatrix} \quad (\text{C.7})$$

$$\mathbf{K}_{c2}^n = k_{pn} \begin{bmatrix} -\cos \psi_n & \sin \psi_n & 0 \\ -\sin \psi_n & -\cos \psi_n & 0 \\ 0 & -1 & 0 \end{bmatrix} \quad (\text{C.8})$$

$$\mathbf{K}_{c3}^n = \text{diag}(k_{pm}, k_{pn}, 0) \quad (\text{C.9})$$

$$\mathbf{K}_{r1}^n = k_{rn}(t) \begin{bmatrix} \sin^2 \psi_{rn} & -\cos \psi_{rn} \sin \psi_{rn} & -\sin \psi_{rn} \\ & \cos^2 \psi_{rn} & \cos \psi_{rn} \\ \text{sym} & & 1 \end{bmatrix} \quad (\text{C.10})$$

$$\mathbf{K}_{r2}^n = k_{rn}(t) \begin{bmatrix} -\sin \psi_{rn} \sin \alpha_r & \sin \psi_{rn} \cos \alpha_r & \sin \psi_{rn} \\ \cos \psi_{rn} \sin \alpha_r & -\cos \psi_{rn} \cos \alpha_r & -\cos \psi_{rn} \\ \sin \alpha_r & -\cos \alpha_r & -1 \end{bmatrix} \quad (\text{C.11})$$

$$\mathbf{K}_{r3}^n = k_{rn}(t) \begin{bmatrix} \sin^2 \alpha_r & -\cos \alpha_r \sin \alpha_r & -\sin \alpha_r \\ & \cos^2 \alpha_r & \cos \alpha_r \\ \text{sym} & & 1 \end{bmatrix} \quad (\text{C.12})$$

$$\mathbf{K}_{s1}^n = k_{sn}(t) \begin{bmatrix} \sin^2 \psi_{sn} & -\cos \psi_{sn} \sin \psi_{sn} & -\sin \psi_{sn} \\ & \cos^2 \psi_{sn} & \cos \psi_{sn} \\ \text{sym} & & 1 \end{bmatrix} \quad (\text{C.13})$$

$$\mathbf{K}_{s2}^n = k_{sn}(t) \begin{bmatrix} \sin \psi_{sn} \sin \alpha_s & \sin \psi_{sn} \cos \alpha_s & -\sin \psi_{sn} \\ -\cos \psi_{sn} \sin \alpha_s & -\cos \psi_{sn} \cos \alpha_s & \cos \psi_{sn} \\ -\sin \alpha_s & -\cos \alpha_s & 1 \end{bmatrix} \quad (\text{C.14})$$

$$\mathbf{K}_{s3}^n = k_{sn}(t) \begin{bmatrix} \sin^2 \alpha_s & \cos \alpha_s \sin \alpha_s & -\sin \alpha_s \\ & \cos^2 \alpha_s & -\cos \alpha_s \\ \text{sym} & & 1 \end{bmatrix} \quad (\text{C.15})$$

$$\begin{aligned} \psi_{sn} &= \psi_n - \alpha_s \\ \psi_{rn} &= \psi_n + \alpha_r \end{aligned} \quad (\text{C.16})$$